

ARTDINK

BASICSTUDIO
「パワフルゲーム工房」

USERS' MANUAL

ユーザーズマニュアル

Introduction

About the User's Manual

The User's Manual provides detailed explanations of the commands used in the various tools included in "BASIC STUDIO Powerful Game Workshop."

The BASIC interpreter "EZ-BASIC" comes with a "Reference Manual" that contains detailed information about all commands, and an introductory manga "Let's get started! EZ -BASIC" that explains EZ-BASIC in an easy-to-understand manner. BASIC beginners should start with "Let's get started! EZ-BASIC" and then read the "User's Manual," using the "Reference Manual" as a dictionary.

Please keep this book in a safe place as it will not be reissued.

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B-SHELL

B-SHELL Features

"B-SHELL" allows you to copy and delete files used in "BASIC STUDIO" and run BASIC programs.

Controls

Controller Explanation

Analog Controller (DUALSHOCK 2)



※"BASIC STUDIO Powerful Game Workshop" supports analog controllers (DUALSHOCK 2). The LED indicator will always be red while the software is running.
 ※Vibration can be turned on/off in the options on the title screen.

■BSHELL Controls

Directional Buttons	Move cursor
Left stick	Move cursor
○ Button	Confirm
△ Button	Open menu
X Button	Return
START+SELECT Buttons	Terminate program (Only while executing EZ-BASIC Program)
SELECT Button	Print Screen Function While connected to popegg

※The SELECT button may not be used for Print Screen during program execution.

Start B-SHELL

Device selection

Once "BASIC STUDIO" has started, select "B-SHELL" from the title screen. This will launch the execution shell "B-SHELL" and display the following screen. This screen displays icons of "devices" that can load and save files used by "BASIC STUDIO," such as the disc inserted in the disc tray and PlayStation 2 memory cards (8MB).



Highlight a device with the cursor and press the circle button to view the "BASIC STUDIO" files stored within the selected device.

Directory Selection

Once you have selected a device, you will see a screen like this, with several icons displayed. These are called "directories," and are like "containers" that hold files that can be used with "BASIC STUDIO."

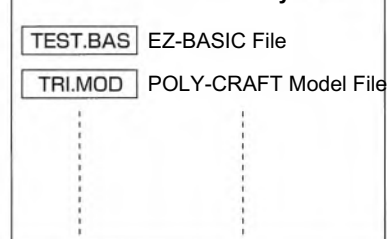
The directories stored in the selected device are displayed on the screen as icons and directory names.

Place the cursor on a directory and press the Δ button to display the directory information and directory menu.

The directory structure can be easily illustrated as follows: The EZ-BASIC file "TEST.BAS" and the POLY-CRAFT model file "TRI.MOD" are stored in a container called the "SYSTEM" directory.



SYSTEM Directory



Directory Menu

When you display the directory menu, the information and menu for the selected directory are displayed.

Directory Information

ディレクトリ名 (Directory Name) The name of the directory.

作成日時 (Creation date) The date and time the directory was created.



Directory Menu

開く (Open)

You can open the selected directory to view the files it contains.

コピー (Copy)

Copies the selected directory and all files to another device. When you select Copy, you will return to the device selection screen. Use the directional keys to select the device you want to copy to and press the O button to copy.



削除 (Delete)

Deletes the currently selected directory and all its contents. Please note that deleted directories and files cannot be restored.

名前の変更 (Rename)

You can change the name of the currently selected directory. If you select Rename, a name input window will appear to enter the new directory name.

"Directories on the disc tray cannot be "deleted" or "renamed."



File selection

When you open a directory, you can view the files contained in it. As with the directory view, if you select a file and press the Δ button, file information and the file menu will be displayed.



File Menu

When you display the file menu, information about the selected file and the menu are displayed.

File Information

ファイル名 (File Name) The file name.



ファイルの種類 (File type)

The file's type. There are the following file types.

Extension	File type	VRS	String variable file	MOD	Model file
BAS	Program file	TEX	Texture file	BGM	BGM File
VRN	Numeric variable file	PAL	Palette file	SED	SE File

サイズ (Size) Size of the saved file.

更新日時 (Update date) The date and time the last time the file was saved.

File Menu

開く (Open)

For program files only, you can run the program by executing the "Open" command.

※To run the program, the program file to be run (and any other files if used) must be on the PlayStation 2 memory card (8MB). Also, please make sure the PlayStation 2 memory card (8MB) is connected to the memory card slot where you created the program.

コピー (Copy)

Copies the selected file. If you select Copy, the following screen will appear.

1) The device selection screen will be displayed. Use the arrow keys to select the device you want to copy.

2-A) Copy to a new directory

If you want to copy to a new directory, select "New Directory" when selecting the destination directory, and enter the directory name to copy to the new directory.

2-B) Copy to an existing directory

If you want to copy to an existing directory, select the destination directory from the directory list.

削除 (Delete)

Deletes the currently selected file. Please note that deleted files cannot be restored.

名前の変更 (Rename)

You can change the name of the currently selected file. If you select Rename, a name entry window will appear, allowing you to enter a new file name.

※Only "copy" can be performed on files in the disc tray.



How to start a sample program

To launch a sample game included on the "BASIC STUDIO DISC," first copy the sample game from the "BASIC STUDIO DISC" to the "PlayStation 2" dedicated memory card (8MB), then launch the copied sample game. Let's start the sample game. Please make sure you have a PlayStation 2 dedicated memory card (8MB) with enough free space inserted into MEMORY CARD slot 1.

Copying a sample game from the BASIC STUDIO DISC to the PlayStation 2 memory card (8MB)

Selecting a Device

"BASIC STUDIO" has started, select "B-SHELL" from the title screen. This will launch the execution shell "B-SHELL" and display the following screen.

This time, we want to copy the sample game included on the "BASIC STUDIO DISC," so select "Disc Tray."



Directory Selection

Next, you will see a screen like this, showing the directory containing the files containing the samples.

The program files and image files used by one sample game are all in one directory. To copy a directory, move the cursor to the directory of the sample game you want to start and press the Δ button to display the directory menu. In this example, move the cursor to the "CARNAGE" directory and press the Δ button.



Copying a Directory

"CARNAGE" directory folder, select "Copy" from the directory menu and set the copy destination to "MEMORY CARD Slot 1." This completes the copy.



Start the sample game from the PlayStation 2 memory card (8MB)

Next, run the program file "CARNAGE.BAS" in the "CARNAGE" directory copied to the PlayStation 2 memory card (8MB) to launch the sample game.

Selecting a Device

As before, select the device containing the file you want to access. The "CARNAGE" directory has been copied to the PlayStation2 memory card (8MB) connected to MEMORY CARD Slot 1. This time, select "MEMORY CARD Slot 1."

View the contents of a directory

Next, open the "CARNAGE" directory to view the files inside. As mentioned earlier, you can view the contents of a directory by displaying the directory menu and then selecting "Open," or by placing the cursor over the directory and pressing the O button.



Launch the sample game from the file menu

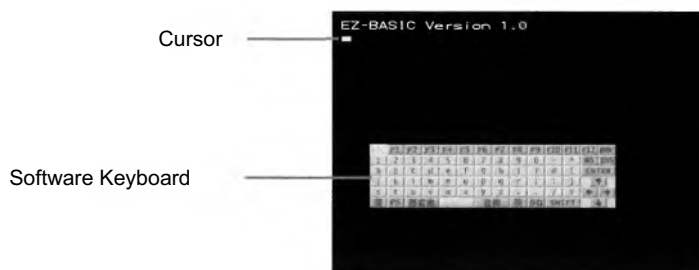
The sample game's executable file is "CARNAGE.BAS" in the "CARNAGE" directory. Select "CARNAGE.BAS" from the file list and then select "Open" from the file menu to launch the sample game.



EZ-BASIC

Starting EZ-BASIC

Select "EZ-BASIC" from the title screen to start the BASIC interpreter "EZ-BASIC." When you start BASIC, the following screen will be displayed.



The flashing square below "EZ-BASIC Version 1.0" is called the "cursor." Also, if a USB keyboard is not connected to the PlayStation 2, the software keyboard will be displayed.

Controls

Controller Explanation

If the USB keyboard is not connected, EZ-BASIC uses the software keyboard displayed on the screen and the controller to input programs. However, if a USB keyboard is connected, the controller is not used to input programs.

Analog Controller (DUALSHOCK 2)



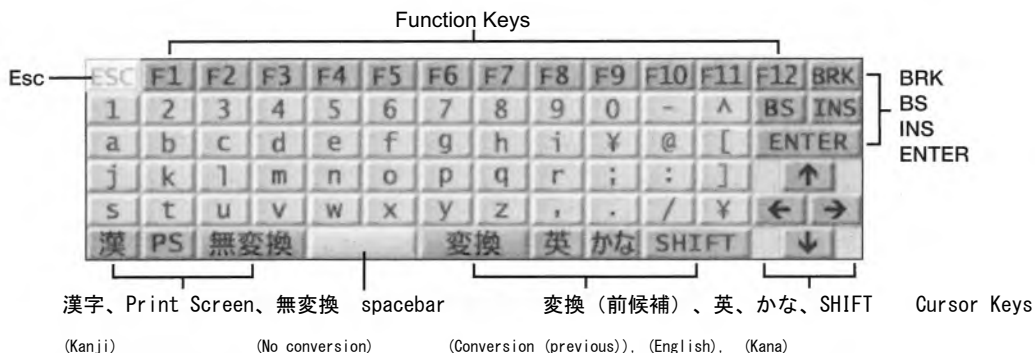
- ※ "BASIC STUDIO Powerful Game Workshop" supports analog controllers (DUALSHOCK 2). The LED indicator will always be red while the software is running.
- ※ Vibration can be turned on/off in the options on the title screen.

■ EZ-BASIC Controls

Directional Buttons	Software keyboard cursor movement
Button	Entering characters
L1 + Directional Buttons	Moving the software keyboard
L2 + Directional Button	EZ-BASIC cursor movement
R1 Button	"SHIFT" key ON/OFF
R2 Button	"ENTER" key
START+SELECT Buttons	Pause/Break key

Software Keyboard

On the software keyboard, you can move the cursor with the directional keys and enter characters by pressing the O button.



*The notation in brackets is displayed when the SHIFT key is ON.

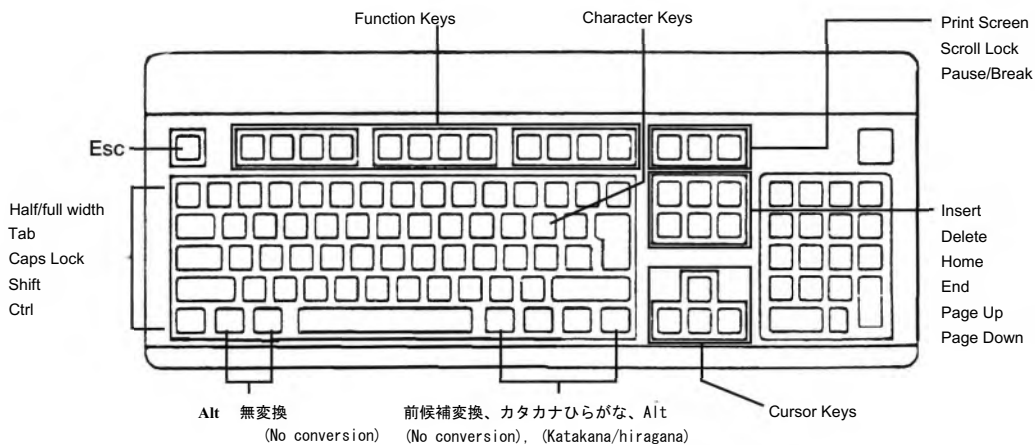
*See the table for key descriptions.

*The software keyboard cannot be used while the program is running.

USB Keyboard

With a USB keyboard connected to the PlayStation 2, you can input programs comfortably. You can input the characters printed on each key on the keyboard.

General Key positions



■ List of special keys

Software Keyboard	USB Keyboard	
Esc	Esc	Clears the text displayed on the screen
漢 (Kanji)	半角/全角	Japanese input mode on/off
—	Tab	Move cursor four characters to the right
英	Caps Lock英数	In Japanese input, turn on full-width alphanumeric input mode
—	Shift+Caps Lock	Switch between uppercase and lowercase input modes
BS	Backspace	Deletes the character right before the cursor.
ENTER	Enter	Direct execution of commands, registration of program statements
SHIFT	Shift + Character key	Switching input characters
space	Space	Input a one character space.
かな (カナ)	—	Switch between Hiragana and Katakana input in Japanese input mode
—	カタカナひらがな	Hiragana input
—	Shi代+カタカナひらがな	Katakana input
—	Alt+カタカナひらがな	Switch between Roman alphabet and Kana in Japanese input mode
PS	Print Screen	Print the displayed screen using popegg.
—	Scroll Lock	Text editing screen on/off
—	Pause/Break	Interrupts a running program, VIEW, or FILES command
BRK	—	VIEW and FILES commands are interrupted
INS	Insert	Switch between insert mode (inserts between the text in front of the cursor and the text on the cursor) and overwrite mode (Replaces text on the cursor)
—	Delete	Erases the character that the cursor is over.
—	Home	Move the cursor to the left end of the line.
—	End	Move the cursor to the right end of the line.
—	Page Up	Moves the text characters above the cursor up.
—	Page Down	Moves the text characters below the cursor down.
Arrow Keys	Arrow Keys	Moves the text cursor.
変換 (前候補)	前候補変換 (次候補)	Converts to kanji in Japanese input mode. Pressing this button again will display the next conversion candidate.
無変換	無変換	In Japanese input mode, it will not convert to kanji, but will convert to full-width hiragana or full-width katakana.
—	Num Lock	Numeric keypad ON/OFF
—	Ctrl + D	Delete from cursor position to end of line
—	Ctrl + C	Copy from the cursor position to the end of the line
—	Ctrl + P	Paste the copied text at the cursor position

Character input

Text input

First, let's try entering text on the screen. To enter text using the software keyboard, use the directional keys to position the cursor on the character you want to enter and press the O button.

If you have a USB keyboard connected, press the key of the character you want to enter. The character you entered will then be displayed on the screen, and the cursor will move one character to the right.

Also, when entering characters, you can enter the same character repeatedly by holding down the O button on a software keyboard or a key on a USB keyboard. This is called "auto-repeat."



Entering characters using the Shift key

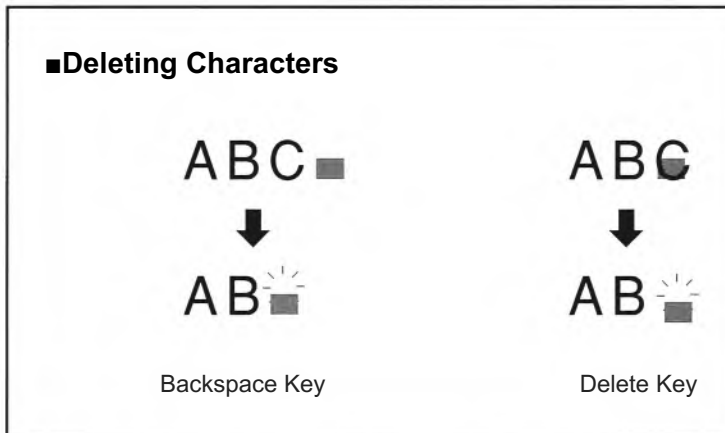
On the software keyboard, if you place the cursor on the [SHIFT] key and press the O button, the characters displayed on the software keyboard will change. This allows you to enter uppercase letters and special symbols in addition to lowercase letters and numbers.

This is equivalent to holding down the [Shift] key and pressing the character input key when using a USB keyboard. When pressing two keys simultaneously like this, this manual will refer to it as [Shift] + [a].

Deleting text

To erase a character, use the [BS] key or the [Backspace] key. Use the cursor keys to move the cursor to the right of the character you want to erase, and press the [Backspace] key to erase the character to the left of the cursor.

If you are using a USB keyboard, you can also press the [Delete] key to delete the character under the cursor.



The [Backspace] key erases the character to the left of the cursor, and the [Delete] key erases the character overlapping the cursor.

Deleting all text onscreen

To erase all characters on the screen, use the [Esc] key. Pressing the [Esc] key will erase all characters displayed on the screen and move the cursor to the top left of the screen.

* Pressing the [Esc] key only deletes text. It does not delete primitives or models, which will be explained later.

Using Japanese input mod

Entering Japanese input mode

If using a controller, press the [Kanji] key on the software keyboard, or if using a USB keyboard, press the [Half-width/Full-width] key to enter Japanese input mode. In Japanese input mode, an icon will appear in the bottom left of the screen.



Hiragana input mode/Katakana input mode

Japanese characters are entered by entering the hiragana characters displayed on the software keyboard. With a USB keyboard, the hiragana characters printed on the keytops are entered. To enter hiragana input mode, press the [Kana] key on the software keyboard or the [Katakana/Hiragana] key on the USB keyboard. To enter katakana, in hiragana input mode, press the [SHIFT] key on the software keyboard followed by the [Kana] key, or the [SHIFT] + [Katakana/Hiragana] keys on the USB keyboard.



Roman alphabet input mode (USB keyboard only)

Enter romaji using the English characters printed on the keytops of the USB keyboard. Entering "NIHONGO" will display "にほんご." To enter romaji input mode, press the [Alt] key + [Katakana Hiragana] key. Note that this input format is not available on software keyboards.



Full-width English character input mode

You can enter full-width alphanumeric characters by entering the alphanumeric characters displayed on the software keyboard. To enter full-width alphanumeric input mode, press the [English] key on the software keyboard or the [Caps Lock Alphanumeric] key on the USB keyboard. Note that in full-width alphanumeric input mode, characters cannot be converted to kanji, hiragana, or katakana.

Kanji Conversion

When you input hiragana or romaji, the characters you input are displayed in black on a white background, and when you press the [Convert] key, the Japanese you input is converted to kanji and displayed. Pressing the conversion key again will display the next kanji conversion candidate. To return to the previous conversion candidate, press the [Shift] key once and then the [Previous Candidate] key on the software keyboard. Press the [Shift] key + [Conversion] key on the USB keyboard. Repeat this process until the kanji you want to enter is displayed, then press the [Enter] key to confirm. Note that kanji, hiragana, katakana, and full-width alphanumeric characters use a space equivalent to two half-width alphanumeric characters.



Full-width hiragana and katakana conversion

If you want to confirm in full-width hiragana or katakana, you can switch between hiragana and katakana display by pressing the [No Conversion] key.

Using BASIC

Entering Commands, Executing

Commands

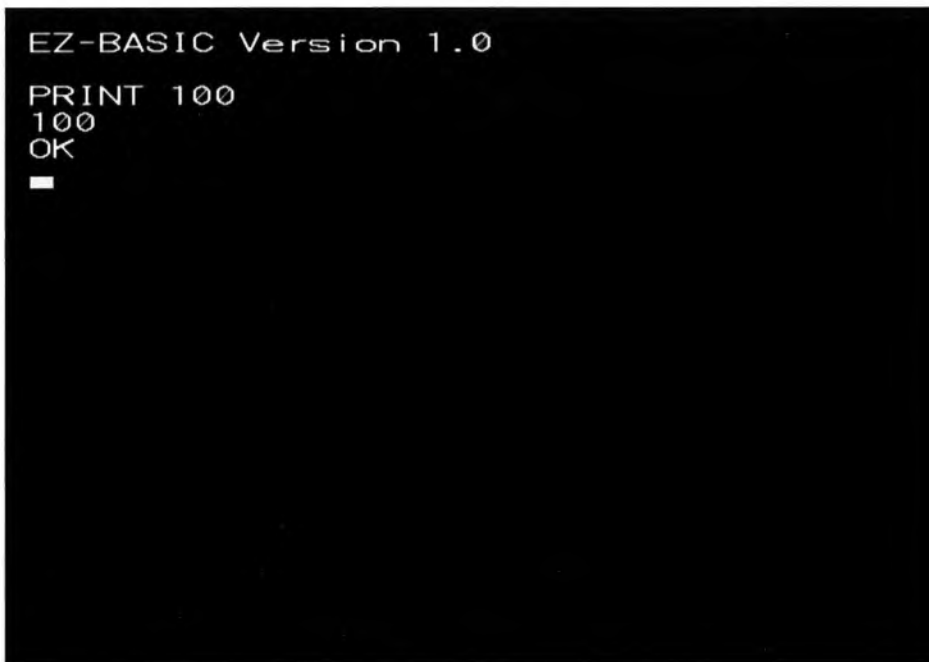
Now that you can enter characters, let's try entering and executing commands. Commands are made up of a combination of alphanumeric characters, such as "PRINT" or "CLEAR," and their functions are easy to understand from their names.

Writing Commands

First, input

`PRINT 100`

and press enter. Include a space between "PRINT" and "100". Please turn off Japanese input mode. The following screen will then appear.



```
EZ-BASIC Version 1.0
PRINT 100
100
OK
█
```

The "PRINT" command displays the text written after it on the screen. In other words, this statement displays the number "100" written after PRINT, and is executed by pressing the [Enter] key. Commands in BASIC are basically written as follows:

COMMAND [space] PARAMETERS

This is how BASIC is made to execute a command. These writing rules are called "syntax," and commands written according to that syntax are called "command statements." The way parameters are written and their meanings vary depending on the command, so please refer to the separate "Reference Manual" for details.

Notes on entering commands

There are a few things to keep in mind when entering commands.

- **Only half-width alphanumeric characters and certain symbols can be used in commands**

"PRINT" and "100" mentioned earlier cannot be entered with "full-width alphanumeric characters," "kanji," "hiragana," or "katakana," which can be entered in Japanese input mode.

- **Commands can be entered in uppercase or lowercase.**

When entering commands, you can use either uppercase or lowercase. "PRINT" and "print" are the same command.

- **How to use spaces**

In the command statement,

```
PRINT100[Enter]
```

If there is no space between the command and the parameter, BASIC will recognize it as a single command, "PRINT100," and it will not be able to be executed.

Also,

```
PR INT 100[Enter]
```

If you enter the above, it will be interpreted as the "PR" instruction having the "INT" parameter, and the instruction will also not be able to be executed.

Be careful, because if you do not enter spaces where they are required or if you enter spaces in unnecessary places, the command will not be executed correctly.

Error Messages

Error Messages

Next, try entering the command incorrectly. Remove the "T" at the end of PRINT and type

```
PRIN 100[Enter]
```

When this is typed, instead of it showing 100,

```
PRIN 100
```

構文の誤りです。(Syntax error.)

is displayed. If you enter an incorrect syntax like this, BASIC will point out the mistake and display a message. This message is called an "error message." Conversely, by reading this error message, you can understand the mistake in the command you wrote. For example, the error message "構文の誤りです。(Syntax error)" indicates that the format

COMMAND [space] PARAMETER

syntax rule is not being followed. In other words, since the command "PRIN" does not exist in BASIC, it was determined that the syntax rule was not being followed. This type of mistake is mainly caused by typos.

Common mistakes

The main mistakes that beginners make are listed below:

- Misspellings in commands
ROAD (incorrect) → LOAD (correct), etc.
- Interchanging characters that appear similar
1 (number one), I (capital letter i), l (lowercase letter L), etc.
- Full-width spaces are used for spaces (blank spaces).

Be careful of these mistakes when inputting code.

Numbers and Strings

Displaying with PRINT

Now that we know how to write commands, let's try executing some of the PRINT statements we used earlier.

Displaying Strings

We earlier displayed numbers, so now let's display letters.

Example input:

```
PRINT "HELLO"[Enter]
```

Execution result:

```
HELLO
```

```
OK
```

Now, as you may have noticed, the word HELLO is enclosed in double quotation marks on both sides.

This is to distinguish it from what is called a "numeric variable" which will be explained later. If you try removing the "" from the previous input example and displaying it, the execution result will be displayed as 0. If you want to display the characters written after the PRINT statement as is, remember to add " on both ends. Characters enclosed in "" are called strings.

Display result of an expression

Next, enter a formula after the PRINT command and display it.

Example input:

```
PRINT 1+2[Enter]
```

Execution result:

```
3
```

```
OK
```

This time, instead of the "1+2" entered after PRINT, the result of the formula, "3", was displayed.

"A formula entered as a parameter of an instruction command is recognized as its calculated value."

This is because of the rule that "1+2" written after the PRINT command is recognized as "3" in BASIC, so "3" is displayed when the command is executed. Conversely, if you want to display "1+2" as is, just enter "1+2" with quotes as shown above.

Numbers and Strings

Data in BASIC can broadly be divided into two categories: "numbers" and "strings."

• Numbers (numeric values)

Numerical values are a set of "digits" that can be used for in various ways such as performing calculations or comparing which is larger. Most parameters in instruction commands are in this numeric format. When an expression is used as a parameter in an instruction command, the calculated value is recognized and evaluated as the parameter. All fractions are displayed converted to decimal notation. The symbols for arithmetic operations used are as shown in the table.

+	Addition
-	Subtraction
*	Multiplication
/	Division
^	Exponentiation

Example input:

```
PRINT 2-3[Enter] (1)
PRINT 3*4[Enter] (2)
PRINT 10/3[Enter] (3)
```

Execution result:

```
(1) -1
(2) 12
(3) 3.33333
```

The types of numerical input and notation are as follows:

Integer part,(decimal point),Decimal part	1.23, 72.031, etc.
Integer part,(decimal point),Decimal part, exponent part	3.7E+10, 2.5E-5, etc.

*The exponent part is expressed as "E (sign) number".

Exponential notation expresses a number by multiplying the integer part by 10 to the power of (sign) number.

Example:

Exponent notation	Normal numeric display
2.34E+6	2340000
4.56E-5	0.0000456

• Strings

A string is a collection of numbers, English letters, Japanese characters, and symbols enclosed in double quotation marks (") on both sides, and is primarily used to display characters. Strings generally cannot be used for arithmetic operations, but as an exception, addition is possible.

Input example:

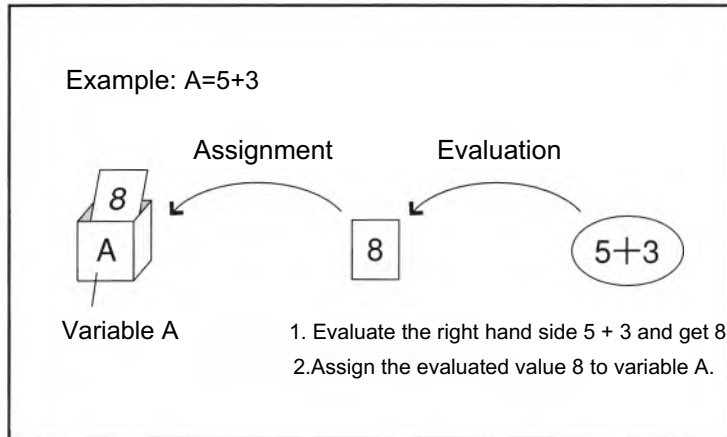
```
PRINT "AB"+"CD"[Enter]
```

Execution result:

```
ABCD
OK
```

Variables

In BASIC, you can temporarily store numbers and strings in box-like things, which allow you to reuse the numbers and strings stored in other commands. These boxes are called variables. There are two types of variables: "numeric variables" which store numbers, and "string variables" which store strings.



The numeric value is put into the box.

Numeric Variables

When programming in BASIC, you often want to use the same numerical value as a parameter for another command. In such cases, you use a "numeric variable," which is a box that stores a numerical value. Let's try using a numeric variable.

Example input:

```
LET A=10[Enter]
```

Execution result:

OK

"LET" is the command, "A" is the name of the numeric variable, and "10" is the value to be placed in the variable.

The "LET" command assigns the value "10" on the right side of the equal sign entered after LET to the variable name ("A" in this case) written on the left side. When this command is executed, the only thing that will change on the screen is the display of "OK," but to confirm that "10" has been assigned to "A," try executing the following command.

Example input:

```
PRINT A[Enter]
```

Execution result:

10
OK

The PRINT statement is used to display the variable "A", "10" is displayed, indicating that "10" has been assigned to "A". In this way, numeric variables are treated the same as numeric values.

Note that the numeric value on the right side of the equal sign is assigned to the variable name on the left.

An error message will appear and the program will not run if (1) a numeric value is on the left side and a variable name on the right side, or (2) there is a formula on the left side. Conversely, by assigning the numeric value "A+1" on the right side to the new "A" on the left side, as in (3), you can add 1 to the value of variable A. Finally, variable names are recognized as the same whether written in uppercase or lowercase, such as "ABC" or "abc."

Also, unlike other commands, the LET command can be omitted and entered as shown in (4).

```
LET 10+A      (1)
LET A/2=10    (2)
LET A=A+1     (3)
B=30          (4)
```

String Variables

Strings can be treated as variables in the same way as numbers. String variables have a \$ at the end of the variable name to distinguish them from numeric variables. The string you want to assign is enclosed in double quotation marks.

Example input:

```
A$="HELLO"[Enter]
```

```
B$="How"+"are"+"You"[Enter]
```

Execution example:

```
PRINT A$,B$[Enter]
```

```
HELLO HowareYou
```

```
OK
```

*By separating the parameters entered after the PRINT command with a comma (,), you can display multiple variables, numbers, strings, etc. at once.

Programs

What is a program?

Commands shown up to this point have been executed by entering them and then pressing the [Enter] key, but by entering a "line number" before the command to indicate the order in which the commands should be executed, it is possible to execute several commands in the order of their line numbers.

Example input:

```
10 A=1[Enter]
20 PRINT A[Enter]
30 B=10[Enter]
40 PRINT A,B [Enter]
```

When you enter the commands in this way, pressing the [Enter] key at the end of the command will not execute it immediately, but will register it as a "program" in EZ-BASIC. To run the registered "program," input RUN [Enter].

Example input:

```
RUN[Enter]
```

Execution output:

```
1
1 10
実行を完了しました。(Execution complete)
```

Program flow

When the above program is executed, the program flow will be as follows:

Line 10: Assign 1 to variable A.

Line 20: The PRINT command displays the value of variable A (1) on the screen.

Line 30: Assign 10 to variable B.

Line 40: The PRINT command displays the value of variable A (1) and the value of variable B (10) on the screen.

Also, programs are executed in the order of their line numbers, so the order in which they are written does not affect them. For example, let's add some more statements to the program above and then run it with RUN.

Example input:

```
35 A=A+10[Enter]
RUN[Enter]
```

Execution output:

```
1
11 10
実行を完了しました。(Execution complete)
```

Now, on line 35 the value of A was increased by 10, so the value of A changed before it was displayed on line 40, resulting in a different value being displayed from the value of A displayed on line 20. Because programs are executed in the order of line numbers, it is important to follow the order of line numbers to check the program flow. Also, leaving an interval of about 10 between line numbers like this makes it easier to insert commands.

END Command

BASIC , execution ends when a program reaches the last line, but by adding an "END" command at the end of the program to end execution, it will become a program in the correct programming manner.
Let's add an "END" command to the program above.

Example input:

```
50 END
```

Execution result:

```
1  
11 10  
実行を終了しました。 (Execution complete)
```

STOP Command

The "STOP" command is similar to the END command. When the "STOP" command is executed, program execution is interrupted. Unlike the END command, the "STOP" command allows program execution to be resumed with the "CONTINUE" command.

Example input:

```
25 STOP
```

Execution result:

```
1  
実行を終了しました。 (Execution complete)
```

CONTINUE Command

Resumes execution of a program that was interrupted by the STOP command.

Example input:

```
CONTINUE[Enter]
```

Execution result:

```
11 10  
実行を終了しました。 (Execution complete)
```

Editing a Program

Confirming an entered program

LIST Command

If you run a program several times, the program you entered will disappear from the screen when the execution results are displayed. To display and check the program you entered again, use the "LIST" command. Here, let's display the program up to the END command we entered earlier.

Example input:

LIST[Enter]

Execution result:

```
10 A=1
20 PRINT A
25 STOP
30 B=10
35 A=A+10
40 PRINT A,B
50 END
OK
```

In this way, the LIST command displays the program from beginning to end in the order of line numbers. However, if the program is too long to be displayed on one screen, the beginning of the program will disappear off the screen. In such cases, you can specify a range of line numbers as shown in (1), and display only the program in the specified range. You can also search for programs that contain a specified string by specifying a string as shown in (2).

Example input:

LIST 100 TO LAST (1)

LIST REF"PRINT" (2)

Explanation:

(1) Displays from line 100 to the end of the program.

(2) Displays all lines containing the string "PRINT" (uppercase and lowercase letters are distinguished).

VIEW Command

The "VIEW" command works in a similar way to the LIST command. When the "VIEW" command is executed, the entered program is displayed in the same way as the LIST command, but if the program exceeds one screen, you can scroll up and down using the cursor keys. Also, although you can specify a line range, you cannot search for programs that contain a specified string like you can with the LIST command.

Controls while using VIEW

Cursor keys up/down..... Scroll the program listing up and down.

BRK (Pause/Break) or Enter.....Return to program editing.

Program modifications and additions

When writing a program, you may need to add or delete statements or correct mistakes. This section explains how to edit a program.

Revising a command

To edit a command that has already been registered as a program, use the LIST command to display the program you want to edit, then move the cursor to the part you want to edit and edit it. Then press the [Enter] key to re-register the command.

* If you do not press the [Enter] key, the modified statement will not be registered in the program.

Editing example:

100 PRINT_ABC_	Place the cursor on the command you want to edit,
↓	
100 PRINT_DEF_	Make corrections
↓	
100 PRINT_DEF[Enter]	Once you have finished editing, press the Enter key to finish.

Adding commands

As explained earlier, you can add a command by entering the command using an available line number.

Deleting commands

There are two ways to delete programs that you no longer need, such as unused statements.

- **Enter only the line number and press Enter.**

you can delete unnecessary lines by entering only the line number and pressing the Enter key.

Example input:

30 [Enter]

Explanation:

Line 30 has been deleted.

- **Use the DELETE command**

Entering only the line number to delete is fine if you only want to delete a few lines of statements, but it is inconvenient if you want to delete many at once. In such cases, you can use the "DELETE" command. By specifying the range of line numbers you want to delete after the "DELETE" command, you can delete multiple lines at once.

Example input:

DELETE 30 TO LAST [Enter]

Explanation:

The commands from 30 to the last line have been deleted.

Program initialization

If you want to erase the entire program you've entered, such as when rewriting a new program, use the "NEW" command. Using the "NEW" command erases all of the programs registered in EZ-BASIC. If you enter a new program while the previous program remains in memory, the newly entered program and the old program will be mixed together and run, causing strange behavior. To avoid this, execute the "NEW" command when entering a new program.

Line numbering

When adding or deleting programs, the line numbers will not increase consistently, making the program difficult to read. In such cases, use the "RENUMBER" command. Let's use the program to which we just added a statement as an example.

Example input:

```
10 A=1
20 PRINT A
30 B=10
35 A=A+10
40 PRINT A,B
50 END
```

A program with irregular line numbering.

RENUMBER[Enter]

Execution result:

```
LIST[Enter]    Display the program with LIST
100 A=1
110 PRINT A
120 B=10
130 A=A+10
140 PRINT A,B
150 END
```

By executing the LIST command after the RENUMBER command, you can see that the line numbers, which had been increasing irregularly, have been corrected to line numbers in increments of 10, starting from line 100.

Comment input

When you are writing a long program, it can become difficult to understand what each part of the program does.

To prevent this, write comments in your program so that it is easy to understand even when you look at it later.

Also, by commenting out a statement, you can disable the statement without deleting it.

Put the "REM" command after the line number and then write your comment. You can also put an "!" (exclamation mark) at the end of a command statement and write your comment after it. You can also use Japanese in your comment text.

Program input example:

```
10 PRINT A
20 REM Display variable A
30 PRINT B !Display variable B
40 !PRINT C
```

Program explanation:

Display variable A

Does nothing because it has a REM statement.

After displaying variable B with the PRINT command, the statements after the exclamation mark (!) are ignored.

The PRINT command is ignored because of the "!". If you delete the "!", the PRINT command will be valid.

Save and load created programs

Saving a program

The program created using the steps up to this point is only recorded in EZ-BASIC's main memory, so if you execute the NEW or QUIT command, or press the PlayStation 2's (On/Standby)/RESET button to turn off the power to the PlayStation 2 or perform a RESET, the contents of the program you entered will be permanently lost. You can save the program you entered as a "file" on a memory card using the "SAVE" command, so that it be preserved even when the PlayStation 2 is turned off.

Saving a program with SAVE

To save a program, use the "SAVE" command. Files saved in BASIC STUDIO are managed together in a container called a directory. Now let's actually use "SAVE" to save to the memory card. In preparation, please connect the memory card to MEMORY CARD slot 1.

Format:
SAVE "DirectoryName/FileName"

Example input:
SAVE "TEMP/TEST" [Enter]

The program you entered has now been saved to the PlayStation 2 dedicated memory card (8MB) connected to MEMORY CARD slot 1. "TEMP" written after the "SAVE" command is the name of the directory where the file will be stored, and "TEST" is the name of the file that will be given when the program is saved.

The SAVE command follows these rules:

- File and directory names can be set using up to 8 alphanumeric characters and underscores (_).
- Use a "/" (slash) to separate the directory name and file name.
- You cannot create a directory within a directory.
- If you do not specify a directory, the file will be stored in the "SYSTEM" directory.
- A maximum of 15 files can be stored in one directory.

Confirming saved data with FILES

To check the files stored on the "PlayStation 2" dedicated memory card (8MB), use the "FILES" command.

Example input:
FILES[Enter]

Execution result:
TEMP <DIR> 2001/04/05
OK

If no parameters are written after the "FILES" command, the directories containing the BASIC STUDIO files stored on the memory card will be displayed. In this case, "TEMP" is the name of the directory, and <DIR> indicates that "TEMP" is a directory.

To view the contents of a directory, enter the name of the directory you want to view the contents of in quotation marks after the "FILES" command, as shown below.

Example input:
FILES "TEMP"[Enter]

Execution result:
TEST.BAS 100Byte 2001/04/05
OK

This shows that the "TEST" file we just saved has been saved correctly. The ".BAS" after "TEST" is what's called an "extension," indicating that the file's contents are a program saved with EZ-BASIC. The directory also can contain files saved with other BASIC STUDIO tools, so these file types are indicated by their extension. See page 6 for information on the extensions of each file.

Setting save target with DEVICE

Up until now, files have been saved to the PlayStation 2 memory card (8MB) connected to MEMORY CARD slot 1. By executing the "DEVICE" command before the SAVE or LOAD command, you can specify the device to use when saving or loading. This "DEVICE" command applies to all types of save and load commands used in EZ-BASIC.

Example input:

```
DEVICE MEMC2[Enter]
SAVE "SHOOT/MAIN"[Enter]
```

Explanation:

Save the program to the memory card connected to MEMORY CARD slot 2 with the directory name "SHOOT" and the file name "MAIN".

The devices that may be specified by DEVICE are as follows:

Parameters of the DEVICE instruction

MEMC1	"PlayStation 2" dedicated memory card (8MB) connected to MEMORY CARD slot 1
MEMC2	"PlayStation 2" dedicated memory card (8MB) connected to MEMORY CARD slot 2

Commands influenced by the DEVICE command

Save	SAVE, VARSAVE
Load	LOAD, VARLOAD, TEXLOAD, PALLOAD, MODLOAD, BGMLOAD, SELOAD
Others	FILES, KILL

Loading a program

To load a saved program file, use the "LOAD" command. The "LOAD" command specifies the file using the same syntax as the SAVE command. You can also specify the load destination using the DEVICE command.

Example input:

```
DEVICE MEMC2[Enter]
LOAD "SYSTEM/TEST2" [Enter]
```

Explanation:

EZ-BASIC file "TEST2" in the "SYSTEM" directory of the PlayStation 2 dedicated memory card (8MB) connected to MEMORY CARD slot 2 will be read.

Exiting EZ-BASIC

If you want to exit EZ-BASIC and start another tool, use the "QUIT" command.

Entering the QUIT command will exit EZ-BASIC and return you to the title screen.

When you execute the QUIT command, the programs registered in EZ-BASIC will be erased.

Program Flow Control

Up until now, programs have executed statements in sequence from the lowest line number, but sometimes you may want to go back to a previous line and execute the same statement again, or change the flow of the program based on different conditions. There are control commands that determine the flow of a program in this way.

Repeating a program using DO~LOOP

For example, if you want to create a program that displays the numbers 1 to 5, it would be very time-consuming to display variable A every time as shown below, or to assign the next value to variable A.

Example input:

```
10 A=1
20 PRINT A
30 A=2
40 PRINT A
50 A=3
60 PRINT A
70 A=4
80 PRINT A
90 A=5
100 PRINT A
110 END
```

Here, we use the DO and LOOP commands. DO and LOOP commands repeatedly execute the statements between them.

Program input example:

```
10 A=1
20 DO
30 PRINT A
40 A = A + 1
50 LOOP
60 END
```

Program explanation:

Assign 1 to variable A

This command does nothing by itself.

Display variable A

Add 1 to variable A

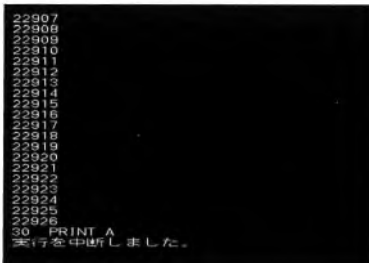
Instead of moving on to the next line, it returns to line 20 where the corresponding "DO" command is located and continues execution.

Execution result:



Forced program interruption

When you run the program above, the numbers will increase at a tremendous rate and flow up the screen. If you look at the program list and follow the flow, you can see that after the statement on line 50 is executed, it will go back to line 20, making it a program that never ends. This state is called an "infinite loop." To interrupt this program, press the SELECT and START buttons on the controller simultaneously. Also, if you have a USB keyboard connected, you may press the "Pause/Break" key. This will interrupt the currently running program.



Setting end conditions

To avoid this kind of infinite loop, you need to set an end condition. Let's try to end the program when the value of variable A becomes 5. To set an end condition, change the LOOP command line in the previous program as follows:

Program input (edited):

```
10 A=1
20 DO
30 PRINT A
40 A=A+1
50 LOOP UNTIL A>5
60 END
```

Program explanation

Assign 1 to variable A

Display variable A

Add 1 to variable A

If variable A is greater than 5, go to the next line without executing the LOOP command. Otherwise, execute the LOOP command.

Execution result:

```
1
2
3
4
5
```

Execution finished.

"UNTIL" runs "LOOP" until the condition written after it is met (in this case, when variable A exceeds 5). There is also "WHILE" which runs "LOOP" as long as the specified condition is met.

For more information, please refer to the Reference Manual .

Logical expressions and Comparators

"A>5" written as the termination condition in "LOOP UNTIL" is called a logical expression. A logical expression compares variables or numbers to determine whether the result is true or false. In the program above, if variable A is 5 or less (e.g. if A is 2) when the logical expression in line 50 is evaluated, the result will be "2>5" and the logical expression will evaluate as false. Thus, LOOP UNTIL is executed as a command to return to DO and continue. If A is greater than 5, the logical expression is true, LOOP UNTIL does nothing, and execution continues to the next line.

The ">" symbol, which compares two values, is called a comparison operator. Comparison operators are as follows:

=	Check if whether left and right are the same
>	Check if left is greater than right
<	Check if left is less than right
>=	Check if left is greater than or equal to right
<=	Check if left is less than or equal to right
<>	Check if left is different from right
><	

Controlling execution speed

FSYNC Command

When you run the program above, the numbers 1 to 5 appear to be displayed all at once, even though the program actually displays them in one by one. This is because the program runs very quickly, making it appear as if the numbers are displayed all at once.

To make it clear that the numbers are displayed in sequentially, use the "FSYNC" command. The "FSYNC" command waits for the specified number of frames since the last "FSYNC" command was executed to redraw the display, which is done 60 times per second. For example, with the following change, it will increase the variable every second.

Program input example:

```
10 A=1
20 DO
30 PRINT A
40 A=A+1
45 FSYNC 60
50 LOOP UNTIL A>5
60 END
```

Program explanation:

Wait for the screen to refresh 60 times (about 1 second)

Also, if you include the "FSYNC 1" command in the main loop like this, it will synchronize with the timing of rewriting the TV screen, which has the advantage of preventing the display from flickering. Be sure to include it.

Conditional branching with IF

Next we will explain the "IF ~ THEN" command, which branches that is executed depending on the set conditions. "IF ~ THEN" is a command that means "if ~ then execute ~", and if the logical expression written between "IF" and "THEN" is correct, the commands after "THEN" will be executed. Below, the previous program is rewritten using an "IF" statement without using "UNTIL" or "WHILE". Note that the "EXIT DO" command is used to exit the loop.

Program input example:

```
10 A=1
20 DO
30 PRINT A
40 IF A>=5 THEN EXIT DO
50 A=A+1
60 FSYNC 60
70 LOOP
80 END
```

Program explanation:

If A is five or more, then exit the loop.

Here, each time the program repeats the process from the DO command on line 20 to the LOOP command on line 70, the value of variable A on line 40 is compared with 5, and when the value of variable A becomes 5 or greater, the logical expression "A>=5" is determined to be true. As a result, the command "EXIT DO" written after "THEN" is executed to exit from the LOOP, and the END command on line 80 is executed to end the program.

Adding conditions to IF

To set multiple conditions to be judged in an IF statement, connect the logical expressions with "AND" or "OR" as shown below. Also, by preceding the logical expression with "NOT", the commands following THEN will be executed if the condition is not met.

Example input:

```
IF A>=0 AND B>=0 THEN PRINT A
IF A>=0 OR B>=0 THEN PRINT A
IF NOT B>=0 THEN PRINT A
```

Explanation

If variable A is 0 or greater and variable B is 0 or greater, display variable A.
If variable A is 0 or greater or variable B is 0 or greater, display variable A.
If variable B is not greater than 0, display variable A.

Setting multiple commands to execute after IF

With the program written previously, only one command can be executed after the "THEN" command. If you want to set multiple commands to execute, leave nothing after "THEN" on line 40 and write the commands you want to execute on the next line. In this case, unlike an IF statement which ends on one line, after you have written the last command you want to execute, you need to enter an "END IF" command on the next line to close the "IF" statement.

Program input example:

```
100 A=1
110 DO
120 PRINT A
130 IF A>=2 THEN
140 PRINT "End loop"
150 EXIT DO
160 END IF
170 A=A+1
180 FSYNC 60
190 LOOP
200 END
```

Program explanation:

If A is 2 or more, execute the following lines
Display "End loop"
Exit the loop
The code executed by the IF statement on line 130 ends here

Execution result:

```
1
2
End loop
End of execution.
```

ELSE

The "ELSE" command can be used to execute a command even if the condition in the IF statement is not met. If the condition is not met, the command entered after the "ELSE" command will be executed.

The "END IF" that closes the IF statement should be placed after the statement that will be executed in the "ELSE".

Program input example:

```
100 A=1
110 DO
120 PRINT A
130 IF A>=2 THEN
140 PRINT "End loop"
150 EXIT DO
160 ELSE
170 PRINT "Continuing loop"
180 END IF
190 A=A+1
200 FSYNC 60
210 LOOP
220 END
```

Program explanation:

If A is 2 or more, execute the following lines
Display "End loop"
Exit the loop
If A is not 2 or more, execute the following lines
Display "Continuing loop"
End of branching by the IF statement

Execution result:

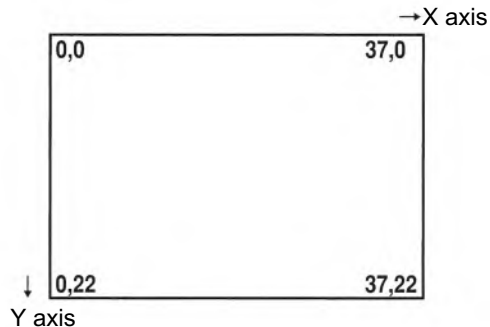
```
1
Continuing loop
2
End loop
```


Control commands

To summarize the control commands, let's create a program that moves text from left to right on the screen. First, we will explain the "LOCATE" command, which specifies the position of the text to be displayed with the PRINT.

LOCATE Command

To specify the location of characters to be displayed with a PRINT statement, execute the "LOCATE" command before PRINT. Using the top left corner of the screen as the base, the x-axis runs from left to right and the y-axis runs from top to bottom. To specify a position on the screen, you specify it in the form of "coordinates," such as "how many characters along the x-axis and how many characters along the y-axis." These coordinates, which use the character as the basic unit, are called "text coordinates." Note that the "LOCATE" command cannot specify coordinates outside the range.



So the program looks like this:

Program input example:

```

10 X=0
20 DO
30 LOCATE X, 11
40 PRINT "O"
50 X=X+1
60 FSYNC 10
70 IF X>=37 THEN EXIT DO
80 LOOP
90 END

```

Program Description:

Assign 0 to the variable X for the X-axis coordinate

DO ~ Start of loop

Move cursor to text coordinate (X, 11)

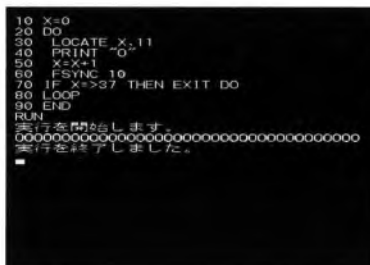
Display a O at text coordinates (X, 11)

Add 1 to the X-axis coordinate variable X to shift the position to the right

Wait for the screen to refresh 10 times

When X is 37 or more (the right edge of the screen), exit the loop.

Execution result:



However, if you leave it like this, the program you just wrote and each "O" displayed by the PRINT statement will remain on the screen.

CLEAR Command

Here we use the "CLEAR" command to erase the characters on the screen. The "CLEAR" command erases all characters on the screen, just like pressing the [Esc] key. However, be careful when using the LOCATE command, as the cursor position will return to the origin (0,0) of the text coordinates. Using the "CLEAR" command results in a program like this:

Program input example:

```
10 X=0
20 DO
30  CLEAR
40  LOCATE X,11
50  PRINT "O"
60  X=X+1
70  FSYNC 10
80  IF X>=37 THEN EXIT DO
90  LOOP
100 END
```

Program Description:

Assign 0 to the variable X for the x-axis coordinate
DO ~ Start of loop
Erases text on the screen. Moves cursor to origin.
Move cursor to text coordinate (X,11)
Display O at text coordinate (X, 11)
Add 1 to the x-axis coordinate variable X to shift the position to the right
Wait for the screen to refresh 10 times
When X is 37 or more (the right edge of the screen), the loop ends.

Execution result:



Functions

IP function

BASIC has instructions called "functions," which change a number or string into another number or string by passing it to the function.

First, let's take the IP function as an example.

Format:

IP (*argument*)

Example input:

```
PRINT IP(1.25)[Enter]
```

Execution result:

```
1
OK
```

The IP function returns only the integer part of the number written in the subsequent "()" parentheses. In other words, the number "1.25" was changed to the integer 1 by the IP function. What is placed in the parentheses can be not only a number, but also a numeric variable. What is placed in the parentheses is called an "argument." The basic syntax for writing a function is as follows:

Function (*argument*)

Some functions have no arguments, while others require several arguments. Also, whether the arguments are numeric or strings varies depending on the function. For details on each function, refer to the Reference Manual.

Comparing Commands and Functions

The syntax of a command, "command[space]parameter", and the syntax of a function, "function (argument)", are very similar. However, there are differences between functions and commands in the following ways:

- **Functions cannot be used alone.**

For example,

```
INT(1.25)[Enter]
```

An error will be displayed if you enter the following. Functions are treated like "variables" rather than as commands. In other words, they are ultimately treated as numbers or strings.

- **Arguments must be enclosed in parentheses.**

Unlike parameters in imperative commands, arguments must be enclosed in parentheses.

Numeric and string functions

Just as there are numeric variables and string variables, there are two types of functions: "numeric" and "string functions."

- **Numeric Functions**

A numeric function returns a number when passed an argument. Therefore, it is treated like a numeric variable. The arguments passed are not limited to numbers; some functions will return a number when passed a string as an argument.

- **String Functions**

String functions are functions that handle strings, and the arguments passed are not necessarily strings.

Program using the PAD function

Now let's create a program that uses the "PAD" function to move text with the controller. The PAD function returns a numerical value representing the strength with which a controller button specified by two arguments is pressed. First, let's take a look at what the PAD function is.

Format:

PAD (*controllerNumber*, *buttonNumber*)

Program input example:

```
10 DO
20 PRINT PAD (1,1)
30 LOOP
```

Execution result:



When you press the right on the directional pad on the controller 1, the number displayed will change between 0 and 255. The larger this number, the harder the input is pressed. For the correspondence between arguments and buttons, please refer to the "Reference Manual."

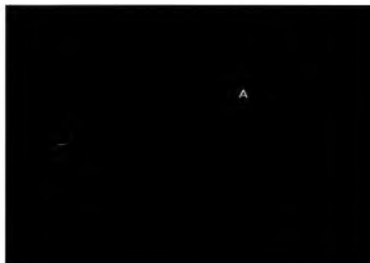
Let's create a program that uses this PAD function to move the letter "A" using a controller.

Program input example:

```
100 DO
110 CLEAR
120 IF PAD(1,1)>0 THEN X=X+1
130 IF PAD(1,2)>0 THEN X=X-1
140 IF PAD(1,3)>0 THEN Y=Y-1
150 IF PAD(1,4)>0 THEN Y=Y+1
160 IF X<0 THEN X=0
170 IF X>37 THEN X=37
180 IF Y<0 THEN Y=0
190 IF Y>22 THEN Y=22
200 LOCATE X,Y
210 PRINT "A"
220 FSYNC 1
230 LOOP
```

Program Description:

Start loop
Clear the text on the screen
If the right directional key is pressed, add 1 to the coordinate variable x.
If the left directional key is pressed, subtract 1 from X.
If the up directional key is pressed, subtract 1 from Y.
If the down directional key is pressed, add 1 to Y.
Ensure that the x coordinate is not off the left of the screen.
Ensure that the x coordinate is not off the right of the screen.
Ensure that the Y coordinate is not off the top of the screen.
Ensure that the Y coordinate is not off the bottom of the screen.
Move the cursor to the text coordinates (X,Y).
Display "A" at the coordinate specified by LOCATE.
Wait for the screen to be replaced once.
Return to DO.

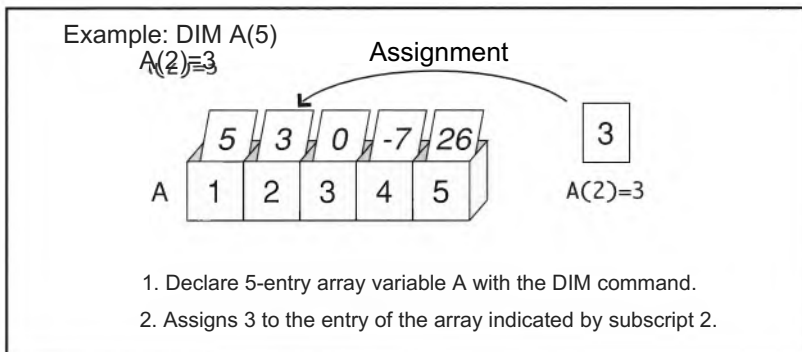


"A" will move around the screen in response to input to the direction keys.

Arrays

Numeric arrays

Simply put, an array is like a collection of variables. While a variable can only have one numeric value or string in its name, an array can have multiple values, distinguished by numbers called subscripts (or indexes) written next to the array name.



First, to use an array, you must declare that you are using it. To declare an array, use the "DIM" command.

Format: DIM *arrayName* (*maximum number of subscripts*)

Example input:

```
DIM AB(10)[Enter]
```

An array called "AB" has been prepared. The number in parentheses indicates how many elements to prepare for the array. Here, 10 was specified, so an array with 10 elements from 1 to 10 has been prepared. Note that an array name and a variable name cannot be the same. After declaration, you can actually enter numbers into it. Arrays are written as follows:

ArrayName(subscript)

The array name is the name of a previously declared array, with a "subscript" in parentheses to the right of it.

Example input:

```
AB(1)=10[Enter]
```

```
AB(2)=20[Enter]
```

```
PRINT AB(1),AB(2)[Enter]
```

Execution result:

```
10 20
```

Now we can see that even with the same array name AB, we can store different values by changing the subscript. The advantage of arrays is that variables can be stored in the subscript portion. This can simplify the program when managing the coordinates of player 1 and player 2, for example.

Program example:

```
100 DIM X(2),Y(2)
110 X(1)=0
120 X(2)=37
130 DO
140  CLEAR
150  P=1
160  DO
170   IF PAD(P,4)>0 THEN Y(P)=Y(P)+1
180   IF PAD(P,3)>0 THEN Y(P)=Y(P)-1
190   IF Y(P)<0 THEN Y(P)=0
200   IF Y(P)>22 THEN Y(P)=22
210   LOCATE-X(P),Y(P)
220   PRINT "A"
230   P=P+1
240  LOOP UNTIL P>2
250  FSYNC 1
260  LOOP
```

Program explanation:

Define an array to represent the coordinates of each player

X coordinate of player 1 (fixed)

X coordinate of player 2 (fixed)

Start of repeat (main loop)

Clear the text on the screen

Assign 1 to the player identification variable P

Start of controller input loop. This DO corresponds to the LOOP on line 240.

If the down arrow key is pressed, add 1 to the coordinate variable Y(P).

If the up arrow key is pressed, subtract 1 from the coordinate variable Y(P).

Ensure that the Y coordinate is not past the top of the screen

Ensure that the Y coordinate is not below the bottom of the screen

Move the cursor to the text coordinates X(P) and Y(P)

Display "A"

Add 1 to the player identification variable P

If the variable P exceeds 2 (after processing up to 2 players), exit the loop

Wait for one screen refresh

Return to DO

Execution result:



Explanation: This program displays "A" on the right and left edges of the screen, and can be moved up and down using the controllers connected to controller port 1 and 2. By using arrays X(P) and Y(P) to specify the coordinates where "A" is displayed, and switching the array subscripts with a variable, the instructions for processing the controller connected to controller port 1 and the instructions for processing the controller connected to controller port 2 are shared.

String arrays

Like numeric variables, character variables can also be treated as arrays. To distinguish string arrays from numeric arrays, add \$ to the array name and enclose the string you want to assign in double quotation marks (").

Input example:

```
DIM A$(10)[Enter]
```

```
A$(1)="HELLO"[Enter]
```

Subprograms

A "subprogram" is used when the same code is used multiple times. By creating a subprogram, it is possible to call and use the same process multiple times from different places in the program. Creating subprograms effectively also simplifies the program as a whole, making it easier to understand the overall flow of the program.

CALL and SUB Commands

The SUB command is used in conjunction with the END SUB command to set up a subprogram (this is called defining a subprogram), and the CALL command is used to call and execute a subprogram defined with the SUB command. For information on how to call and write subprograms, please refer to the Reference Manual. Subprogram names, like variables, can only contain alphanumeric characters.

Sample

Programs can be simplified by making frequently used processes subprograms. Let's rewrite the previous program using subprograms.

Program input example:

```
10 DIM X(2),Y(2)
20 X(1)=0
30 X(2)=37
40 DO
50  CLEAR
60  CALL PLAYER(1)
70  CALL PLAYER(2)
80  FSYNC 1
90 LOOP
1000 SUB PLAYER(P) !Player movement control
1010 IF PAD(P,4)>0 THEN Y(P)=Y(P)+1
1020 IF PAD(P,3)>0 THEN Y(P)=Y(P)-1
1030 IF Y(P)<0 THEN Y(P)=0
1040 IF Y(P)>22 THEN Y(P)=22
1050 LOCATE X(P),Y(P)
1060 PRINT "A"
1070 END SUB
```

Program explanation:

Start of main loop

Clear the text on screen

Run the subprogram "PLAYER" , passing 1 as the argument.

Run the subprogram "PLAYER" , passing 2 as an argument.

Main loop end

A subprogram that handles player movement

If the down directional button is pressed, add 1 to the variable Y(P).

If the up arrow directional button is pressed, subtract 1 from the variable Y(P)

Ensure that the Y coordinate is not above the upper edge of the screen.

Ensure that the Y coordinate is not below the lower edge of the screen.

Move the cursor to the text coordinates (X(P), Y(P)).

Display "A"

Explanation: The part that processes controller input and displays "A" is defined in the subprogram called "PLAYER" starting from line 1000. This makes the main loop span just line 40 to line 90, making the flow easier to understand.

2D Graphics

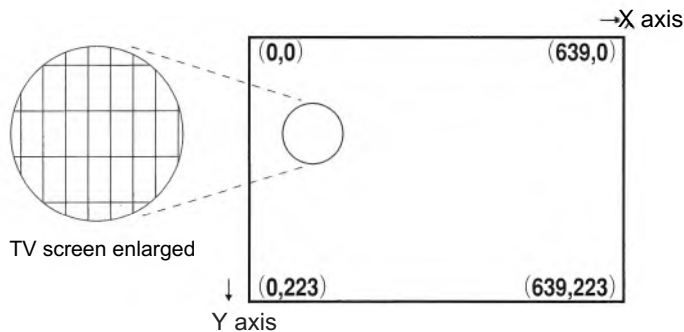
In a typical game, not only text is displayed, but also characters and objects are displayed as pictures as well. In EZ-BASIC, you can display lines, rectangles, and triangles, and you can also display graphics created with IMAGE-SKETCH on the screen. These 2D graphic objects are collectively called "primitives."

2D graphics coordinate system

The "text coordinates" used in the LOCATE command represent the coordinates of a television screen as 38 characters by 23 characters. In primitives, the position of the primitive is specified in a coordinate system called "screen coordinates."

Screen coordinates

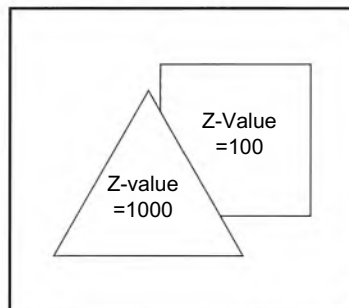
The display of TV and monitor screens are treated as a collection of small squares called pixels. In other words, you can think of the entire screen as being like graph paper. In EZ-BASIC, the TV screen is treated as graph paper divided into 640 pixels horizontally and 224 pixels vertically. The top left corner of the screen is the origin (0,0), and from this origin, the x-axis runs from left to right, and the y-axis runs from top to bottom. To specify a position on the screen, you specify it in terms of "coordinates," as in "how many pixels along the x-axis and how many pixels along the y-axis." These coordinates, with the pixel as the basic unit, are called "screen coordinates."



As you can see in the enlarged image, each pixel is twice as tall as it is wide, so please note that even if you increase the X and Y coordinates equally, the displayed object will not move in a 45-degree downward and right direction.

Z-Value

There is another coordinate in the screen coordinate system that is not visible on the screen, called the Z-value. This represents depth, with the backmost part of the screen being 0 and the front being 16777200. This determines which primitive will be displayed on top when multiple primitives overlap. Note that even if the Z-value becomes smaller and the primitive moves further back, the size of the primitive will not become smaller. The only difference is which one will be displayed in front.



Basic shape display

Line primitive display

Now let's display a basic shape - a line primitive. A line primitive is displayed as follows. First, start with the "LINEDISP" command, which displays the line primitive.

Input example:

```
LINEDISP 1[Enter]
```

Execution example:



Now a single line, or line primitive, is displayed in the top left of the screen. The "1" after the "LINEDISP" command is an identification number called the primitive number, so the previous command means "display line primitive number 1." There are 1000 primitive numbers for each primitive, and it is possible to set coordinates and colors for each.

Specifying coordinates

The "LINEPOS" command is used to set the coordinates of a line primitive. The line primitive displayed earlier should still be displayed, so let's use the "LINEPOS" command to move it to the center of the screen.

Format:

```
LINEPOS PrimitiveNumber, StartX, StartY, EndX, EndY
```

Input example:

```
LINEPOS 1,300,100,340,124,0[Enter]
```

Execution example:



This moves the line primitive to the center of the screen. The start and end coordinates are the coordinates of the two ends of the displayed line primitive, and the line will be drawn between these two points.

If you look at the top right of the screen, where the line primitive was previously, you will see that the line primitive that was displayed there has disappeared. Primitives such as line primitives move when you change the coordinates, so there is no need to delete the one that was previously displayed.

Setting color

"LINECOL" is the command to change the color of the line primitive that was displayed earlier. Let's try changing it right away.

Format:

LINECOL *PrimitiveNumber, Red-value, Green-value, Blue-value, alpha-value*

Example input:

```
LINECOL 1,0,255,0,127[Enter]
```

When you do this, the line primitive will change from white to green. The "LINECOL" command specifies the color of the specified line primitive using the intensity of the three primary colors of light. The three primary colors of light are red, green, and blue, and the intensity is set from 0 to 255 to create a color. These are called "RGB values."

alpha-value

The final parameter of the LINECOL command, the alpha value, indicates the transparency of the primitive. 0 is completely transparent, and 128 is opaque. High transparency allows primitives to show through from lower z-values.

Erasing a line primitive

To remove a line primitive, use the LINEOFF command.

```
LINEOFF 1[Enter]
```

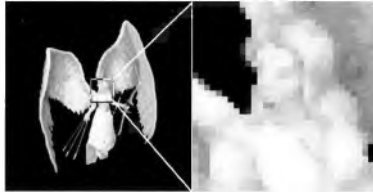
Now the line has disappeared from the screen. As seen so far, remember that all commands that operate on primitives must first specify the primitive number.

Sprites

To display graphics created with IMAGE-SKETCH on the screen, you use "Sprite Primitives." First, we will explain the graphic materials that are the source material for Sprite Primitives.

Graphic file composition

As mentioned in the section on screen coordinates, the screen is made up of a collection of tiny dots called "pixels." Graphic materials created using the graphics tool "IMAGE-SKETCH" are also made up of the color displayed for each pixel.



EZ-BASIC handles graphic materials using a method called "index color."

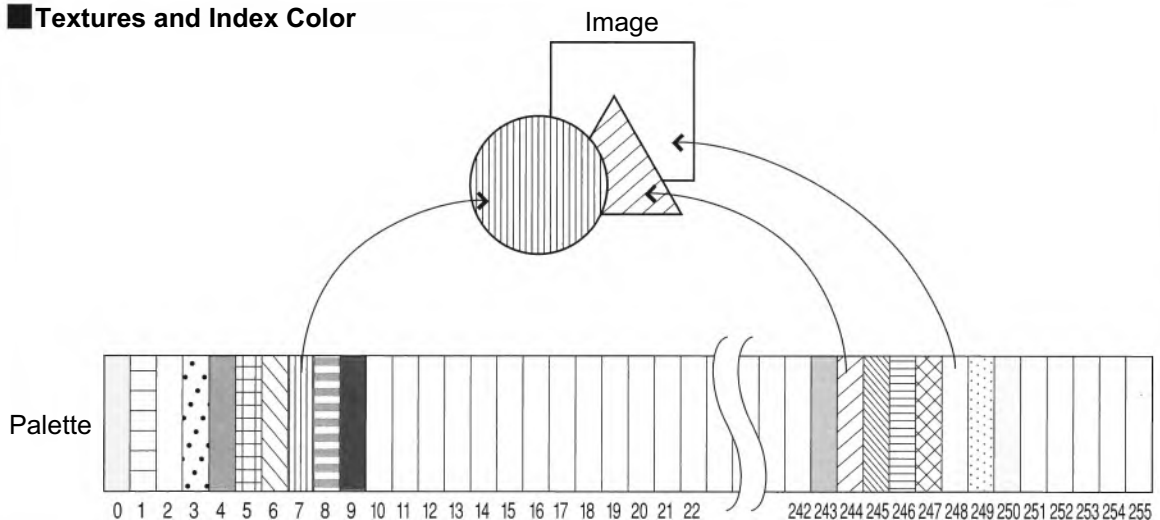
*Pixels for graphic materials in IMAGE-SKETCH are square, unlike screen coordinates. However, data created with IMAGE-SKETCH will not appear vertically elongated when displayed in EZ-BASIC .

Index color

"Index color" is a method of selecting a set of 256 or 16 colors from 16777216 colors using RGB values, and then drawing a picture using those selected colors. For this reason, files created with IMAGE-SKETCH are divided into two files: a "texture file" and a "palette file." A "palette file" contains 32 pieces of data containing the 256 selected colors, e.g. color 0 being white, color 1 being yellow, and so on up to 255. In other words, you can think of it as a palette with 256 color data items. This is the same in 16-color mode.

A "texture file" is a collection of up to 16 pages of "texture images" that contain image data. A "texture image" is like a "draft of an image" that specifies which entry from the palette to paint each pixel with. Changing the palette will change the color painted on the texture, so you can create characters with different colors just by changing the palette data.

■ Textures and Index Color



Sprite display procedure

As explained earlier, sprite primitives consist of two parts: a "texture image" and "palette data." To display the data created with "IMAGE-SKETCH" on the screen, follow the steps below.

Step 1: Read the "texture file" and "palette file" using "TEXLOAD" and "PALLOAD".

Step 2: Use "SPRTEX" and "SPRPAL" to assign the "texture image" and "palette data" to a sprite primitive (using a primitive number 1 to 1000).

Step 3: Use "SPRDISP" to display it on the screen, and "SPROFF" to remove it from the screen.

Other adjustments such as adjusting the display position in "SPRPOS" and "SPRPOS2"

*If you are using a pre-loaded texture, you can display it by simply following step 3. Please refer to page 126 for the contents of the default texture.

Loading graphic data

• TEXLOAD

This is not necessary if you are using pre-loaded graphic materials, but if you want to use data created with IMAGE-SKETCH, you will first need to load the texture file you created.

Example input:

```
TEXLOAD "TEX1"[Enter]
```

Explanation: Load a texture file with the file name "TEX1" saved in texture format.

"TEX1" saved in IMAGE-SKETCH has now been loaded. A texture file contains up to 16 texture pages saved together.

If you load a file that only uses specific texture pages, only the texture pages that are used will be overwritten, and any texture pages not used by the loaded texture file will remain the same as they were before the file was loaded.

• PALLOAD

Similarly, use the "PALLOAD" command to load the palette file you created.

Example input:

```
PALLOAD "PAL1"[Enter]
```

Explanation: Loads the palette file with the file name "PAL1" .

Now that we've loaded the texture image and palette data, we need to assign them to a sprite primitive.

Applying loaded data to a sprite primitive

• SPRTEX

This assigns a portion of the loaded texture page to a sprite primitive. The syntax is as follows:

Format:

`SPRTEX PrimitiveNumber, TexturePage, TexStartX, TexStartY, TexEndX, TexEndY`

Example input:

```
SPRTEX 1,1,0,0,31,31[Enter]
```

Explanation: For sprite primitive 1, the area enclosed by a rectangle with a diagonal line running from coordinates (0,0) to (31,31) on texture page 1 is set as the texture.

• SPRPAL

Assigns palette data corresponding to a texture to a sprite primitive.

Format:

SPRPAL *PrimitiveNumber, PaletteNumber, PaletteOffset, alpha-mode*

Example input:

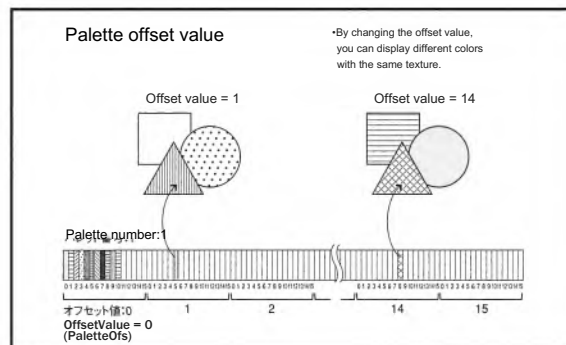
SPRPAL 1,3,0,1[Enter]

Explanation:

Assign palette number 3 to sprite primitive number 1.

* About palette offset

All palettes have 256 colors, so when the texture is in 16-color mode, the 256 colors are divided into 16 blocks and the pattern to use is selected from among them. In this case, the palette offset is '0', so the first 16 colors are used. Note that when using 256-color texture data, always set the palette offset to 0.



Regarding Alpha-mode

Changing the alpha mode changes the way the transparency is calculated. The effect of each mode is as follows:

alpha mode	Calc. method	Effect
0	Subtract	RGB values are subtracted from what they overlap. For example, if a blue sprite (0,0,255) overlaps white (255,255,255), it will be displayed as yellow (255,255,0).
1	Translucent	If they overlap, normal semi-transparent processing will be performed. To become translucent, the palette must have an alpha value set.
2	Add	RGB values of sprites overlapping are added together. For example, a blue sprite (0,0,255) overlapping a red background (255,0,0) will be displayed as purple (255,0,255).

Displaying sprite primitives

SPRDISP

To display a sprite primitive, use the "SPRDISP" command, just like with line primitives. To specify the size, set the drawing range with the SPRPOS2 command.

Example input:

SPRDISP 1[Enter]

Explanation:

Displays sprite primitive 1.

Execution result:



SPROFF

Similarly, sprites can be removed using "SPROFF"

Input example:

SPROFF 1[Enter]

Explanation

Removes sprite primitive 1 from the screen.

Changing a sprite primitive's position

SPRPOS

The SPRPOS command specifies the display position of a sprite primitive. The position is specified by the coordinates of the top left corner of the sprite. The size of the sprite will be the same as when it was created with IMAGE-SKETCH.

Format:

SPRPOS *SpritePrimNumber*, *ScreenCoordX*, *ScreenCoordY*, *Z-value*

Example input:

SPRPOS 1,320,112,0[Enter]

Execution result:



SPRPOS2

To scale a sprite, use the SPRPOS2 command. Unlike the SPRPOS command, this command sets start and end coordinates, the line connecting which forms the diagonal of the rectangle to which the sprite is scaled. You can also display an inverted image by adjusting the relative positions of the start and end coordinates.

Format:

SPRPOS2 *SpritePrimNumber*, *StartCoordX*, *StartCoordY*, *EndCoordX*, *EndCoordY*, *Z-value*

Input example:

SPRPOS2 1,340,124,300,100,0[Enter]

Execution result:



■ SPRPOS2 Table

Start Coord.	End Coord.	
Top left	Bottom right	Sprite is presented as drawn in IMAGE-SKETCH
Bottom left	Top right	Sprite is displayed vertically mirrored
Top right	Bottom left	Sprite is displayed horizontally mirrored
Bottom right	Top left	Sprite is both horizontally and vertically mirrored

■ Background color

BGCOL

With BGCOL the background color can be set.

Format:

BGCOL *RED*, *GREEN*, *BLUE*

Input example:

BGCOL 180,85,25[Enter]

Explanation:

The background becomes a sunset orange.

3D Graphics

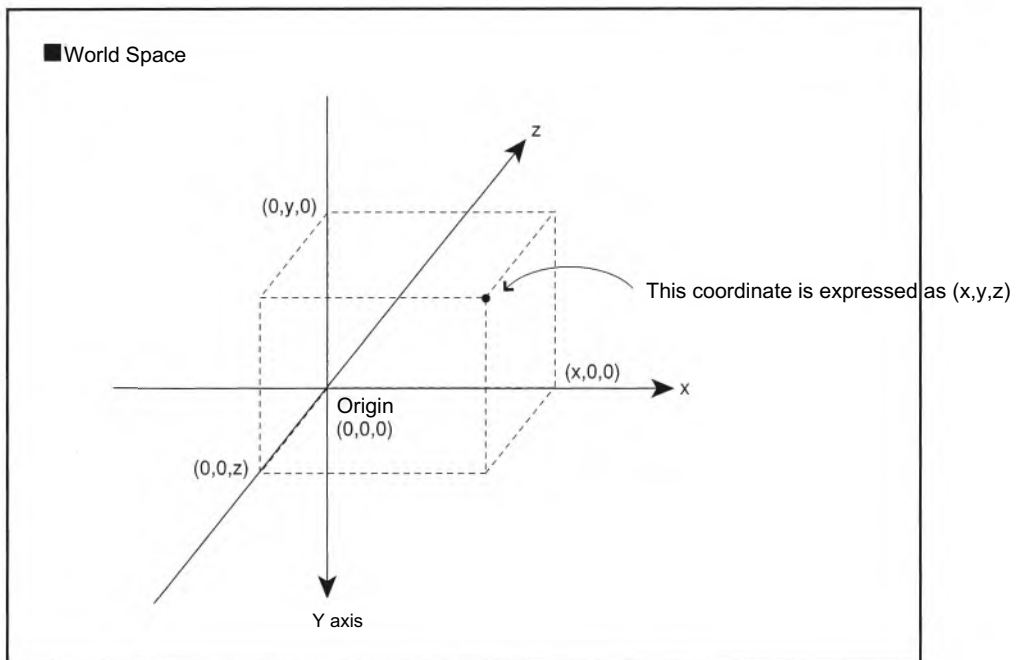
Many modern games feature characters that move around the screen in a three-dimensional manner. Unlike 2D graphics, which use the screen as the basis of coordinates, 3D graphics deal with three-dimensional space, so it takes some getting used to, but once you're proficient, you can do many operations, such as viewing models from various angles or rotating them. EZ-BASIC also has many 3D functions that are easily used in games, making it easier to use than 2D graphics in some cases.

To display a model

As mentioned earlier, to display a character on the screen using 3D graphics, you set up a "model" and a "camera" that captures the model in a space called "world space". That said, there's no need to think about it too hard; just think of this "User's Manual" that you are looking at right now as the "model", your eyes as the "camera", and the space around you as the "world space".

World space coordinate system

In the previous explanation, we came across the unfamiliar term "world space." This is a space that represents position using three components - length, width, and height, just like the real world we live in.



As shown in the figure, in world space the x-, y-, and z-axes extend from the origin (0,0,0) and intersect perpendicularly to each other. As with screen coordinates, positions in world space are expressed as "in the direction of the x-axis..., in the direction of the y-axis..., in the direction of the z-axis...". These are called "world space coordinates". Note that because world space is a virtual space, unlike text coordinates and screen coordinates, there are no minimum values. In EZ-BASIC, a length of 1 in world space coordinates corresponds to 1 cm in the modeling tool "POLY-CRAFT".

Model

If a sprite is like a picture drawn on paper in this world, then a model is like a physical replica. In order to display a model, it must first be placed in world space.

Camera

Once you have placed the model, you need to set up a camera to fit it on the screen and point it in the direction of the model. The camera position is an important factor, as the orientation and size of the model that can be seen will change depending on the relative positions of the camera and model.

Procedure to display a model

To display a model, follow the steps below. The model placement in step 1 is similar to the process for displaying sprites, so it should be easy to understand.

Step 1: Place the model in world space

- A. Use "MODLOAD" to load the "model file" into the model buffer. If the model uses texture , load the texture file and palette file used in the model file with "TEXLOAD" and "PALLOAD" respectively.
 - B. Assign the "model data" loaded with "MODSET" to one of the model numbers 1 to 1000 .
 - C. Use "MODDISP" to make the model visible.
- Optionally specify the position with "MODPOS" and the orientation with "MODROT" .

Step 2: Set up the camera

- D. Use "CAMPOS" to specify the camera position in world space.
- E. Use "CAMVEC" or "CAMROT" to determine the camera direction.

The model is now displayed on the screen. There are also other environment settings such as "Light" which illuminates the model and "Fog" which makes the model disappear into mist as it gets further away.

Model placement in world space

Let's start the process of displaying a model. First, place the model in world space.

Load the model buffer

• MODLOAD

Objects made in "POLY-CRAFT" are saved as a model file, which contains up to 10 models, each assigned a number from 1 to 10. EZ-BASIC has eight buffers that can store these 10 model files when they are loaded. For this reason, when loading a model file, you must specify the buffer number in addition to the file name.

Format:

MODLOAD *FileName*, *ModelBufferNumber*

Input example:

MODLOAD "TEST",1[Enter]

Explanation:

Load the data stored in the file named "Test" and store it in model buffer 1.

• TEXLOAD, PALLOAD

See page 43 for information on loading texture and palette data to use for textures on your models.

Apply loaded model data to a display model

• MODSET

Model data stored in the model buffer can be assigned to display models numbered 1 to 1000 using the "MODSET" command.

Format:

MODSET *DisplayModelNumber, ModelBufferNumber, ObjectNumber*

Input example:

MODSET 1,1,8[Enter]

Explanation:

Assign the model data in slot 8 (of 10) of the first model buffer to display model 1.

Making the model visible

• MODDISP

Just like with sprites and other primitives, models are displayed by executing the "MODDISP" command. Note that the "MODDISP" command only displays the specified display model; it will not be displayed on the screen unless the camera is pointing in the direction of the model.

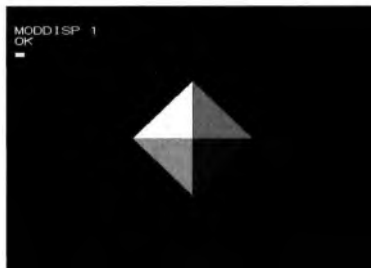
Format:

MODDISP *DisplayModelNumber*

Input Example

MODDISP 1[Enter]

Execution result:



Explanation:

Displays display model 1.

• MODOFF

The "MODOFF" command hides a model that was made visible by the MODDISP command.

Format:

MODOFF *DisplayModelNumber*

Example input:

MODOFF 1[Enter]

Explanation:

Hide display model number 1.

Positioning and rotating a model in world space

The "local space" of a model

When you create a model with "POLY-CRAFT", the x-, y-, and z-axes are displayed on the screen, indicating that the model has a coordinate system. This coordinate system is unique to each model you create and is called "local space coordinates." Local coordinates are used as a reference when specifying the position of a model, etc.

• MODPOS

The "MODPOS" command is used to specify the position of the display model in world space. Using the MODPOS command changes the position of the model to the specified world space coordinates.

Format:

MODPOS DisplayModelNumber, WorldXCoord, WorldYCoord, WorldZCoord

Example input:

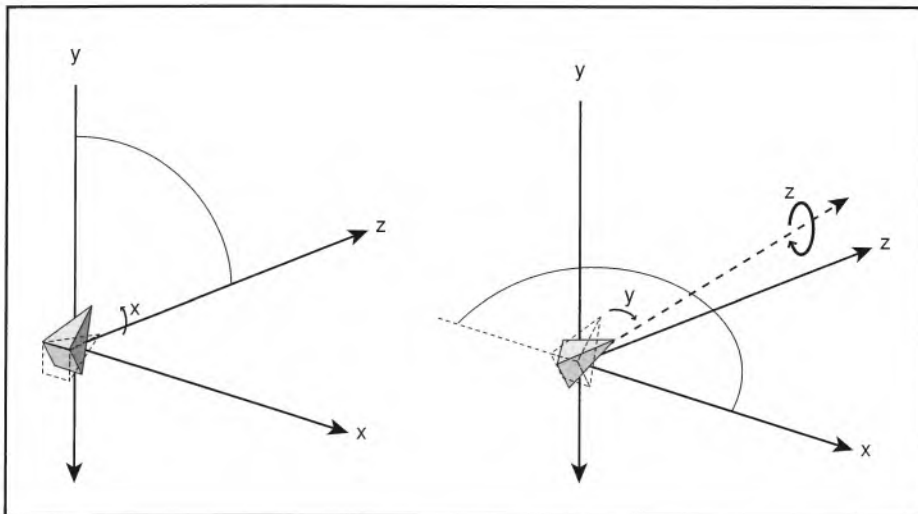
MODPOS 1,10,20,30[Enter]

Explanation:

Place display model number 1 so that the origin of its local coordinate system coincides with (10, 20, 30) in world space.

• MODROT

"MODROT" command specifies the orientation of the model for display. The orientation is determined as shown in the figure. The angle is specified in radians, where 360 degrees is 2π .



Explanation:

When a model is placed in world space, it is positioned with the model's local space coordinate axes and the world space coordinate axes pointing in the same direction. To specify the orientation using the "MODROT," think of it like this: First, rotate by angle X around the X axis. Next, rotate by angle Y around the Y axis. Finally, twist by angle Z around the axis in the direction the model is facing. Note that the orientation is specified with the origin of the local coordinate system as the center.

Format:

MODROT DisplayModelNumber, AngleX, AngleY, AngleZ

Program input example:

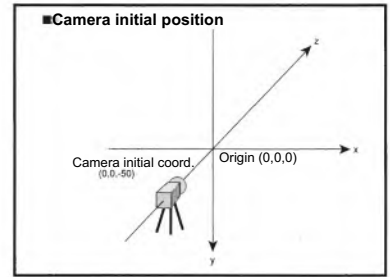
MODROT 1,0,PI/2,0[Enter]

Explanation:

The model with display model number 1 is oriented to face the positive direction of the y-axis.

Camera settings

Once you have placed the model in world space and made it visible, the next step is to set the camera. When we used the MODDISP command earlier, the model placed at the origin was displayed on the screen because the camera was positioned away from the origin and facing in the direction of the origin. The position of the model displayed on the screen is determined by the position and orientation of the camera.



Setting the camera position

• CAMPOS

To change the camera position, use the "CAMPOS" command. The position is specified in world space coordinates, just like specifying a model's position.

Format:

CAMPOS *CameraNumber, WorldXCoord, WorldYCoord, WorldZCoord*

Input example:

```
CAMPOS 1,50,0,0[Enter]
```

Explanation:

Positions camera number 1 at world space coordinates (50, 0, 0).

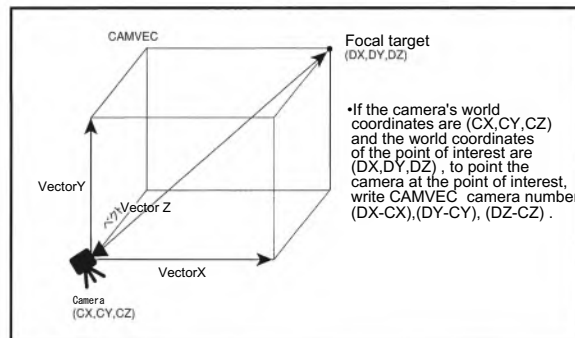
There are two cameras, and camera number 1 is usually used.

Camera orientation

Next, set the camera orientation. There are two ways to set the camera direction: "CAMVEC" and "CAMROT".

• CAMVEC

The "CAMVEC" command sets the camera direction using the vectors X, Y, and Z. The vectors can be visualized as follows:



• CAMROT

The "CAMROT" command determines the camera direction by angle. Unlike the CAMVEC, it allows you to twist the camera.

Format:

CAMROT CameraNumber, AngleX, AngleY, AngleZ

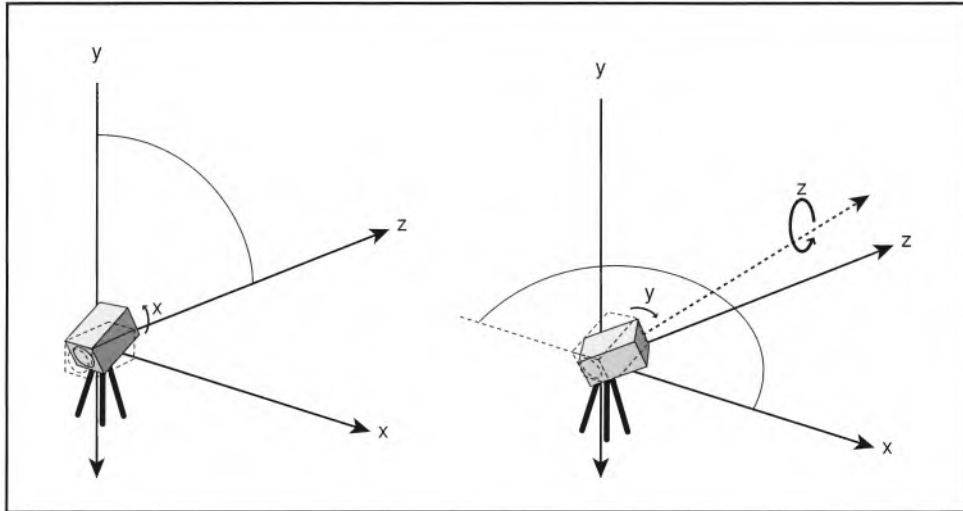
Example input:

CAMROT 1, 0, $\pi/2$, $\pi/2$ [Enter]

Explanation:

Camera 1 is facing in the positive direction of the x- axis, twisted 90 degrees.

The camera rotation is executed as follows:



Specifying the camera direction is the same as specifying the model direction. In the initial state, the camera is facing in the positive direction of the Z-axis in world space. To specify the direction with the "CAMROT" command, think of it as shown in the diagram. As shown in the diagram, first rotate by angle X around the X axis. Next, rotate by angle Y around the Y axis. This determines the direction of the camera's line of sight. Finally, add a twist by angle Z about the viewing direction's axis. This completes the specification of the camera direction.

Environmental Preferences

3D graphics, in addition to positioning models and setting up cameras, you can also set up the environment, such as light sources, ambient light, and fog.

Types of light

You can cast colored light onto placed models. There are two types of light that can be used in EZ-BASIC: "Parallel light sources" like sunlight, and "Ambient light" which sets how bright it is even in places where there is no light. Note that these settings do not affect 2D primitives or text.

• LIGVEC

Sets the direction of the light that illuminates the world space. You can use up to three lights and set a direction for each. The light direction is specified as a vector like CAMVEC .

Format:

LIGVEC *LightNumber, VectorXComponent, VectorYComponent, VectorZComponent*

Example input:

LIGVEC 1,100,0,0[Enter]

Explanation:

Sets the light direction of light number 1 to the positive direction of the X axis.

• LIGCOL

The "LIGCOL" command sets the color of the light. Set each of the three lights with RGB values.

Format:

LIGCOL *LightNumber, RED, GREEN, BLUE*

Example input:

LIGCOL 1,0,0,255 [Enter]

Explanation:

Light number 1 will be blue.

• AMBCOL

This sets how bright areas that are not illuminated by lights will be. By setting the ambient light to (0,0,0), you can create a pitch-black effect like outer space.

Format:

AMBCOL *RED, GREEN, BLUE*

Example input:

AMBCOL 0,255,0[Enter]

Explanation:

Sets the ambient light to green.

Fog

Fog creates the effect of a model disappearing into mist as it moves far away from the camera. You can set the color and distance of the fog.

• FOGCOL

Set the fog color. By using the same color as the background, the model will appear to disappear as it moves away.

Format:

FOGCOL *RED, GREEN, BLUE*

Example input:

FOGCOL 255,0,0[Enter]

Explanation: The fog color will be red.

• FOGSET

"FOGSET" command sets the fog effect range.

Format:

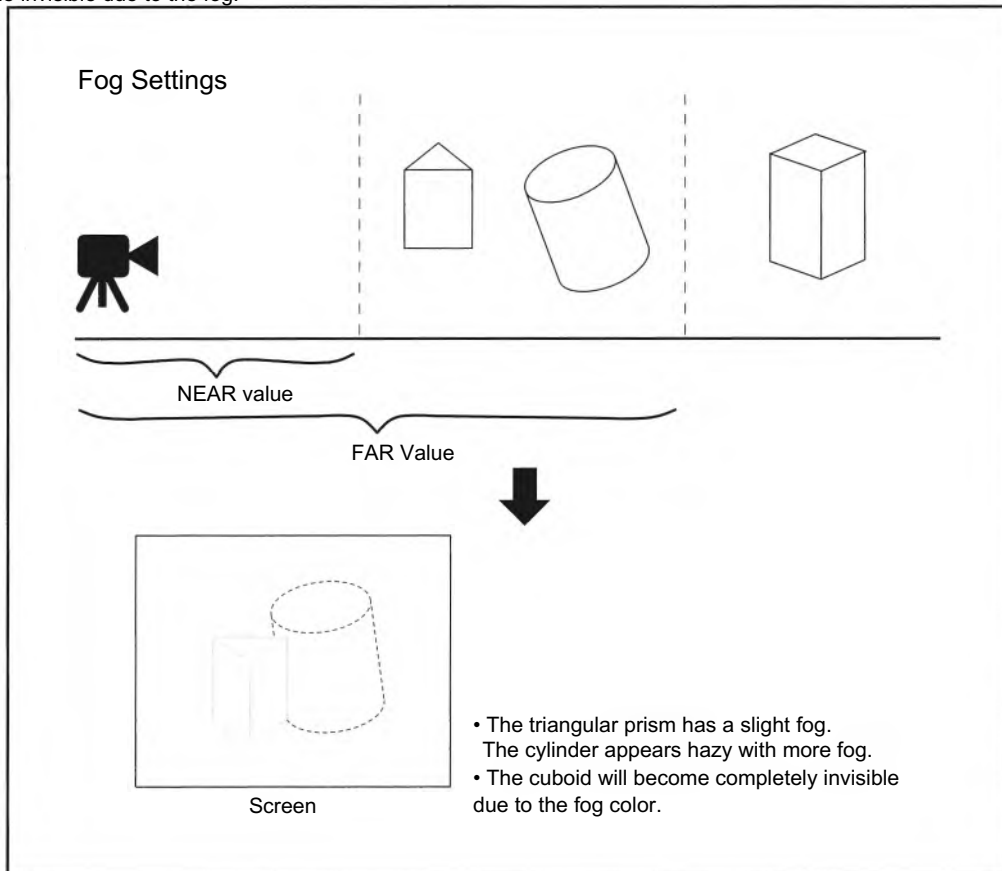
FOGSET *Distance at which fog begins, distance at which fog is completely set*

Example input:

FOGSET 200, 1000 [Enter]

Explanation:

When the model is 200 away in world coordinates, fog will begin to appear, and when it is 1000 away, the model will become invisible due to the fog.



EZ-BASIC coordinate systems

Up until now, EZ-BASIC has used three coordinate systems to display text, primitives, and models. How are these coordinate systems combined and displayed on the screen?

Display of the 3 coordinate systems

First, let's review the three coordinate systems used in EZ-BASIC.

• Text Coordinate System

The text coordinate system is used for text input or PRINT command output. As explained on page 32, this coordinate system represents the entire television screen as 38 x 23 characters, with the character size being the smallest unit.

• Screen coordinates

Screen coordinates are the coordinate system used to display primitives, and are expressed as 640x224 pixels, using pixels of the display as the smallest unit. Unlike text coordinates, screen coordinates have a depth called the Z-value.

• World space coordinates

World space coordinates create a three-dimensional space in which models and cameras are positioned. Unlike the previous two coordinate systems, world space does not have a specific value for the smallest unit. Furthermore, each model placed in world space has its own local space. This is the coordinate system from when the model was created.

So how are these three spatial coordinates combined to display the screen?

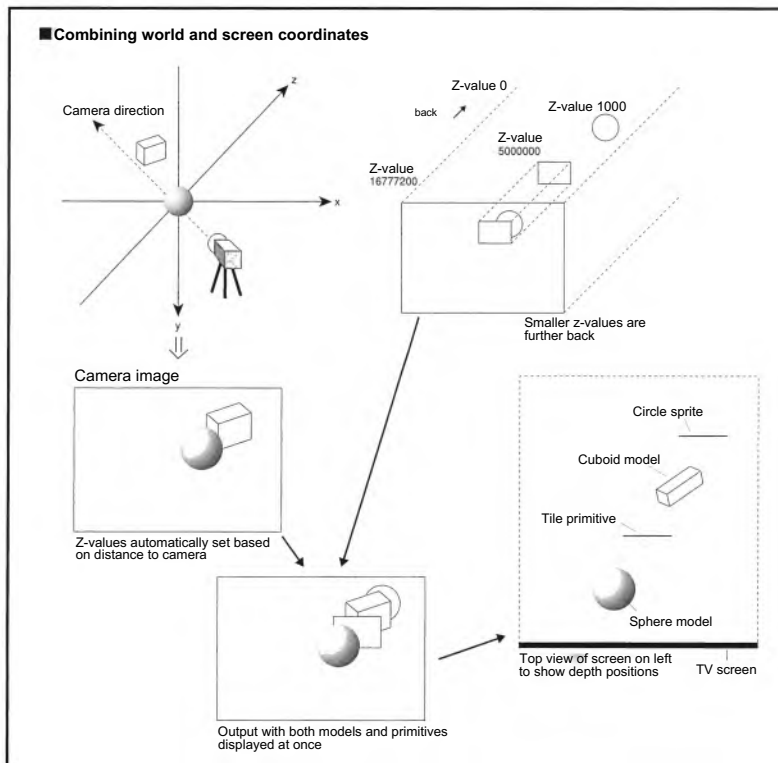
The text screen is displayed in the foreground (no depth)

Of the three systems above, only the text coordinate system does not have depth information. Therefore, text is always displayed in front of the screen. Text written with the PRINT command will never be hidden by models or primitives.

Depth in screen and world space coordinates

In screen coordinates, each primitive has a depth value called a z-value, and when primitives overlap, the one with the larger z-value is displayed preferentially.

In world coordinates, the z-value is automatically determined as depth based on the relative positions of the camera and the displayed model. Combining the concepts of depth in these two coordinate systems results in the following diagram.



Depending on the z-value of the primitive and the distance between the model and the camera, the primitive may appear to slice through the model, as in the following example.

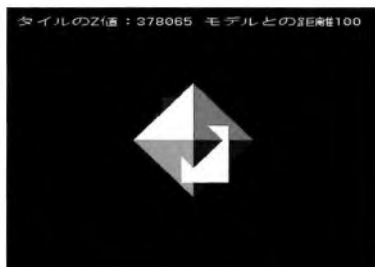
Program input example:

```
100 INIT
110 TILEDISP 1
120 MODDISP 1
130 PZ=350000
140 CAMPOS 1,0,0,-100
150 DO
160 IF PAD(1,13)>140 OR PAD(1,13)<114 THEN
170 MZ=MZ-PAD(1,13)/32+4
180 END IF
190 IF PAD(1,15)>140 OR PAD(1,15)<114 THEN
200 PZ=PZ+PAD(1,15)-127
210 END IF
220 IF PZ<0 THEN PZ=0
230 IF PZ>16777200 THEN PZ=16777200
240 MODPOS 1,0,0,MZ
250 TILEPOS 1,300,100,380,148,PZ
260 CLEAR
270 LOCATE 0,0
280 PRINT "Tile Z-value: ";PZ;"Model distance: ";MZ+100
290 FSYNC 1
300 LOOP
```

Explanation:

Use the left analog stick to move the model to change the distance from the camera, and the right analog stick to change the Z value of the tile primitive. The distance from the model and the Z value of the tile primitive are displayed at the top of the screen.

Example output:

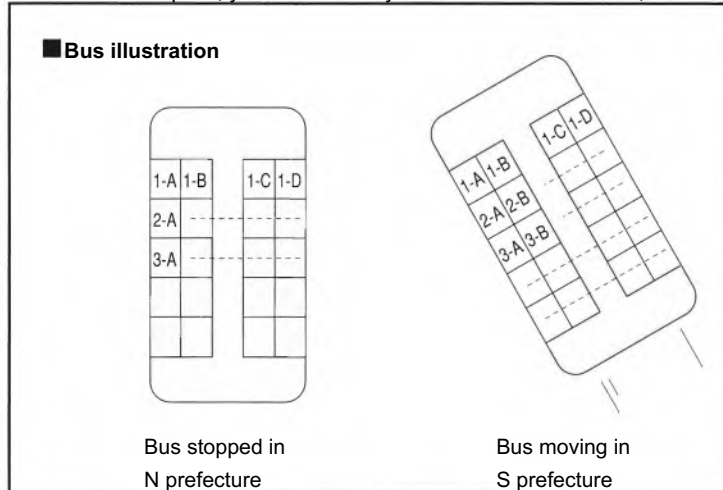


World and local space

The term "local space" was used when discussing MODPOS, but it is not yet clear how this connects to "world space." For example, let's consider a model replaced by a "bus." The bus you are riding on can move freely around the world space, "Japan" (though it cannot fly, of course). This "space" is called "world space."

The location of the bus can be indicated by a world space coordinate called an "address." However, if you want to ask someone to move your seat on the bus, you specify it using a "seat map" like 1-A instead of an address. This seat map corresponds to "local space coordinates." Therefore, the seat you are sitting in can be expressed using either a "Japanese address" or a "bus seat map."

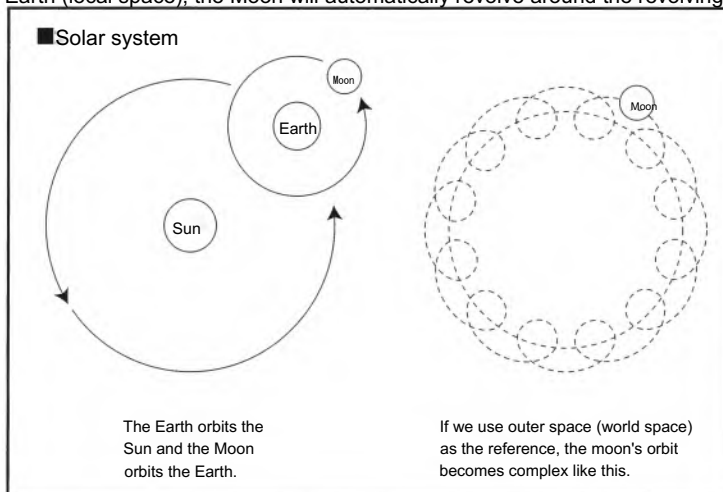
However, if you represent your seat with an address, the address will change every time the bus moves. If you instead represent your seat with a bus seating chart, it will not change no matter where the bus is currently traveling or which way it is facing. By using this kind of local space, you can smoothly move two related models, such as "you" on a "bus."



In this way, the local coordinates do not change even when the bus moves and changes direction. By using the local coordinates of the bus, we can easily handle the position and direction of passengers, which depend on the position and direction of the bus.

Parent-child model links

"MODJOIN" allows you to create parent-child relationships between models, such as between a bus and passengers. By creating a parent-child relationship, you can easily specify the movement of the model which depends on a parent model. The relationship between the Sun, Earth, and Moon is easy to understand. The Earth revolves around the Sun, and the Moon revolves around the Earth. If we consider the Moon's movement based on outer space (world coordinates), it will move as shown in the illustration, making it difficult to reproduce. Here, we use the "MODJOIN" command to create a parent-child relationship between the Earth and Moon. Then, by simply creating a circular orbit in a space based on the parent model, the Earth (local space), the Moon will automatically revolve around the revolving Earth.



Sound

EZ-BASIC allows you to easily play songs and sound effects created with the sound tool "SCORE-NOTE."

Music

To play music created in "SCORE-NOTE," follow the steps below.

Step 1: Use the "BGMLOAD" command to load the BGM file you created into the BGM buffer.

Step 2: Play the loaded song with the "BGMPLAY" command. End playback with "BGMSTOP."

Step 3: Adjust the background music volume with the "VOLUME" command.

• BGMLOAD

To play music data, you need to load a BGM file into the BGM buffer. There are four BGM buffers, allowing you to hold four music files at once.

Format:

BGMLOAD *"File name", BGMBufferNumber*

Example input:

BGMLOAD "SAMBA",1 [Enter]

Explanation: Load the BGM file "SAMBA" into BGM buffer 1 .

• BGMPLAY

The "BGMPLAY" command plays the song stored in the BGM buffer. By setting the song data to loop using "SCORE-NOTE," the song can be made to loop automatically even after it reaches the end. If a new "BGMPLAY" command is executed during playback, the song that was playing will stop and the specified song will play.

Format:

BGMPLAY *BGMBufferNumber*

Example input:

BGMPLAY 1[Enter]

Explanation:

Plays the background music stored in BGM buffer number 1.

• VOLUME

To adjust the volume of the song being played, use the VOLUME command. The maximum volume is 255.

Format:

VOLUME *Volume*

Example input:

VOLUME 64[Enter]

Explanation:

Play music at volume 64.

• BGMSTOP

"BGMSTOP" stops the currently playing song.

Format:

BGMSTOP

Example input:

BGMSTOP[Enter]

Explanation:

Stops the currently playing song.

Sound effects

Sound effects made with "SCORE-NOTE" are compiled into a single file. The "SELOAD" command loads an SE file, which can then be played with "SEPLAY."

SELOAD

Load the SE file created with "SCORE-NOTE" .

Format:

SELOAD *"File name"*

Example input:

SELOAD "NOISE" [Enter]

Explanation:

Load the sound effect file "NOISE" .

SEPLAY

Plays a sound effect. Specify the sound effect number from the loaded file "SCORE-NOTE" file, and left and right volume. The sound effect will automatically stop playing when it finishes.

Format:

SEPLAY *SoundEffectNumber, LeftVolume, RightVolume*

Example input:

SEPLAY 27,45,60 [Enter]

Explanation:

Sound effect number 27 will be played at a volume of 45 on the left and 60 on the right.

IMAGE SKETCH

Controls

Controller explanation

- The DUALSHOCK 2 analog controller connected to port 1 is used.

DUALSHOCK2 Analog controller



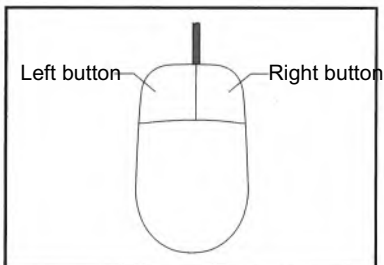
* "BASIC STUDIO Powerful Game Workshop" is compatible with the analog controller (DUALSHOCK 2). The LED indicator will always be red while the software is running.

Left stick	Used to move cursor.
Directional buttons	Used to move cursor.
Button	Draw, select icons, etc.
X Button	Cancel drawing, pick up colors, etc.
Button + Left stick	Drag
+ Directional buttons	Drag
Button	Option ON/OFF
□ Button	Switch between tool/palette view
L2,R2 Buttons	Change palette number
R1 Button	Change to zoom/normal mode
SELECT Button	Printscreen button (if popegg connected)

USB Mouse explanation

- You can connect a USB mouse to the USB port on the PlayStation 2 console and use it.

* Not compatible with the Playstation mouse.



Left button	Used for drawing, selecting icons, etc. (Same function as the circle button.)
Right button	Used to cancel operations, retrieve drawing colors, etc. (Same function as the X button.)

- When the text mentions the operation to "drag," hold down the button and move the cursor.

USB Keyboard Explanation

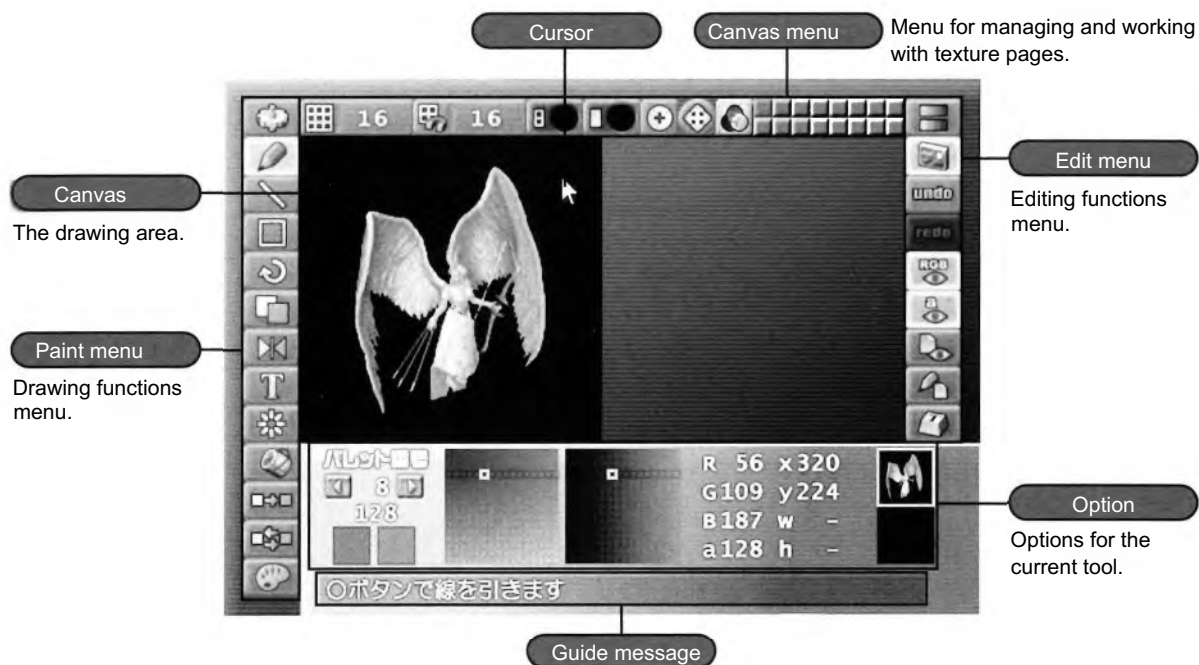
- You can connect a USB keyboard to the USB port on the PlayStation 2 and use it.
It can be used to enter text or to enter file names.
- Printscreen function: When the printer "popegg" is connected, you can press the Printscreen key to print out the currently displayed screen.

How to read the screen

"IMAGE-SKETCH" Screen

IMAGE-SKETCH is a tool for creating images. The images you create can be used as textures for models and sprite primitives.

- See page 100 for instructions on how to use the image you created as a texture for your model.
- See page 43 for information on how to use the images you create as sprite primitives.



Images

Images

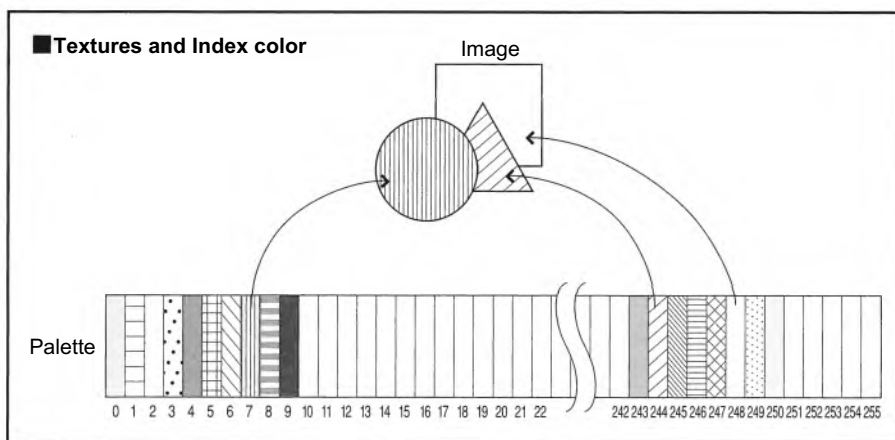
- For textures, you can create images with dimensions of 512x256 pixels (in 16-color (4bit) mode) or 256x256 pixels (in 256-color (8-bit) mode).
On the canvas, there is a work area of the same size at the bottom of the screen. (Please note that the work area is not saved.)
- There are 16 texture pages, one is used for each image.
- You can use one full-color image as a temporary workspace.
Full color images can be loaded from sample data.
- 16 images are saved together in one TEX format file. Full color images cannot be saved.

Color mode

- "IMAGE-SKETCH" has an **index color mode** and a **full color mode**.

Index color mode

- Index color mode allows editing of images and palettes for use in textures.



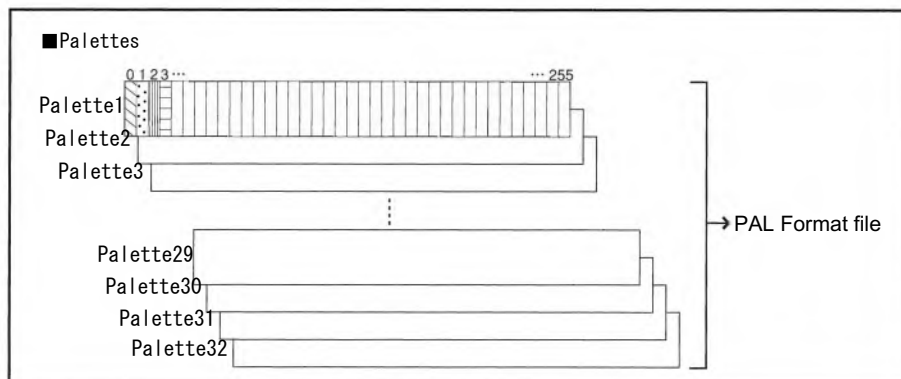
- Index color mode is a mode that uses 256 colors selected from full color (16 million colors) to express an image.
A color consists of 256 levels of red, green, and blue components, and 128 levels of alpha value (transparency).
- Index color mode has two modes: 16-color (4-bit) mode, and 256-color (8bit) mode.
- In the 16-color (4-bit) mode palette, the colors used by the texture are specified in units of 16 colors.

Full color mode

- A temporary working area for creating high-quality images.
- You can convert it into an indexed color image and then use it as a texture image.
- The full color image itself cannot be saved.

Palettes

- You can use 32 palettes in index color mode.
- 32 palettes are saved together in one PAL file.



Index color mode

Selecting one of the texture pages from the page selection buttons in the canvas menu (see p.82) will switch to index color mode.

System



- Save and load images and palettes.
- Place the cursor on the icon and press the O button to open the system menu.

新規作成 Create new

- Discards any data being edited and initializes the palette and texture pages.

読み込み Load

- Loads an image file or palette file.
- First, select the device where the desired file is stored.
The device can be selected as a CD-ROM, a PlayStation 2 dedicated memory card (8MB) inserted into MEMORY CARD slot 1, or MEMORY CARD slot 2.



- Next, select the directory where the desired file is stored.



- Select the desired file.
Select the desired texture file (TEX format) or palette file (PAL format) from the file list.



- Select the data stored in the file.
- The buttons are marked to correspond to the areas where data is stored in the file.
*For image data, there are 16 buttons corresponding to the 16 texture pages.
*In the case of palettes, there are 32 buttons corresponding to the 32 palettes.
- The data marked on the button will be loaded.
If there is data you do not want to load, uncheck the corresponding button.
Areas that are not loaded will retain the data currently being edited.



Load sample

- You can load the sample data included in the "BASIC STUDIO DISC."
- Select the sample type or sample data and load.
- For images, also load the corresponding palette.
*The texture page and palette you are currently editing will be overwritten by the texture page and palette containing the samples.

テクスチャーを保存 Save texture

- Save the texture page to a file.
 - First, select the texture page(s) you want to save.
Check the button corresponding to the texture page you want to save.
 - Next select the device you want to save to (see page 63) .
 - Finally, select the directory and file you want to save.
- If you want to create a new directory, select "New Directory" and give it a name.
If you already have a file you want to save over, select it.
If you want to create a new file, select "New File" and give it a name.



• Directory and file names

Select the characters and use the circle button to enter the file name.

Once you have finished entering the file name, press OK.

Pressing Delete will delete the character immediately before the cursor.

The file name can be up to 8 characters long.

* Directory names can be up to 8 characters long.

* If a USB keyboard is connected to the USB port on the PlayStation2 console, you can enter the file name from the connected USB keyboard.

* On the canvas, there is a working area at the bottom of the screen that is the same size as the texture to be saved.
Please note that this working area will not be saved.

パレットを保存 Save palette

- Save the palette to a file.
- First, select the palette number you want to save.
Check the button corresponding to the palette you want to save.
- Next, select the device you want to save to (see p.63).
- Finally, select the directory and file you want to save to, just like when saving a texture.

If you want to create a new directory, select "New Directory" and give it a name.

If you already have a file you want to save over, select it.

If you want to create a new file, select "New File" and give it a name.

終了 Exit

- Closes IMAGE-SKETCH and returns to the title screen.

Pen



- Draws a line on the canvas in the foreground color.
The current drawing color is displayed in the Palette Options (see page 75).
You can select a different drawing color by hovering the cursor over the palette in the Palette Options and pressing the O button.
You can change the RGB and alpha values of the drawing color by editing the palette (see page 71).
- Press the x button on the canvas to select the color the cursor is pointing at as the drawing color.
- You can set the pen thickness in the tool options (p.75) .

Line



- Press the O button on the canvas to specify the first point where you want to start drawing a line.
Press the O button again to specify the next point and draw a straight line between those two points.
Furthermore, if you press the O button repeatedly, you can continue drawing lines.
- Press the x button while drawing a line to stop drawing the line.
- Before you start drawing, you can use the x button to select the color at the cursor as the drawing color.
- You can set the line thickness in the tool options (p.75).

Shape tool



- Draw shapes such as ellipses and rectangles within a specified area.
- Drag on the canvas to specify the area you want to draw.
- Use the x button to select the color at the cursor as the drawing color.

Rectangle



- Draws a rectangle in the specified area.

Fill rectangle



- Fills a rectangle in the specified area.

Ellipse



- Draws an ellipse in the specified area.

Fill ellipse



- Fills an ellipse in the specified area.

Transform

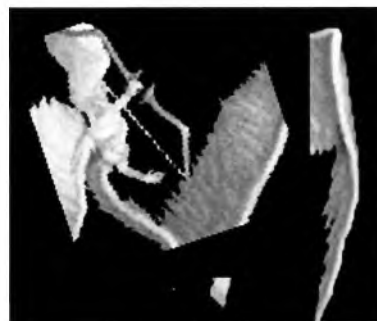


- Rotate or scale the area you drag on the canvas.
Drag within the specified area to rotate or scale.
Press the O button outside the specified area to confirm.
- Press the X button to cancel the operation.

Rotate



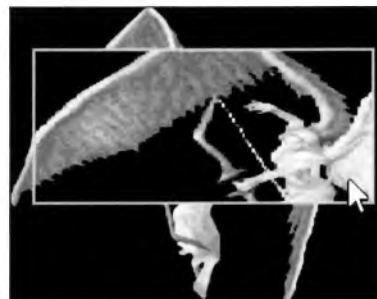
- Rotate the specified image area.



Enlarge - shrink



- Enlarge or shrink the specified image area.



Copy



- Drag on the canvas to copy or move the specified area.
- You can copy between texture pages by pressing a Texture Page button.

However, you cannot copy from the full color area to a texture page or between 16-color (4b) mode and 256-color (8-bit) mode.

- When copying, press the X button to cancel the copy or move.

Copy



- Copies the specified area on the canvas.
- Specify the area, place the cursor on the copy destination, and press the O button to paste the copy.

Merge Copy



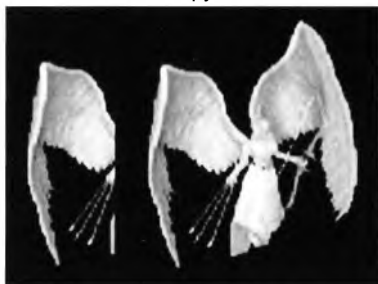
- Copies a specified area on the canvas, optionally excluding a color.
- Optionally specify colors not to copy.
 - 単色 "Single color": You can specify each color on the palette one by one.
 - ライン "Line": You can specify the color on the palette by line.

クリア "Clear": Clears the specified color.

反転 "Invert": Inverts the specified colors and the unspecified colors.

- Specify the range on the canvas, place the cursor on the copy destination, and press the O button to execute the merge copy

Copy



Merge copy



Move



- Move the specified image area.

Reflect



- Drag on the canvas to reflect the specified area.

Mirror (horizontal)



- Flips the specified area horizontally.

Mirror (vertical)



- Flips the specified area vertically.

Text



- You can draw text images on the canvas.
 - Press the circle button on the canvas to start entering text.
 - Enter text in the optional text window.
 - When you have finished entering text, press the O button again on the canvas or press the Enter key on the USB keyboard to confirm the text you entered.
 - Pressing the X button while entering text will cancel the text entry.
- フォント "Font": Set the text font.
サイズ "Size": Sets the size of the text.



*If a USB keyboard is connected to the USB port of the PlayStation 2, you can input characters from the USB keyboard.

*You can input letters and symbols.

Page initialization



- Initialize the page you are currently editing.
- Choose whether new images will be in 16-color (4-bit) or 256-color (8-bit) mode.

Fill

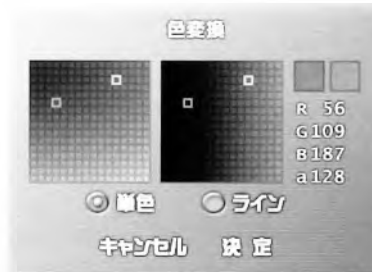


- The area on the canvas where you press the circle button will be filled with the drawing color.
- You can use the x button to select the color at the cursor as the drawing color.

Color conversion



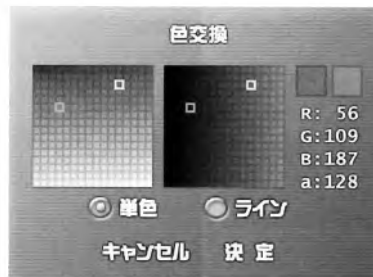
- Drag on the canvas to select a specific color within the area and convert it to a different color.
- Select the color you want to convert and the color you want to convert it to.
 - 単色 "Single color": You can set each color on the palette one by one.
 - ライン "Line": You can set the color on the palette by line.



Color exchange



- Drag on the canvas to specify a range to exchange the color within that range with another color.
- Select the color you want to exchange.
 - 単色 "Single color": You can set each color on the palette one by one.
 - ライン "Line": You can set the color on the palette by line.



Palette operations



- Pressing the circle button on the icon will open the palette operation menu.
- You can move to each menu by pressing the O button on the menu button for each palette operation.
- Select "OK"(決定) to confirm the changes made in the palette operation. Select "Cancel" (キャンセル) to cancel the changes made in the palette operation.

Palette editing

- Edit the colors on the palette.
- Select the palette number, move the cursor to the palette color you want to edit, and press the O button.
- Drag the RGB and alpha sliders to change the color.



Palette copy

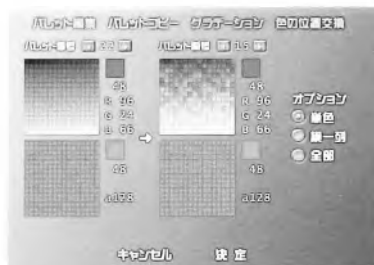
- Copies the colors on the palette to the specified palette.
- Select the palette number, move the cursor to the palette color you want to copy, and press the O button.

"Single color"(単色): You can set each color on the palette one by one.

"Line"(ライン): You can set the color on the palette by line.

"All"(すべて): Selects all colors on the palette.

- Select the destination palette number, move the cursor to the destination color, and press the O button.



Gradation tool

- Use the O button to create a gradient between two colors on the selected palette.
- Select the palette number for which you want to create a gradient.
- Select two colors on the palette with the circle button to create a gradient between them.



Vertical (縦方向): Creates a gradient in the vertical direction of the palette.

Horizontal (横方向): Creates a gradient in the horizontal direction of the palette.

Swap color positions

- Use the circle button to swap the two colors you selected on the palette.
- Select a palette number, move the cursor to the color you want to swap positions with, and press the O button.



"Single color" (単色): You can set each color on the palette one by one.

"Line" (ライン): You can set the color on the palette by line.

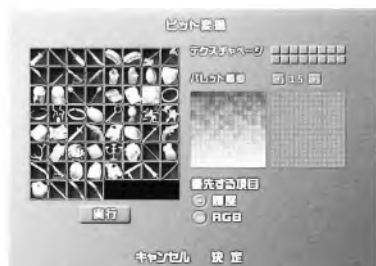
- Select the palette number you want to exchange with, move the cursor over the color you want to exchange with, and press the O button.

Bit conversion

Bit conversion



- Convert the color mode of the texture.
16 color (4bit) mode to 256 color (8bit) mode, or from 256 color (8bit) mode to 16 color (4bit) mode.
- Specify the area of the image on the canvas you want to convert.
- Specifying an image range opens the bit conversion menu.



- Select the palette number you want to convert and the texture page to which you want to write the converted image.
- When converting from 256-color (8bit) mode to 16-color (4bit) mode, also select a row on the palette.
"Luminance": Priority is given to luminance during bit conversion.
"RGB": RGB values are given priority during bit conversion.
- Select "Execute" to display the image after the bit swap.
- Select "OK" to execute the bit conversion and overwrite the specified texture page. Select "Cancel" to cancel the bit conversion.

Magnifier mode



- Magnifies an area of the canvas.

Normal



- Displays the entire canvas image.

Magnifier mode



- You can enlarge a portion of the canvas on the screen and process the image there.
- You can choose between two magnification levels.

Undo



- Undo the previous action.

Redo



- Reverses the previous undo operation.

RGB Display



- Toggle color visibility.

Alpha display



- Toggles whether alpha values are applied or not when displaying.

Option display



- Toggles the visibility of options.

Option swap



- Switch between tool and palette options.

Keyboard display

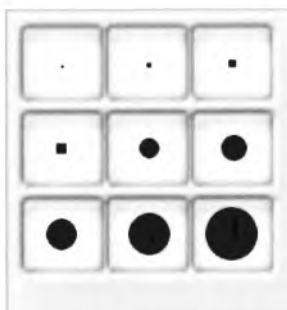


- Toggle display of the software keyboard.

Tool options

Pen width

- Specify the line thickness when drawing.
- Used when drawing pens, lines and shapes.



Palette options

Palette options

- Displays the current drawing color.
 - Place the cursor over a color on the palette and press the O button to change the current drawing color.
- You can also choose the palette you want to use.

*In 16 color (4 bit) mode, use the circle button to select the 16-color palette line to use from the palette you are using



Navigator

Navigator

- The image currently being edited is displayed in a smaller size.
- The area enclosed by the frame is the part displayed on the canvas.
- You can change the display range by dragging the frame.



Full color mode

Selecting the full color button next to the page selection buttons in the canvas menu (see p.82) will switch to full color mode.

System



新規作成 (New)

- The basic operation is the same as in index color mode (see p.63).

読み込み (load)

- The basic operation is the same as in index color mode (see p.63).

終了 (Quit)

- Exit IMAGE-SKETCH

Pen



- The basic operation is the same as in index color mode (see p.66).
- You can select the drawing method in the tool options (see p.81).

Pen



- Same as index color mode (see p.66).

Brush



- Draws lines with smoother edges than pen mode.

Airbrush



- Draws lines that look like blurred brush strokes.

Line



- The basic operation is the same as in index color mode. (see p.67)
- You can select the drawing method in the tool options.

Shapes



- The basic operation is the same as in index color mode. (see p.67)
- You can select the drawing method in the tool options.

Rectangle



- Same as index color mode.

Rectangle fill



- Same as index color mode.

Ellipse



- Same as index color mode.

Ellipse fill



- Same as index color mode.

Transform



- The basic operation is the same as in index color mode (see p.67).

Rotate



- Same as index color mode.

Scale



- Same as index color mode.

Copy



- The basic operation is the same as in index color mode (see p.68).
- You can select the drawing method in the tool options (see p.81).

Copy



- Same as index color mode.

Move



- Same as index color mode.

Reflect



- Same as index color mode (see p. 68).

Reflect (horizontal)



- Same as index color mode.

Reflect (vertical)



- Same as index color mode.

Text



- The basic operation is the same as in index color mode (see p.69).
- You can select the drawing method in the tool options (see p.81).

Initialize page



- Initializes the currently edited image.

Fill



- The basic operation is the same as in index color mode (see p.68).
- Depending on the threshold, colors close to the drawing color will also be included in the fill.
"Threshold"(しきい値): Sets the range of colors with values close to the drawing color.

Filter



- Apply various effects to the image
- Optionally, configure various filter settings.
- Drag on the canvas to apply the effect to a specified area.

Texture



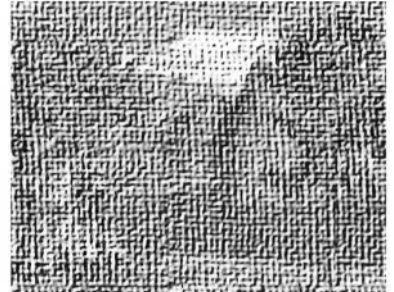
- Applies a three-dimensional effect to the image, giving it a bumpy appearance.

"Pattern": Sets the effect pattern.

"Color 1": Sets the color 1 to be used for the effect.

"Color 2": Sets the color 2 to be used for the effect.

"Strength": Sets the amount of texture depth.



Emboss

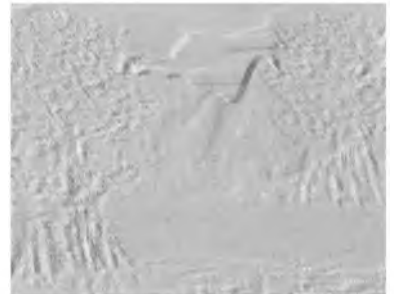


- Creates a sculptural effect on the image.

"Angle": Specifies the direction in which the effect is applied

"Height": Sets the engraving depth.

"Strength": Sets the amount to apply.



Blur



- Blur the image.

"Normal": Blurs the entire image.

"Horizontal": Blurs horizontally.

"Vertical": Blurs vertically.

"Strength": Sets the degree of blurring.



Noise



- Generates noise data on the image.

"Strength": Sets the amount of noise.

"Grayscale noise": Check this to draw noise in grayscale.



Cloud pattern



- Applies a cloud-like pattern to the image.
- "Color 1": Sets the color to be used for the cloud pattern.
- "Color 2": Sets the color used for the cloud pattern.
- "Normal": Draws cloud patterns in normal mode.
- "Add": Adds the cloud pattern to the current image.
- "Subtract": Subtracts the cloud pattern from the current image.
- "Opacity": Sets the opacity of the cloud pattern.



Gradient



- Draw using a gradient of any two colors of your choice.
- First, select the two colors you want to use in the gradient.
- Drag on the canvas to specify the area of the gradient, then drag to determine the direction of the gradient.
- "Normal": Draws the gradient normally.
- "Add": Adds the gradient to the current image.
- "Subtract": Subtracts the gradient from the current image.
- "Opacity": Sets the opacity when applying the gradient.



Color adjustment



- Adjust the color of the image.

Hue/Saturation



- Drag on the canvas to change the hue and saturation of a specific area.
- Optionally, set the Hue and Saturation values.
- "Hue": Sets the color component.
- "Saturation": Sets the vividness of the color.
- Select "OK" from the options or press the O button on the canvas to activate the settings.

Brightness/contrast



- Drag on the canvas to change the brightness and contrast of a specific area.
- Optionally, set the brightness and contrast values.
- "Brightness": Sets the brightness of the color.
- "Contrast": Sets the color contrast.
- Select "OK" from the options or press the O button on the canvas to activate the settings.

Color settings



- Edit the current drawing color.
- Press the circle button on the icon to open the menu.
- Drag the RGB and Alpha sliders to change the color.

Color reduction

Color reduction



- Reduce the color depth of the full-color image to create a texture image and palette.
- Select the palette number where the palette will be stored.
- Use the O button to select the texture page that will store the texture image to be created.
- Press "Execute" to display the image after color reduction.

"Yes": Dithering is performed when reducing colors.

"None": No dithering is performed when reducing colors.

- Click "OK" to perform color reduction, or "Cancel" to cancel.



Magnifier mode



- Enlarge a portion of the canvas.

Normal



- Display the entire canvas.

Magnifier mode



- You can enlarge a portion of the canvas on the screen and process the image there.

- You can choose between two magnification levels.

UNDO



- Reverses the last operation.

REDO



- Reverses the last undo.

Option display



- Hides or shows the options.

Swap options



- Switches between tool and palette option views.

Keyboard display



- Displays or hides the software keyboard.

Tool options

Pen width

- Specify the line thickness when drawing.
- Used with pen, line and shape drawing.



Drawing mode

"Normal": Draws with the drawing color.

The drawing part will be overwritten with the drawing color.

"Add": Draw or copy in additive mode.

The drawn area will be colored by adding the current color and the drawing color.

"Subtract": Draw or copy in subtraction mode.

The drawn area will be the current color minus the drawing color.

Normal



Add



Subtract



Transparency

- Set the transparency of the drawing.
- Drag the slider to change the opacity.

100%



60%



30%



Palette options

Palette options

- You can select a color from the palette and set it as the current color.
- Pressing the O button on the palette will set the color the cursor is pointing at as the current color.
- In full color mode, the alpha value of each color does not work.

Navigator

Navigator

- The image currently being edited is displayed in a reduced size.



Canvas menu

Grid



- Displays a grid on the canvas.
- Press the circle button on the icon to show or hide the grid.
- You can change the grid width by pressing the O button on the number shown.



Snap



- When snap is enabled, area specification, drawing, etc. will be in increments of the set value.
- Press the circle button on the icon to turn the snap function on or off.
- You can change the increment value by pressing the O button on the setting value.

Background color Index color mode only



- Displays the color you set as the canvas background. Does not affect the image you are editing.
- Displaying the background color allows you to see the alpha status of the image.
- You can change the background color by pressing the circle button on the icon.

Grid color



- Change the color of the grid.
- You can change the grid color by pressing the circle button on the icon.

Full/Same size display



- Display the image at full size or reduced size.
- Press the circle button on the icon to switch between full-size and reduced size display.

Move display area



- Drag the icon to scroll the image.

Page

Full color area

- Press the O button on the icon to switch to full color mode.
- The full color image will be displayed and can be edited.

Texture page

- Pressing the O button on any button will enter index color mode.
- The contents of the specified texture page will be displayed on the canvas and can be edited.

Full color area



Texture pages

POLY CRAFT

What is POLY-CRAFT?

"POLY-CRAFT" is a tool for creating "model data" to be used in BASIC STUDIO. Here you can create models and apply textures to them.

The 3 modes

"POLY-CRAFT" can be broadly divided into three modes.

• Model Creation Mode

This mode creates the shape of a model. You can load and edit an existing model, or create a model from scratch using basic shapes such as spheres and rectangular prisms.

• Texture Map Mode

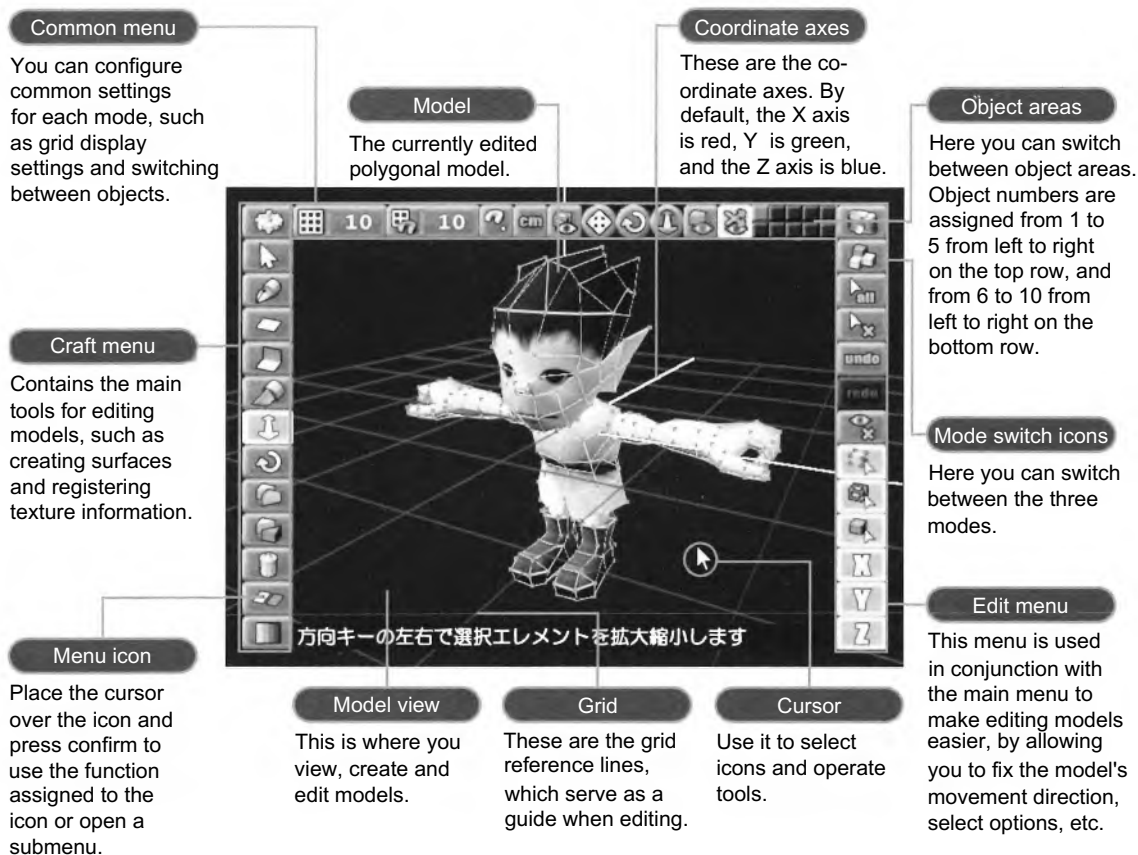
This mode allows you to apply an image called a texture to the surface of a model or color a surface directly.

• Collision Detection Mode

You can set collision detection boundaries (colliders) for models. By setting up colliders, you can easily detect collisions between models in "EZ-BASIC".

How to read the screen

POLY-CRAFT creates models using the following screen. The basic screen layout remains the same whether you are in "Model Creation Mode," "Texture Map Mode," or "Collision Detection Mode."



Model

Polygon



Point

Line

The Model View displays a model made up of surfaces called polygons. When editing a model, you can further edit it by manipulating the "lines" that make up the sides of the polygons and the "points" that make up the vertices. These "polygons," "lines," and "points" are collectively called "elements."

Model data structure

In "BASIC STUDIO," up to 10 models can be grouped together and treated as a single file. In "POLY-CRAFT," 10 models are placed one by one in an "object area," and each model can be edited by switching between object areas.

Controls

Controller explanation

Analog controller(DUALSHOCK 2)



* "BASIC STUDIO Powerful Game Studio" is compatible with the analog controller (DUALSHOCK 2). The LED indicator will always be red while the software is running.

Directional buttons	Move cursor	L1 Button	Shift function
Left stick	Move cursor	Circle + Left stick	Drag
Circle button	Confirm	SELECT Button	PrintScreen(only with popegg connected)
X button	Cancel	L2 Button	Ctrl function
Circle + Directional button	Drag		

USB Mouse

A USB mouse can be used in POLY-CRAFT. The operation method for a USB mouse is as follows:

Movement	Move cursor
Left button	Confirm, select
Right button	Cancel

* Not compatible with the PlayStation mouse.

Common menu

The common menu allows you to save the model you created, and set display and operation settings common to each mode.

System menu



In the System menu, you can save and load the created model, set the lighting, etc. When you select the System menu, the following submenus are displayed.

新規作成 Create New

This will create a model from scratch, discarding all currently edited model objects, textures, and palettes.

読み込み Load

Load the model file from the PlayStation 2 dedicated memory card (8MB).

When you select Load,

- 1) First, select the device.
- 2) A list of directories in which BASIC STUDIO files are saved will be displayed. Select the directory and file you want to load. Note that you can load model files, texture files, and palette files here.
- 3) Finally, select whether to load all data saved in the file or load each object area or page individually.

サンプル読み込み Load sample

Load a sample model included in "BASIC STUDIO." The included data will be displayed to select the type of model you want to load. Please note that when you load a sample, data currently stored in the object areas, texture pages, and palette number data utilized by the sample will be discarded.

モデルの保存 Save Model

Saves the data of the model you are currently editing. When you select Save,

- 1) First, set which object area you want to save.
- 2) Next, select the device to save to.
- 3-A) If you want to overwrite an existing file, select the directory and then the file to overwrite.
- 3-B) If you want to save to a new file, after selecting the directory, select New File and enter the file name, or select New Directory and enter the directory name and file name.

*About the save file capacity limit

In "POLY-CRAFT," there is a limit to the file size. The limit is determined by the total size of all object areas being saved. If the capacity limit is exceeded, try saving it again after dividing the object areas you are saving into smaller ones so that the limit is not exceeded.

テクスチャーの保存 Save texture

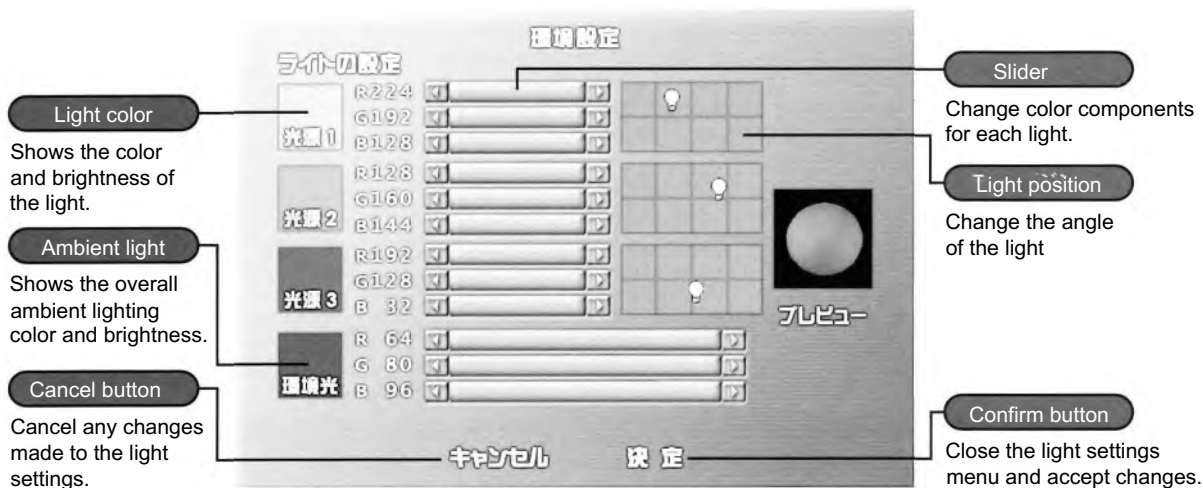
Save the texture page. After switching texture pages or taking snapshots, save the texture here. To save, first specify the texture page to save, then specify the save destination in the same way as when saving a model.

パレットの保存 Save Palette

Save the palette. After switching palettes or taking a snapshot, save the palette here. As with models and textures, specify the palette you want to save, then specify the save destination.

ライトの設定 Light settings

You can set the lighting for the model you are editing. The light settings include the light color and light direction.

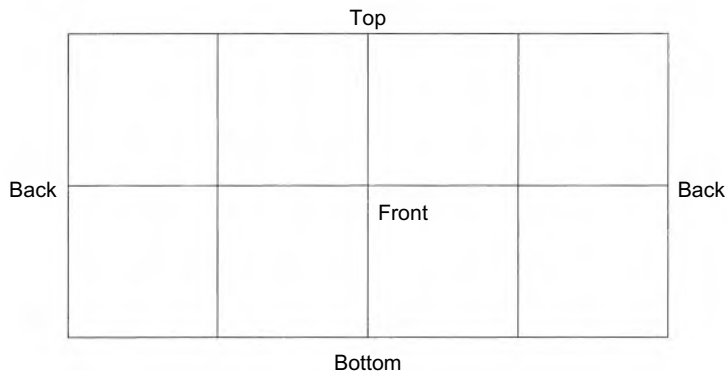


Adjusting the light color

To the right of the light color are three sliders for adjusting the color components of the light. Drag the slider for the color component you want to change.

Changing the light direction

To change the direction of the light, move the cursor to the desired position in the light position and press the circle button. The direction in the model view will be as follows:



Adjusting the ambient light color

This is the setting for the light that shines on the entire scene. Use the sliders to change the light color.

Display color settings

Selecting the display color setting allows you to set the display color of various displays on the model view. Place the cursor on the item you want to change and press the circle button to select it, then change the color using the slider at the bottom of the

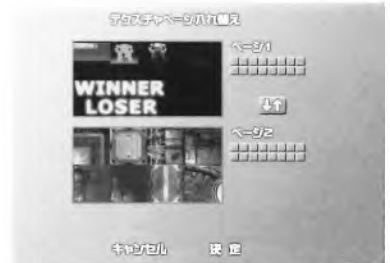
Points	Changes the display color of points.
Lines	Change the display color of lines.
Selected point	Changes the color of selected points.
Selected line	Changes the color of selected lines.
Selected face	Changes the color of selected polygons.
Background	Change the background color.
Grid	Change the color of the reference grid.
X axis	Change the color of the X axis.
Y axis	Change the color of the Y axis.
Z axis	Change the color of the Z axis.



テクスチャーページの入れ替え Exchange texture pages

You can replace the currently loaded texture page. The texture information attached to the model will also be updated, so you can replace the texture page without re-applying the model's texture.

Select the two texture page icons you want to swap and press the Swap button to swap them. The texture information referenced by the model will also be updated at the same time as the swap. For example, if you swap texture pages 1 and 4, any model that had textures from texture page 1 will automatically have textures from texture page 4 applied (and vice versa).



パレットデータの入れ替え Exchange palette data

You can swap palettes just like you swap texture pages.

終了 Quit

Quit POLY-CRAFT and return to the main menu.

Grid display



The grid is a grid of reference lines that serve as a guide for creating a model. The Grid menu allows you to toggle the grid display on and off and set the grid width.



Grid display switch

Click this icon to turn the grid display ON/OFF.



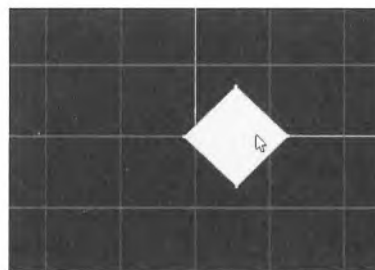
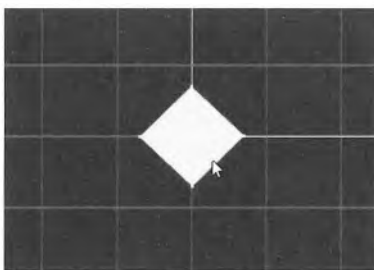
Change the display grid size

Change the size of one grid square. Clicking this icon will open an input window, where you can enter the size of one grid square. The unit of size is determined by "Scale."

Grid snap



The grid snap function allows you to move elements in units of the displayed grid. Moving in grid units makes it easier to adjust the position of the model.

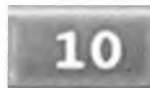


Move in grid units.



Grid snap function ON/OFF

Pressing this icon will toggle between enabling and disabling grid snapping.



Change the snap width

Changes the reference movement width when the snap to grid function is ON .

Scale

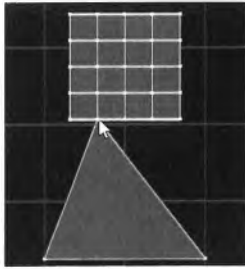


You can change the scale of a model. The scale you set will be reflected in the size of the model when used in EZ-BASIC. There will be no change in POLY-CRAFT.

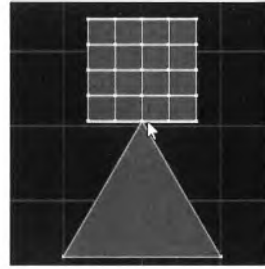
Point snap



When point snapping is enabled, points will snap to other points when you drag them, but lines and polygons will not snap to points when you move them.



You can move a point from one to another.



Move view



To translate the model view, place the cursor over the move icon, hold down the circle button, and move with the directional keys or left stick.

Rotate view



To rotate the model view, place the cursor over the rotation icon, press and hold the circle button, and rotate using the directional keys or left stick.

Scale view



Zoom in and out to scale the model view. To scale the view, place the cursor over the view scale icon, press and hold the Circle button, and use the directional keys or left stick to scale.

View direction



The View Direction menu allows you to view the model from a specific direction, such as looking along the X axis.



Top view

Face the camera from above (along y-axis).



Side view

Face the camera from side (along x-axis).



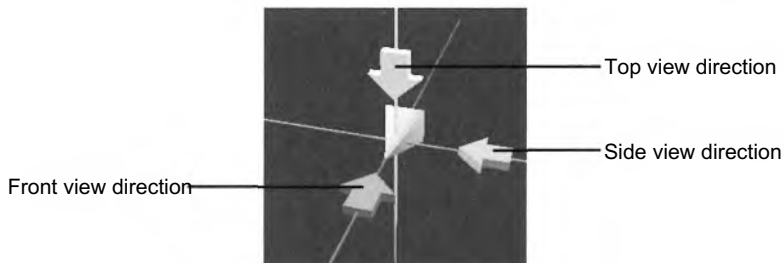
Top view

Face the camera from front (along z-axis).



Fit

Adjusts the view so that the selected element fits perfectly within the screen.



Projection method

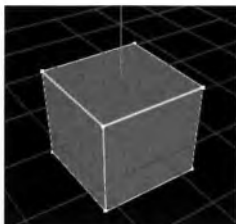


You can switch between displaying the model in perspective projection or parallel projection. If you place the cursor over the projection icon and press the O button, a menu will open. Move the cursor to the icon of the projection you want to change and then release the O button.



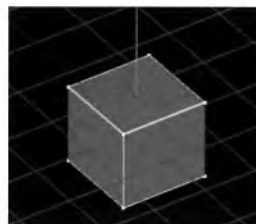
Perspective projection

Displays objects of the same size on the screen so that they appear smaller when they are far away and larger when they are close, as with normal vision.



Parallel projection

Objects are displayed in parallel projection, which displays objects of equal size as the same size regardless of distance.



Model display

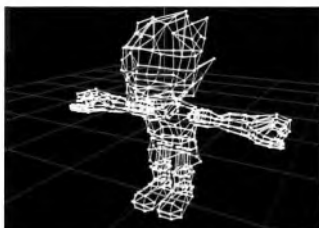


Select the display method for the model. There are four display methods for the model: Wireframe, Solid Polygon, Textured Polygon, and Textured Polygon (No Wireframe).



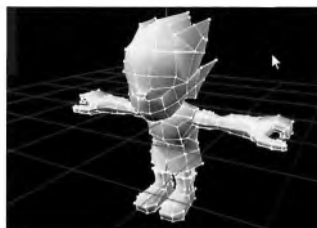
Wireframe

Displays the model using only points and lines. Because there are no polygons, lines and points behind the model are visible.



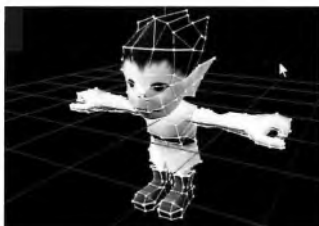
Solid Polygon

Displays the model in its untextured state.



Textured polygon

Displays the model with the texture applied.



Textured polygon (no wireframe)

Displays the model as textured, without showing points and lines.



Model object area switches



By placing the cursor on this button and pressing the circle button, you can switch the active object area. The object areas are numbered 1 to 5 from left to right on the top row, and 6 to 10 from left to right on the bottom row.



The object area currently displayed in the view (red)



Object area with model data (light gray)



Object area without model data (dark gray)

Model Creation Mode

Craft menu

The Craft menu contains icons for creating and manipulating models, such as creating polygons. For icons with submenus, the last selected icon will be displayed in the Crafting Menu.

Selection menu



Selecting the selection icon enters a mode where you can select points, lines, and polygons displayed in the view. Selected elements change color and become available for editing with other commands.



Direct selection

Place the cursor directly on the point, line, or polygon you want to select and press the O button to select it. To cancel the selection, place the cursor on an empty area in the view and press the O button.



Area selection

When you drag within the model View, a rectangular selection area is displayed, and all elements within this area will be selected.



Select contiguous polygons

Place the cursor on any polygon and press the circle button to select all polygons connected to the specified polygon. Polygons that share a line are considered to be connected.

*You can toggle between selected and deselected states of elements by holding down L1 (or the Shift key on a keyboard) and selecting an element.

*You can add selected elements by holding down L2 (or the Ctrl key on a USB keyboard) and selecting an element.

Creating polygons



Pressing the O button in the model view will create a point at the cursor's position. Once four points have been created, a polygon will be created with the four points as vertices. Also, pressing the X button once the third vertex has been created will create a triangular polygon.

Create basic shape



If you place the cursor over the basic shape icon and press the circle button, the basic shape menu will be displayed. You can create a plane, a cube, a sphere, or a rotational solid.



Plane

When you select the plane icon, the plane creation screen will appear. Here, you can set the reference plane, edge length, etc., and press the Create button to create the plane.

- Reference plane
Select whether to create the plane on the XY plane which includes the X and Y axes, the YZ plane, or the XZ plane.
- Side length
Enter the length of the plane's side as a number.
- Number of divisions
Set how many sides the created plane polygon will be divided into.
- Center coordinates
Specify the location of the center of the plane using a numerical value.





Rectangular prism

When you select the cuboid icon, the creation screen will appear. Here, you can set the length of each side, the number of divisions, etc., and press the Create button to create the cuboid.

- Side length
Enter the length of each side.
- Number of divisions
Sets the number of divisions along each axis.
- Center coordinates
Specify the coordinates of the center of the prism.



Sphere

Selecting the sphere icon will open the sphere creation screen. Here, you can set the diameter, number of divisions, etc. and press the Create button to create the sphere.

- Reference axis
Specify the axis that will be the reference for creating the sphere.
- Diameter
Specifies the diameter of the sphere.
- Number of latitude divisions
Specifies how many divisions the sphere will be sliced into along the axis.
- Number of divisions of longitude
Specify how many times to divide the sphere, longitudinally.
- Center coordinates
Specify the coordinates of the center of the sphere.





Rotational solid

You can create a rotated solid around an axis. Selecting the icon will switch to the rotational solid creation screen. Draw the line to rotate and press the Create button to create the solid.

- Line drawing area

Draw the line that will rotate around the axis here. To draw a line, press the Create Line button, move the cursor into the drawing area, and press the O button to create a point. If you create a point again, a straight line will be drawn connecting the points. To stop drawing a line, press the X button in the line drawing area.

- Reference axis

Sets the axis around which the line will rotate.

- Creating lines

Press the Create Line button and then create a line in the line drawing area.

- Number of divisions

Sets how many divisions to rotate around the axis when rotating the line.

- Delete points

Press the Delete Point button, then move the cursor to a point that makes up the line and press the O button to delete the point. When you delete a point, the straight line connected to the point will also be deleted.

- Moving points

You can modify the line by dragging the points drawn in the drawing area.

- Point Snap

When drawing a line or moving a point, you can set so that it will snap to an existing point.

- Side length

Sets the size of the line drawing area.



Line drawing area

Polygon extension icons

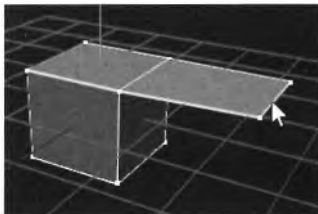


The Polygon extension menu allows you to create a new polygon from a selected line or polygon.



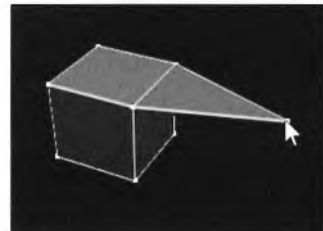
Extend Quad Faces

Select and drag a line to create a new quadrilateral polygon along the dragged direction.



Extend Triangular face

Select and drag a on line to create a new triangle terminating at the cursor, with the selected line as a base.



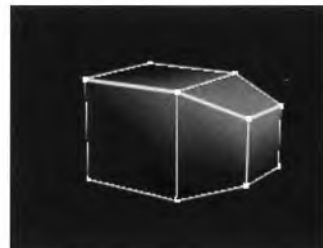
Extrude

If you select a polygon and drag the cursor left and right, the selected polygon will be extruded into a box shape.



Bevel

If you select a polygon and drag the cursor left and right, the selected polygon will be extruded into a trapezoid.



Tools menu



Knife

The polygons that overlap the line connecting two points dragged on the view will be cut. If a selection area is specified, only the selected polygons will be cut.



Point suction

Moves the selected point so that it overlaps with the point specified by the cursor. Overlapping points are treated as one point.



Triangle merging

Select two adjacent triangular faces and combine them to form a quadrilateral.

Move icon



Here you can move and scale points, lines and polygons.



Move

You can move selected points, lines, and polygons. Place the cursor over a selected element and drag it to move the element. You can also select the element to move here.



Scale

You can scale selected lines and polygons. Place the cursor over a selected element and drag left or right to scale the element. You can also select the element to scale here.

Rotate



Rotates the selected element. There are four rotation axes: line of sight, x-axis, y-axis, and z-axis. Place the cursor on the selected element and drag left or right to rotate it. You can also select the element to rotate here.

Copy



The Copy menu allows you to create a duplicate of a polygon. To create a duplicate, you first "copy" the selected polygon to store it in memory, then use "paste" to paste it into the model view.

If you want to create a copy in another object area, switch the object area and then click "Paste." At this time, the difference in scale will be ignored.



Copy

The selected polygons are stored in memory.



Paste in place

A duplicate will be created in the same location in the model view.



Paste to specified location

You can select the location in the model view where you want to create the duplicate. When you select "Paste in place," an image of the model to be duplicated will be displayed, so move it to the location you want to duplicate it and press the circle button to create the duplicate.

Make mirror



This creates a mirror image of the selected object relative to a reference plane. You can select the reference plane from the XY, YZ, or XZ plane. If you set it to join points close to the reference plane, the mirrored duplicated model will be connected at the reference plane. You can set how close to the reference plane the objects must be to join.



Delete



Enters delete mode to delete elements. If an element is already selected, pressing the delete icon will delete the selected element. After entering delete mode, you can also directly specify an element to delete. If you specify a point or line to delete, all polygons that share the specified point or line will be deleted.

Invert, double-side faces



Polygons have a front and a back. Normally, polygons are invisible from the back. To make a polygon visible from the back, you need to use Invert Faces to switch the front and back of the polygon, or use double-sided faces to create another identical polygon on the back.



Invert

Enters a mode to swap the front and back of polygons. If there are already selected polygons, pressing the invert icon will invert all selected polygons. You can also invert polygons by directly specifying them while in invert mode.



Double-sided

This mode creates another inverted polygon behind the selected polygon, so that the polygon is visible when viewed from the back. If there are already selected polygons, all selected polygons will be made double-sided. In double-sided mode, you can also directly specify polygons to make them double-sided.

Surface processing



Changes the polygon display. There are two display methods: "Gouraud shading", which displays adjacent polygons smoothly, and "Flat shading", which does not.



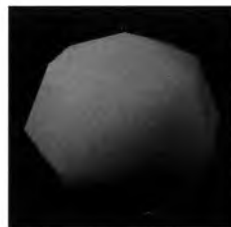
Flat shading

Enters a mode to change the polygon display method to flat shade. If there are already selected polygons, pressing the flat shade icon will change the selected polygons to flat shade. You can also specify polygons directly in flat shade mode.



Gouraud Shading

Enters the mode to change the polygon display method to Gouraud Shade. If there are already selected polygons, clicking the Gouraud Shade icon will change the selected polygons to Gouraud shading. You can also specify polygons directly in Gouraud shading mode.



Edit menu

The edit menu allows you to restrict the plane of operations to make it easier to move polygons in a direction, make it easier to select elements, etc. There are also auxiliary commands available, such as canceling the previous edit.

Snapshot



You can save the model currently displayed in the Model View as a 256x256 pixel image. When you click the snapshot icon, a 256x256 pixel frame will appear in the Model View, so move the frame to fit the model you want to capture.

Pressing the Circle button will display the color reduction settings screen along with an image of the model within the frame. After specifying the color reduction settings and the destination for storing the reduced image and palette, press the Execute button to reduce the colors to 256. After the color reduction, pressing the Confirm button will overwrite the snapshot on the texture page and palette.

To save a snapshot image, select "Save Texture" from the system menu.

• Dither

Sets how to express parts that cannot be expressed with 256 colors when reducing the number of colors.



Mode switch button

You can switch between three modes: "Model Creation Mode," "Texturing Mode," and "Collision Detection Mode."

Model creation mode button



Texturing mode button



Collision detection mode button



Select all



Selects all elements in the object area currently being edited. However, elements hidden by "Hide Elements" in the edit menu will not be selected.

Deselect



Deselects all elements.

Undo



This cancels the last operation, such as moving or deleting a polygon. However, if you switch the object area, the last operation cannot be undone.

Redo



Reverses the last undo action.

Display options



Hide an element

Hides the selected polygons. Hidden polygons are not visible in the model view and cannot be selected.



Show all

Shows all hidden polygons.

Editing options

Make editing easier by limiting the elements that can be selected and restricting movement along specific axes.



Point Selection

Switches point selection ON /OFF. When set to OFF, point selection is disabled.



Line Selection

Switches line selection ON/OFF. When set to OFF, line selection is disabled.



Polygon Selection

Toggles polygon selection ON/OFF. When set to OFF, polygon selection is disabled.



X-axis Editing

Enables/disables editing along the X axis, such as moving and enlarging. If disabled, edits along the X axis will not be performed.



Y-axis Editing

Enables/disables editing along the Y axis, such as moving and enlarging. If disabled, edits along the Y axis will not be performed.



Z-axis Editing

Enables/disables editing along the Z axis, such as moving and enlarging. If disabled, edits along the Z axis will not be performed.

Texture Mapping Mode

In Texture mode, you can apply textures to the models you have created, or change the color of models that do not have textures applied (called solid models).



Craft menu (Texture settings)

In the Texture Map mode craft menu, you can apply textures and more.

Selection



The selection menu works the same as in "Model Creation Mode" (see p.91).

Texture menu



Here you can specify the texture to be pasted and the palette.

Texture specification screen



To specify a texture:

- 1) Change the size of the frame displayed on the texture page to the desired texture area.
- 2) Drag the frame to fit the area you want to use and press the OK button.
- 3) The area surrounded by the frame can now be used as a texture.

*Texture size can be changed in 8 pixel increments.

*When using repeat, the size can be selected from 8, 16, 32, 64, 128, 256, or 512.

Fit



This adjusts the texture size to fit the size of the selected polygon. The size to which the texture is adjusted will be the size of the polygon displayed in the model view. For example, even if the polygon is square, if it is tilted in the view model will appear elongated. Clicking fit changes the texture size to fit this elongated size.



Move, Scale

Move and scale the texture.



Move

Clicking the icon will switch to texture movement mode, where you can drag the texture to the desired location.



Scale

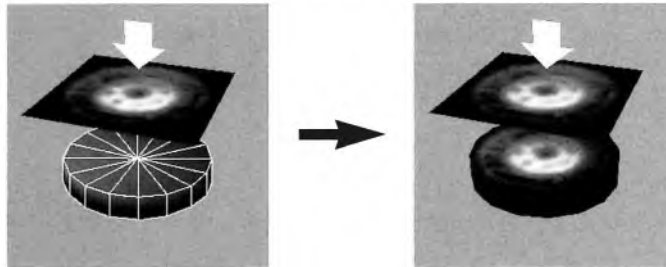
Click the icon to scale the texture. Drag the edges of the texture guide to change the size of the texture.

Set texture



When you press the Set texture button, the texture will be pasted directly onto the currently selected polygon, following the texture guide displayed on the screen.

The texture will be applied by simply placing the texture guide on the model currently displayed in the model view.



This is what the texture looks like from the side.

Delete texture



This will remove the texture from the polygon and return it to a solid polygon mode. If there are already selected polygons, selecting the Delete Texture icon will delete the texture from all selected polygons. You can also delete polygons by directly specifying them while in texture delete mode.

Texture display



Toggles the display of the texture guide on the screen.

How to apply a texture

To apply a texture, follow the steps below:

Step 1: Preparing the texture page

Load the texture file and palette file to prepare the texture to be pasted.

Open the texture file and palette file from the "System Menu".

Next, open the texture selection screen from the texture menu.

If you set the texture page and palette number you want to use here, the contents of the texture page will be displayed on the texture selection screen.



Step 2: Place the texture you want to apply on the texture guide

On the displayed Texture page, resize and move the frame to specify the area you want to use as the texture. Once you have specified the area to use, the specified texture will be displayed in the Texture Guide.

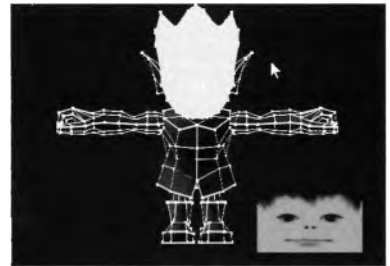


Step 3: Specify the polygons to which you want to apply the texture

Use "Selection" to specify the polygons you want to texture.

Step 4: Point the model in the direction you want to apply the texture

Rotate the model so that it faces the direction you want to apply the texture. For example, when applying a texture to a person's face, make sure the face model is facing forward. In this case, it is recommended that you use parallel projection as the projection method. If you use perspective projection, the applied texture may be distorted.



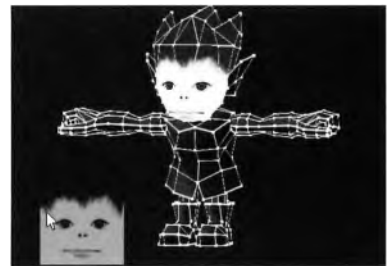
Step 5: Change the texture guide size to match the target polygon.

The texture displayed on the texture guide will be pasted onto the model exactly as displayed, so adjust the size and position of the texture guide to match the size of the polygon.



Step 6: Paste

Once you have finished adjusting the size and position of the texture guide, press the Set texture button to apply the texture.



Craft menu (Surface menu)

In the surface menu at the bottom of the craft menu, you can change the surface settings of polygons.

Surface



The surface icon allows you to set polygon textures for your model. If you hold down the circle button on the surface icon, a submenu will appear.



Fill

Changes contiguous faces of the same color to the current texture setting.



Brush

Specifies a single polygon and changes it to the current material settings.



Dropper

Makes the current material the same as the specified polygon.

Material settings



Sets the current material.

Color

The color of the surface. Adjust the color using the sliders.

Transparency

Adjust the transparency with the slider.

Reflectance

This adjusts the reflectivity (specularity) to light. Changing the reflectivity will change the appearance of the surface.

RGB Toggle



Transparency Toggle



Specular Toggle



When changing surface information, you can switch whether to change the RGB values, transparency, and reflectivity information. For example, by disabling the RGB values and leaving the transparency and reflectivity enabled, you can change the transparency and reflectivity without changing the color.

Edit menu

The edit menu is the same as the "Model Creation Mode" except that some functions are missing.

Switch mode (see p.96)



Select all (see p. 96)



Deselect (see p. 96)



Undo (see p. 96)



Redo (see pg. 96)



Display options (see p. 96)



X-axis editing (see p. 96)



Y-axis editing (see p. 96)

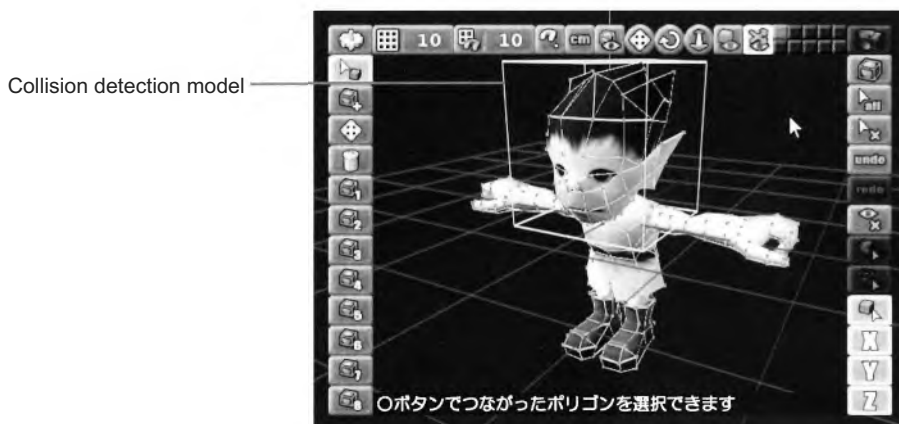


Z-axis editing (see p. 96)



Collision Detection mode

In Collision Detection Mode, you can set the collision detection regions (models) used in "EZ-BASIC".



To set up a collision detection, select a base polygon and click the Create Collision Detection icon. A rectangular prism that surrounds the selected polygon will be set as the collision detection. You can change the size of the collision detection region and move it, but you cannot rotate it. Up to eight collision detection regions can be created in one object area.

Craft menu

Select



The selection menu works the same as in "Model Creation Mode" (see p. 91).

Create collision area



Creates a collision detection model as a rectangular prism that surrounds the selected polygons.

Move, scale



The collision model can be moved and scaled, but cannot be rotated.



Move

You can move the collision detection model. To move it, move the cursor over the collision detection you want to move and



Scale

You can scale the size of the collision detection model. To scale, place the cursor over the collision detection area you want to scale and drag left or right.



Move face

You can move the faces of a collision detection model along an axis. Place the cursor over the face of the collision detection model you want to move and drag it left or right to move the face in parallel.

Delete collision model



Delete the collision detection model. Place the cursor over the collision detection region you want to delete and press the circle button.

Show/hide collision areas



Toggles visibility for each of the eight collision detections.

Edit menu

The edit menu is the same as the "Model Creation Mode", except that some functions are missing.

Switch mode (see p. 96)



Select all (see p. 96)



Deselect (see p. 96)



Undo (see p. 96)



Redo (see p. 96)



Display options (see p. 96)



X-axis editing (see p. 96)



Y-axis editing (see p. 96)



Z-axis editing (see p. 96)



SCORE

NOTE

What is SCORE-NOTE?

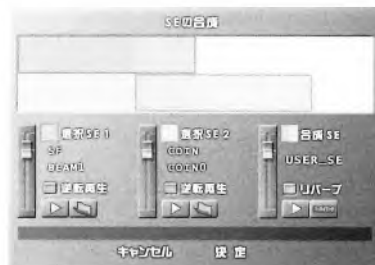
"SCORE-NOTE" allows you to create music and sound effects to be used in BASIC STUDIO. The music and sound effects created here can be used in EZ -BASIC to enhance your game in the same way as graphics and models.

SCORE-NOTE's Structure

SCORE-NOTE is broadly divided into two tools: a sequencer that lets you create songs by placing notes on a sheet of music and adjust the volume of each instrument part using a mixer, and an SE editor that lets you select sound effects to use from a selection of sound effects or combine two sound effects together.



The sequencer has a score mode for creating music, as well as a mixer mode that allows you to change the volume balance of each track.



The SE Editor allows you to select and combine the sound effects you want to use from the available sound effect materials, as well as to create new sound effects by combining sound effects together.

Controls

Controller explanation

Analog controller (DUALSHOCK 2)



* BASIC STUDIO Powerful Game Workshop is compatible with the analog controller (DUALSHOCK 2).
The LED indicator will always be red while the software is running.

Controls - SCORE

Directional buttons	Move cursor
Left stick	Move cursor
Circle button	Confirm, select
X button	Cancel, get note
Circle + Directional buttons	Drag
Circle + Left stick	Drag
SELECT button	PrintScreen function (with popegg connected)

USB mouse explanation

By connecting a USB mouse, you can use SCORE-NOTE with a mouse. The operation method for a USB mouse is as follows:

Mose movement	Move cursor
Left click	Confirm, select
Right click	Cancel, get note

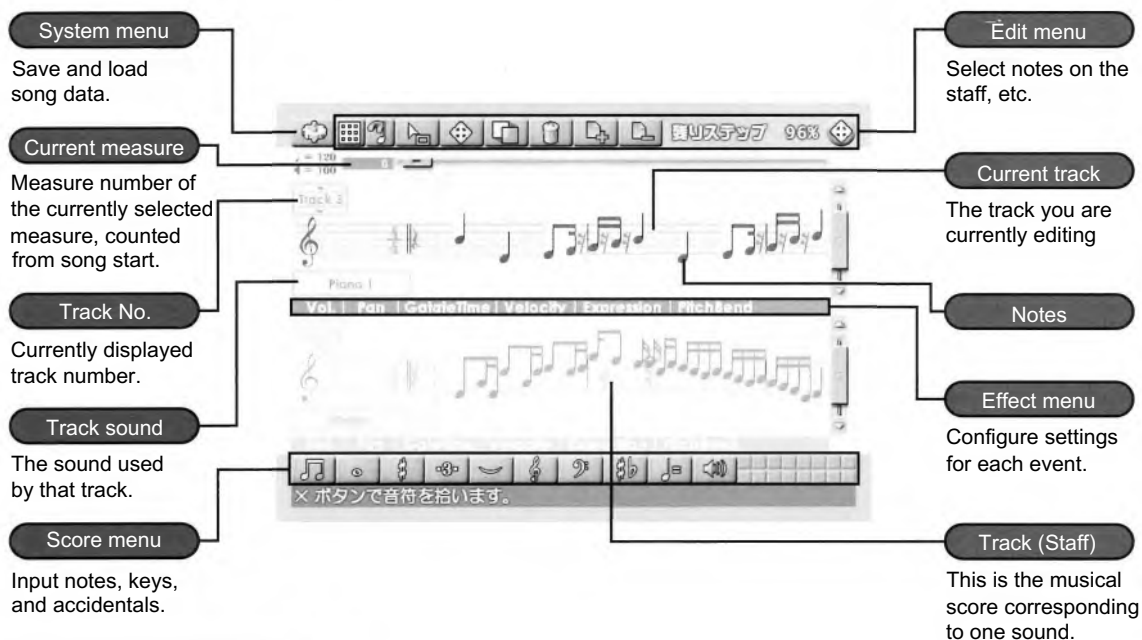
* A "Playstation mouse" cannot be used.

Sequencer

The sequencer is divided into "Score Mode," in which you create songs by placing notes on a staff; "Play Mode," in which you play back the songs you've created; and "Mixer Mode," in which you control the volume balance of each instrument part.

Screen layout

In score mode, you compose music using the following screen.



About each menu

Some of the Edit menus and Score menus, arranged vertically, have submenus, such as the Note menu. Submenus are displayed by placing the cursor over a menu icon and holding down the Circle button. To select an icon within a submenu, hold down the Circle button, move (drag) the cursor to the submenu icon you want to select, and then release the Circle button.



Press the menu icon to display the submenu.



While holding down the O button, move the cursor to the desired submenu, then release.

System menu

The system menu allows you to save and load data you have created.

Create new

Create a new song. Any data currently being edited will be discarded.

Load

Loads song data from the PlayStation 2 dedicated memory card (8MB). When you select Load,

- 1) First, select the device.
- 2) A list of directories containing BASIC STUDIO files will be displayed. Select the directory containing the file you want to load.
- 3) When you select a directory, a list of files in the directory will be displayed. Select the BGM file from this list.

Load sample

Load the sample songs included in "BASIC STUDIO." The included songs will be displayed. Select the song you want to load.

Save

Save the song you are currently editing as a BGM file. When you select Save,

- 1) Select the device to save to.
- 2) If you want to overwrite an existing file, select the directory and then select the file you want to overwrite.
- 3) If you want to save to a new file, after selecting the directory, select "New File" and enter the file name, or select "New Directory" and enter the directory name and file name.

Exit

Exit SCORE-NOTE and return to the title screen.

Edit menu

In the edit menu, you can select and copy notes displayed on the score.

Grid



In musical notation, the relative pitch of a note is determined by the vertical position of the note on the staff, and the time it plays is determined by the horizontal position. The Grid menu displays time reference lines (grids) on the staff to serve as a guide when placing notes. Note that notes are placed in grid units.

Grid



Grid display

Toggles the grid visibility on/off.



Grid size

Switches the grid width. The grid width is based on note lengths, and you can choose from the following five widths.



Quarter note



Eighth note



Sixteenth note



32nd note



64th note

When placing notes using the Note menu, the notes are placed based on the grid. If you are creating a slow song, it will be easier to input notes if you set the grid size to quarter notes. When the grid is hidden, notes are placed based on the length of sixteenth notes.

Selection menu



This is used to select notes placed on the score. You can choose from three note selection methods in the submenu.



Area selection

When selecting a range, you can specify the notes you want to select by placing the cursor on the note and pressing the O button. You can also select all notes in a rectangular area enclosed by dragging.



Time selection

Drag on the staff to select notes by time: all notes within the specified time will be selected, regardless of pitch, as long as they are in the track you are editing.





Pitch selection

Selects all notes of the specified pitch in the track.



Move menu

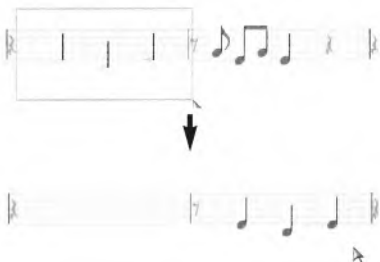


Allows moving and copying of selected notes.



Move

Moves the selected notes. If there are notes at the destination, they are erased and the selected notes are overwritten.



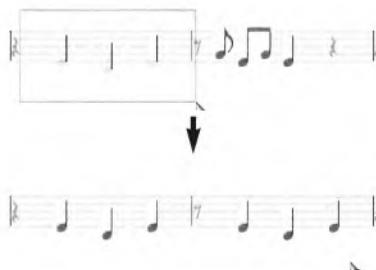
Copy

Copies the selected notes and stores them temporarily in memory.



Paste

Paste the notes stored in memory by copying them to the desired location. If there are notes at the paste destination, they will be overwritten with the new notes.



Delete



This deletes notes placed on the staff. If notes are already selected before pressing the Delete icon, all selected notes will be deleted. You can also delete notes by selecting them after pressing the Delete icon.

Measure edit menu



The Measure edit menu allows you to delete or insert measures (bars), and you can choose to edit bars on the currently edited track or on all tracks.



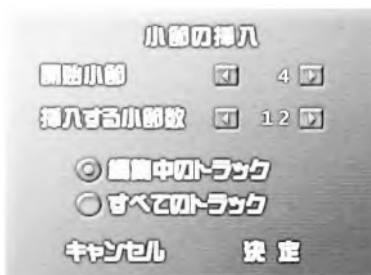
Insert measures

You can insert blank measures on a measure-by-measure basis. Click the Insert Measure icon to open the Insert measure window. Specify the number of measures to insert and the position where you want to insert them. You can choose to insert measures only on the track you are currently editing, or on all tracks.

Start bar - Specify the bar number position where you want to insert.

Number of bars to insert - Specifies the number of measures to insert.

Current track - Inserts bars only into the track you are currently editing.



All Tracks - Inserts bars on all tracks.



Delete measures

Delete individual measures. The measures after the deleted measure will be moved forward. To delete measures, click the Delete Measure icon, and the Delete Measures window will appear. Select the range of measures you want to delete.

Start measure - Set the first measure to delete.

End measure - Set the final measure of the range to delete.

Current track - Deletes measures from the currently edited track only.



All Tracks - Deletes measures from all tracks.

Remaining steps

In SCORE-NOTE, the total amount of data that can be used for a song is limited. This capacity is measured in units called "steps" and is consumed by placing notes and applying effects. The remaining capacity that can be used is displayed as "remaining steps."

When the remaining steps reach 0%, you will no longer be able to place any more notes. You can move the music score by dragging the cursor over the music score movement icon.

Scroll score



You can move through the music score by dragging the cursor over the scroll score icon.

Entering music

Use the Score menu to compose music by entering notes on the staff.

Score menu

The score menu at the bottom of the screen contains menus that allow you to create sheet music by placing notes, accidentals, etc. on the staff.

Mode select



Switch between each mode.



Score Mode

Switch to Score mode to compose music.



Play Mode

Switch to play mode to play songs.



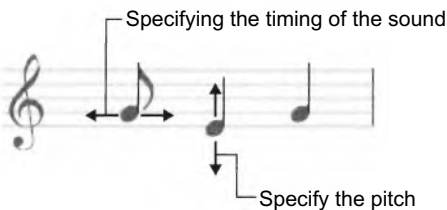
Mixer Mode

Switches to mixer mode, where you can adjust the volume of each track during playback.

Notes menu



You can place notes on the staff. Use the submenu to select the desired note length. The cursor will then change to the selected note, so place the cursor (note) where you want to place the note and press the O button. Specify the pitch of the note by positioning it up and down on the staff, and the timing of the note by positioning it left and right. Rests are automatically created when you place a note.



You can specify the pitch by moving up and down, and the timing of the sound by moving left and right.

To delete a note you have entered, use the selection tool in the Edit menu to select the note you want to delete, and then click the Delete icon.

*If you place the cursor over a note and press the X button, you will automatically switch to placing notes of the length of the hovered note.

Chord input

To input a chord, you can stack notes of different pitches above and below the note that has already been input.

For example, to create a C chord, stack the notes C, E, and G. Note that you can input up to 10 note chords.



About song length

The maximum number of measures that can be input into the SCORE-NOTE is 128, but if you create a complex song that uses lots of data, a warning message will appear because there is too much data, and you will not be able to input notes, etc. You can avoid this by deleting notes or effect data.

About pitch

Each voice has a maximum pitch that it can produce. If you try to play an instrument that originally has a low pitch, such as a bass, above its maximum pitch, the instrument will play the highest note it can produce.

Accidentals menu



Accidentals are symbols such as # and b that can be added to notes. To add an accidental, first select the target note, then select the accidental to add from the Accidental menu. To remove the accidental, simply add the same accidental again. There are five types of accidentals:



Sharp

Raises the note a half step.



Flat

Lowers the note a half step.



Natural

Removes any modification of the note due to key or prior accidentals.



Double sharp

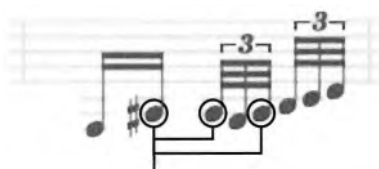
Raises the note a whole step.



Double flat

Lowers the note a whole step.

*Accidentals effect not only for the note with the accidental, but also for notes of the same pitch in the measure after the accidental is used.



The circled notes are all raised by a half step.

Triplets

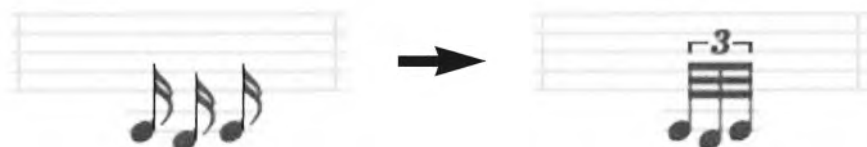


A group of notes obtained by dividing a note such as a quarter note or eighth note into thirds is called a triplet. To create a triplet, select three consecutive notes that are half the length of the note you want to divide into thirds (eighth notes if you want to divide a quarter note into thirds), and then press the triplet icon. To cancel a triplet, select the triplet and press the triplet icon again.

*It does not matter if the pitch is different.

*It is not possible to create triplets across measures.

*Dotted notes cannot be made into triplets.



To create a triplet by dividing a quarter note into thirds, create three eighth notes on the staff, select them using the region selector, and then click the triplet icon.

Tie



Ties two notes of the same pitch. Notes connected with a tie are joined in length, and will sound with one articulation for the duration of the two notes. You can also tie multiple notes together, in which case they will sound with one articulation for the duration of the tied notes. To remove a tie, select the tied notes and click the tie icon.

*Only notes of the same pitch can be connected with ties.

*There can be no rests between the two notes being tied.

*It is okay to cross measures.



Clef menu



Setting the clef determines the absolute pitch of the notes on the staff. Press the clef icon to display the available clefs in a submenu, then select the clef you want to set. The cursor will then change to the selected clef, so select the track you want to set the clef for. Note that you cannot change the clef in the middle of a song.



Treble clef.



Treble clef an octave higher.



Treble clef an octave lower.



Bass clef.



Bass clef an octave higher.

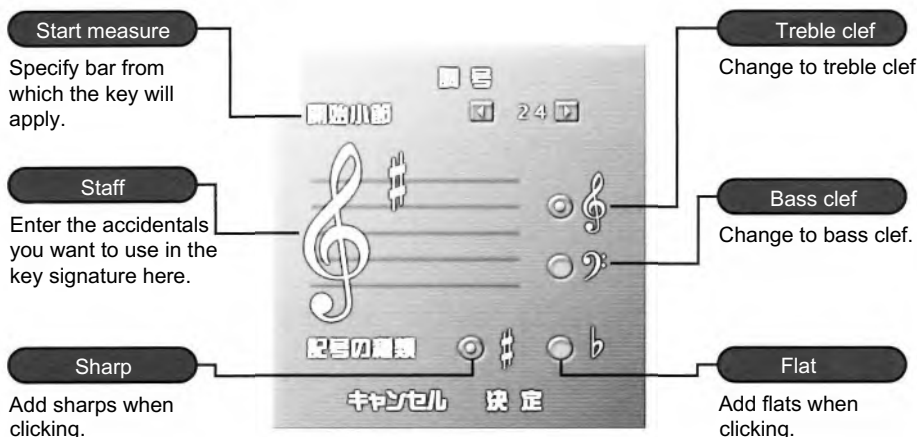


Bass clef an octave lower.

Key signature



You can specify a key such as C major or A minor by setting the key signature. Pressing the key signature icon will display the key signature window, allowing you to select the starting measure, clef, and alterations for the key signature. Use the cursor to add or remove sharps and flats to the key. To delete an alteration, place the cursor on the alteration you want to delete and press the O button. The key signature you set will apply to all tracks. You can also change the key at a particular bar. If you change the key in the middle of a song, the key will change from the starting bar to the end of the song.



Tempo



You can set the tempo for the entire song. Click the tempo icon to display the numeric tempo input window, where you can enter the desired tempo value.

*Tempo is a measure of how many quarter notes are played per minute. For example, if the tempo is 120, then 120 quarter notes will be played per minute.

Master volume



Set the overall volume of the song. Press the Master Volume icon to open the volume input window, then enter the volume you want to set. The higher the Master Volume number, the louder the volume.

Score section

The score section displays the melody on a staff, as well as two tracks, including track numbers and instruments. You can also apply effects to each note here.

Scrollbar (horiz)

This indicates your current position in relation to the entire score, and you can move the score by dragging the bar.

Track (staff)

The score for one instrument.

Track number

The currently displayed track number.

Scrollbar (vert)

Dragging scrolls the music up and down, revealing notes that are not visible above or below.

Effect menu

Effect menu for the upper track.

Event window

This is the state of the event value set in each effect menu.

Track number



Shows the currently displayed track number. You can also change the track displayed by pressing the buttons above and below the track number. Each track can play one instrument.

Instrument



These are the tones assigned to the track. Place the cursor over a tone and press the O button to select a tone from a list of available instruments.

Instrument tone selection screen

Selectable tones (gray)

Unselectable tone (red)

Tone in use in another track (blue)



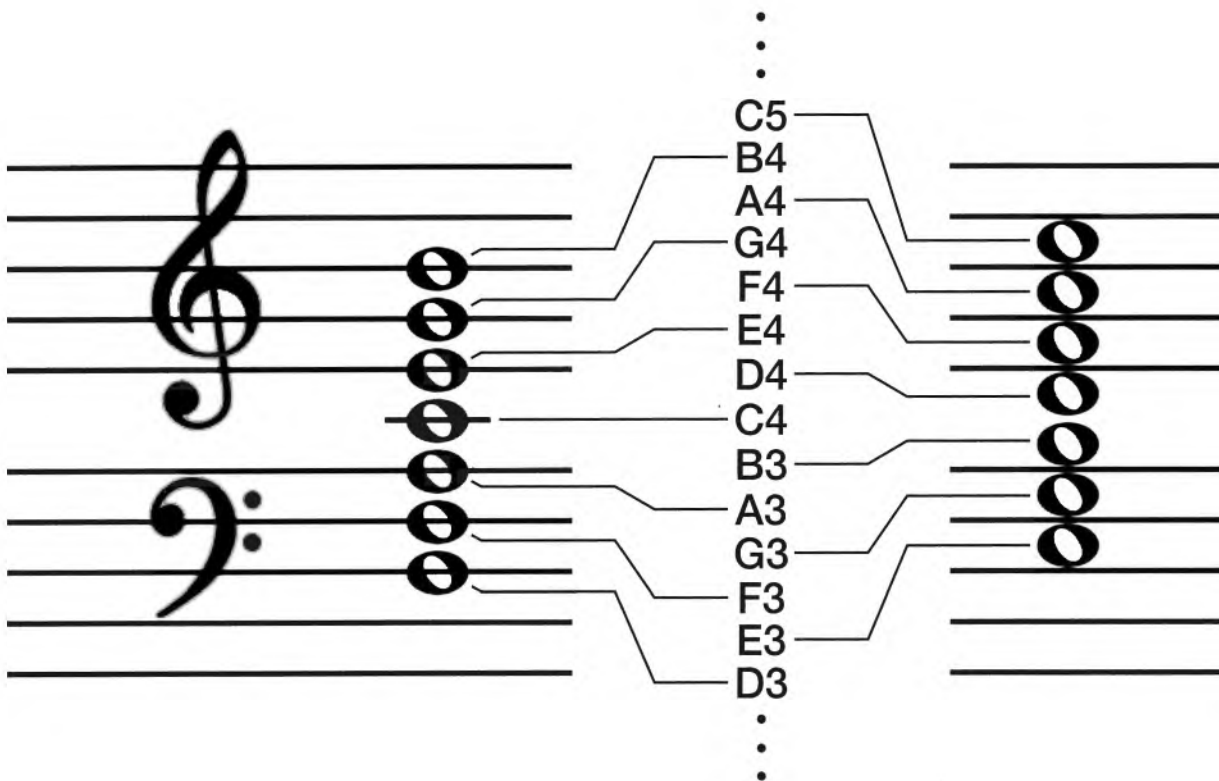
To select a tone, move the cursor to the name of the tone you want to use and press the O button. SCORE-NOTE has a set total capacity for instruments that can be used in one song. The amount of space each tone uses varies, and tones that cannot be selected due to capacity limitations are displayed in red. You can also check the remaining capacity to see how much of the currently used tone capacity is in relation to the limited capacity.

About drum instruments

"Standard Drum1," "Standard Drum2," "Orchestra Drum," and "Electronic Drum," shown in the instrument list, are drum sets that combine rhythm instruments. Unlike regular instrument sounds, drum sets change the type of instrument that is played depending on the pitch of the note. For example, on a track with the Standard Drum2 sound, a note in the C position will produce a crash cymbal, and a note in the D position will produce a ride cymbal.

• Note position and note name on the staff

The drum tone correspondence table shows the note positions that correspond to the drum rhythm instruments by note names. The note names in the table correspond to the musical staff as follows:



In this way, the notes on the staff can be represented as A to G, followed by a number to indicate the octave.

• Drum instruments and associated tones

Standard Drum 1

Basic drum set

Instrument name	Pitch
Crash cymbal 1	A3
Ride cymbal	F3
High tom	D3
Crash cymbal 2	C#3
Mid tom	B2
Open high hat	A#2
Low tom	G2
Closed high hat	F#2
Snare drum	D2
Base drum	C2

Standard Drum 2

The rhythm instrument configuration is the same as Standard Drum 1, but it is easier to handle on a bass clef staff.

Rhythm instrument name	Pitch
Ride cymbal	D4
Crash cymbal 1	C4
Crash cymbal 2	B3
Open high hat	A3
Closed high hat	G3
High tom	F3
Mid tom	E3
Low tom	D3
Snare drum	C3
Base drum	A2

Orchestra Drum

Has a richer range of instruments than the standard set.

Instrument name	Pitch
High wood block	F6
Low wood block	E6
Cortales pad	C6 ~C4
Concert cymbal	B3
Tom	G3 ~A#3
Tambourine	F#3
Low tam	F2 ~F3
Toma (strong)	E2
Tom (soft)	D#2
Concert snare drum	D2
Off-snare drum	C#2
Concert bass drum 1	C2
Concert bass drum 2	B1
Gong	D1~A#1
Snare roll	C#1
Off-snare roll	C1
Concert bass drum (soft)	B0
Concert bass drum (strong)	A#0

Electronic Drum

An electric drum with handclaps and other elements.

Instrument name	Pitch
Ride cymbal	D4
Crash cymbal 2	C4
Crash cymbal 1	B3
Open high hat	A3
Closed high hat	G3
High tom	F3
Mid tom	E3
Low tom	D3
Snare drum	C3
Hand clap	B2
Bass drum	A2

Effect menu

The Effects menu allows you to open windows for various event functions.

- **Vol (Volume)**

Set the volume for each track. Selecting the Volume menu will open the Volume window. Drag the Vol slider on the left to set the volume. The Volume menu will manage the volume of the entire track at once.

- **Pan**

Set the panning for each track. Panning is the left-right volume balance of the sound produced by that track. Use the Pan slider on the left to adjust. When the slider is moved up, the proportion of the volume output from the right increases, and when it is moved down, the proportion of the volume output from the left increases.

- **Expression**

The Expression menu also sets the volume for each track, but unlike the Volume menu, it allows you to make fine adjustments to the volume within a track, allowing you to add emphasis to your song.

Volume graph



In the Expression window, select the desired setting from the menu at the bottom.

- **Numeric input**

Enter the volume as a number.

- **Pen**

By dragging on the volume graph, the volume will change according to the cursor movement.



- **Line**

By dragging on the volume graph, the volume will change to a linear interpolation between the two points.



- **%**

The current volume setting in the track to the percentage entered as 100%. When entering a percentage, you can change the value without changing the relative relationship between each event value.

- **+**

Adds the specified amount to all volumes in the track.

- **-**

Subtracts the specified amount from all volumes in the track.

- **MAX**

Sets the volume limit. Values exceeding the limit will be changed to the limit you set.

- **MIN**

Sets the minimum volume limit. Values below the minimum limit will be changed to the minimum limit you set.

• Gate Time

This indicates the time from when a note is sounded (note on) to when it stops sounding (note off). A short setting allows for staccato sounds, while a long setting allows for legato sounds.

Velocity level
Intensity of the sound.

Note on
Timing for start of the note.

Note off
This is the timing at which the sound stops. Drag here to change the Gate Time .

Gate time
Duration from start to end of sounding.

• Velocity

You can set the volume strength when a note is sounded (note on) on a per-note basis. By adjusting the strength of each note, you can add accents to your song. You can make similar settings in Expression, but Velocity sets the volume strength for each note.

Velocity level
Intensity of the sound. Velocity controls this level.

Note on
Timing for start of the note.

Note off
This is the timing at which the sound stops.

Velocity is set for each individual note, so the graph where notes overlap will look like this.

• Pitch Bend

You can make fine adjustments to the pitch of a note. Using the pitch bend, you can bend the note while it is being played. For example, by adjusting the pitch bend slightly up and down on a violin, you can create a vibrato effect. The pitch bend can change the pitch by up to two steps up or down.

Play mode

In Play Mode, you can play back the songs you've created, and you can also configure loop settings here.

Pointer

The vertical line moves across the score as the song progresses.



Play menu

Control playback of the song.

Play menu



Rewind

Move the pointer back in 1 measure steps.



Start playback

Plays the song from the beginning.



Stop

Stop playback and return to start of song.



Play from specified location

Playback starts from the pointer.



Pause

Pause playback.



Fast forward

Fast forward the pointer by one measure.



Repeat play

This will loop the song. Click the repeat play icon to open a window where you can enter the start and end measures of the repeat. If you save the song with loop playback set here, it will also loop when you use BGMPPLAY in EZ-BASIC.

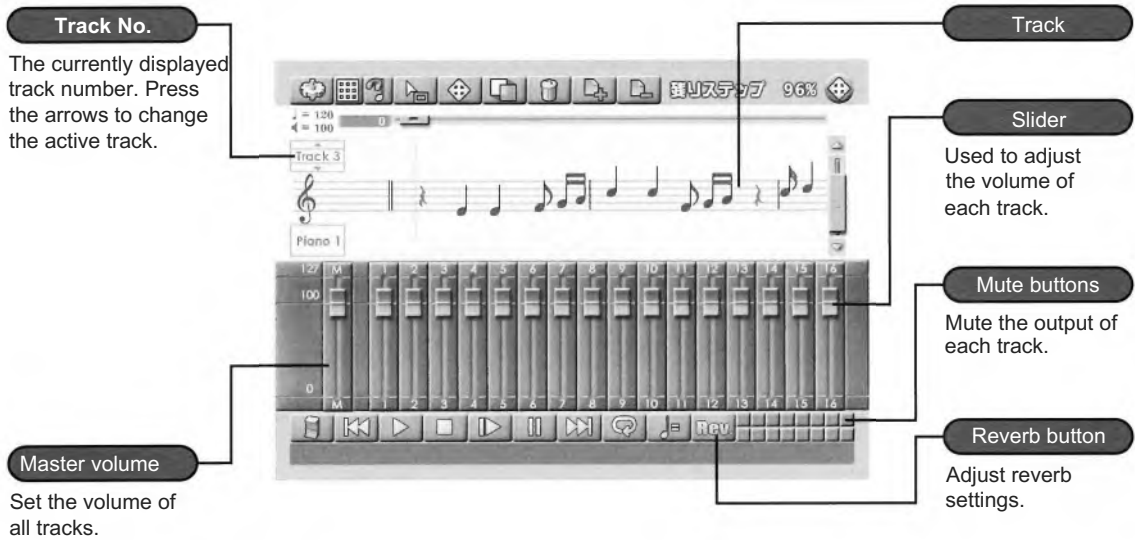


Mute buttons

The tracks whose mute buttons are pressed will not be played during playback. The mute buttons correspond to tracks 1 to 8 from the top row to the right, and 9 to 16 from the bottom row to the right.

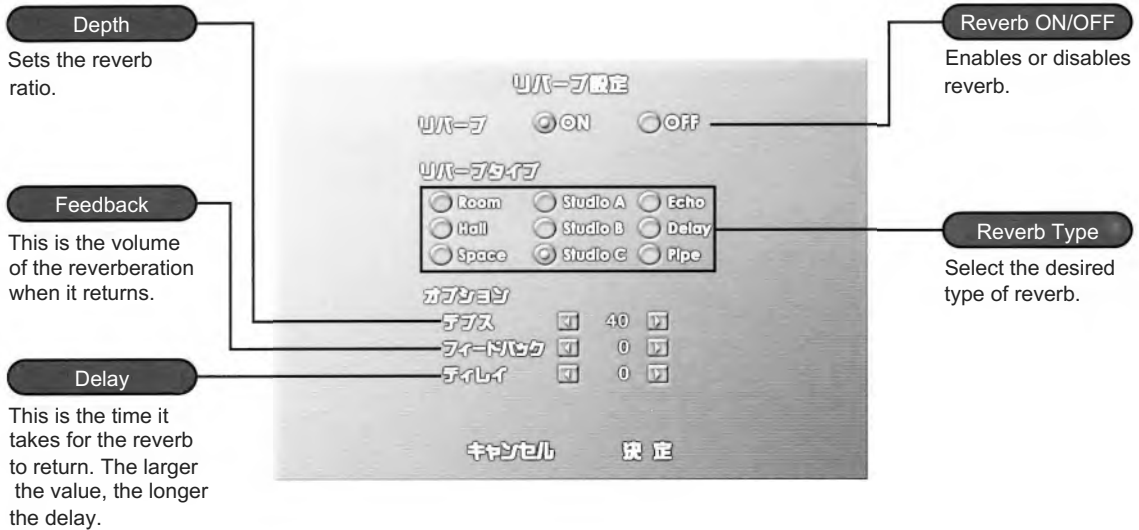
Mixer mode

In Mixer mode, you can adjust the track volume of each track while playing the song.



Reverb settings

Pressing the reverb button will open the reverb settings window, where you can modify the type of reverb.



Reverb is the reverberation that occurs when sound is played in a room or hall. To set reverb, select the reverb you want to set from the Reverb Settings window. Depending on the type of reverb, you can also set depth, feedback, and delay.

Volume settings

Just like in sequencer mode, you can also set the volume in mixer mode. The volume settings are divided into three parts, and the final volume is the value that is output when these parts are multiplied together. In mixer mode, you can only set the master volume and track volume.

Master volume setting

The Master volume sets the volume of all tracks. By adjusting the Master volume, you can adjust the final volume without affecting the volume balance of each track or the intonation of the song. To set the Master Volume, use the Master Volume slider.

Setting the track volume

Track volume is used to balance the volume of each track. By changing the track volume, you can change the volume of the track without losing the intonation of the track. To change the track volume, adjust the slider corresponding to each track in Mixer Mode.

Expression settings

Expression allows you to change the volume of each track as the song progresses. This allows you to add dynamics to your song. To set Expression, open the "Expression" window in Sequencer mode and set it there.

Mute

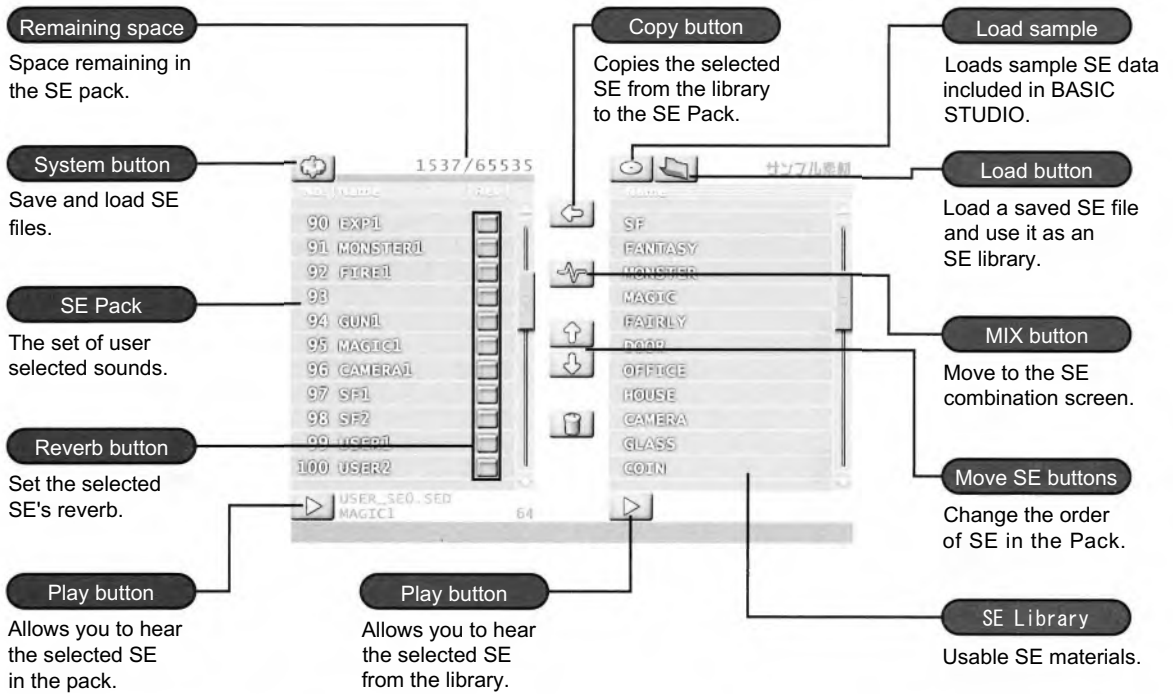
Pressing the mute button will mute the track during playback. Use this when you want to play just one track to check it.

Play menu

For information about the Play menu, see page 121.

SE Editor

EZ-BASIC uses sound effects by loading SE files that contain a number of sound effects (SE), such as explosion and impact sounds. The SE Editor allows you to select pre-prepared sound effects or make combinations of these sound effects, and save them all together in a single EZ-BASIC data file.



System menu

You can save the SE pack you created as an SE file, and load SE files.

Create New

A new SE pack will be created. Any data currently being edited will be discarded.

Save

Save the SE pack you are currently editing as an SE file. When you select Save,

- 1) Select the device to save to.
- 2) If you want to overwrite an existing file, select the directory, and then select the file you want to overwrite.
- 3) If you want to save to a new file, after selecting the directory, select "New File" and enter the file name, or select "New Directory" and enter the directory name and file name.

Load

Load an SE pack file from the memory card. When you select Load,

- 1) First, select the device.
- 2) A list of directories where BASIC STUDIO files are saved will be displayed. Select the directory containing the file you want to load.
- 3) When you select a directory, the files in the directory will be displayed in a list. Select the SE file you want to load from here.

Exit

Exit the SE Editor and return to the title screen.

Making an SE Pack

To make an SE pack, select the SEs you want to use from the pre-prepared materials and register them in the pack.

Loading and Selecting SEs

The SE Editor comes pre-loaded with sound effects categorized by genre. At the top of the sound effect material list are the "Load Sample" and "Load" buttons, which you can use to load sound effects. To load a sample, press the "Load Sample" button, and the genres of the included sound effects will be displayed in the sound effect material list. Simply move the cursor over the genre of the sound effect you want to use and press the circle button.



Sound effects for the selected genre will then be displayed. Select the sound effects you want to use here. You can listen to the selected sound effects by pressing the play button below the sound effect list.

You can also use SE files stored on a PlayStation 2 dedicated memory card (8MB) as material and include them in a newly created SE pack. In this case, press the "Load" button, select the MEMORY CARD slot containing the SE file you want to use from the SE material list, and load the SE file in the same way as loading an SE file from the system menu.

Selecting the SE slot

Next, select the number in the SE Pack list to record the SE you want to use. This number will be used when specifying SEs in EZ-BASIC. After selecting the recording destination, press the Copy button and the SE selected in the SE Material List will be recorded into the SE Pack.

* There is a limit to the number of SEs that can be included in an SE pack. Up to 128 SE materials can be included, but due to the capacity limit of the SE pack, it may not be possible to record even if there are fewer than 128. You can check the remaining capacity of the SE pack in the top right corner of the screen.

Deleting sound effects

To delete an SE included in the SE pack, select the desired SE from the pack list and press the Delete button.

Reverb setting

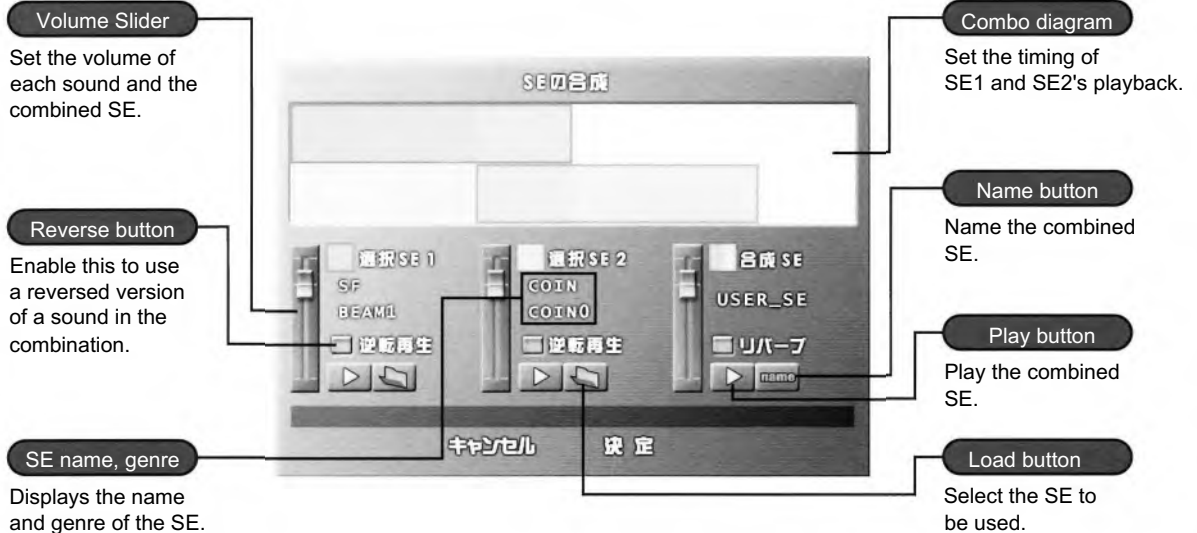
To add reverb to a sound effect, select it from the sound effect pack list and press the reverb button. You can also listen to the currently selected sound effect by pressing the play button below the sound effect pack list.

Move SE within the SE Pack

To change the recording number of an SE within an SE pack, select the SE you want to change and press the SE Move buttons. You can shift the recording number of the SE one by one.

SE Combination

You can create a new SE by combining two SEs. To combine SEs, select the recording destination from the SE pack list and press the MIX button.



Selecting the SE to be combined

Select the SE to be combined. Press the Load button to select the SE you want to use for composite. You can listen to the selected SE by pressing the Play button below.



SE Combination adjustments

When combining, you can adjust the volume and timing of the two SEs.

- Timing settings
The two source SE are displayed on the combination diagram. By dragging each block, you can adjust the timing of the two SEs.
- SE adjustment
You can adjust the volume of each sound effect by adjusting the volume slider next to the selected sound effect, and you can set the volume of the combined sound effect by adjusting the slider next to the combined sound effect.
- Reverse playback
You can play the selected SE in reverse and combine it.
- Reverb
Apply reverb to the combined sounds.

Checking the combined SE and setting the SE name

You can listen to the combined SE by pressing the play button below the combined sound. If you are happy with the result, press the name button to give the synthesized SE a name, then press the confirm button. The synthesized SE will be recorded to the recording destination selected on the SE edit screen.

初期 データ

Initial Data

EZ-BASIC comes with some sample materials pre-loaded.

The pre-loaded sample materials are shown below.

For other sample materials, please check and use them by using the "Load Sample" function in each tool.

Initial texture page data

1	Fonts (letters, symbols, card symbols, etc.)
2	Fonts (letters, symbols, card symbols, etc.)
3	Menus, switches, icons, etc.
4	Menus, switches, icons, etc.
5	Pieces (Go, Chess, Shogi, Mahjong tiles, etc.)
6	Pieces (Go, Chess, Shogi, Mahjong tiles, etc.)
7	128 32x32 px. characters for vertical shooters
8	128 32x32 px. characters for vertical shooters
9	128 32x32 pixel characters for RPGs
10	128 32x32 pixel characters for RPGs
11	128 32x32 pixel characters for action games
12	128 32x32 pixel characters for action games
13	Textures for default models
14	Textures for default models
15	Textures for default models
16	Textures for default models

Regarding palette data, the corresponding palette is pre-loaded for the initial texture page.

Initial sample models

Model buffer 1 - Material models
1: Diamond box
2: Square hanging
3: Cylinder
4: Triangular prism
5: Red triangular prism
6: Green triangular prism
7: Blue triangular prism
8: Black triangular prism
9: Sphere
10: Black ball

Model buffer 2 - Material models
1: Box
2: Red box
3: Green box
4: Blue box
5: Black Box
6: Red wire box
7: Blue wire box
8: Border box
9: Border box
10: Border box

Model buffer 3 - Terrain models
1: Soil box
2: Grass box
3: Sand box
4: Water surface box
5: Stone wall box
6: Wood
7: Bush
8: Rock
9: Stone pillar
10: Wooden box

Model buffer 4 - Item models
1: Treasure chest
2: Drum
3: First aid kit
4: Heart
5: Medicine bottle
6: Dollar bags
7: Bomb
8: Crystal
9: ?
10: Sky

Model buffer 5 - Effect models
1: Yellow beam
2: Blue energy ball
3: Iron ball
4: Meteorite
5: Missile
6: Pillar of Fire
7: Water column
8: Shockwave
9: Explosive Ball
10: Smoke ball

Model buffer 6 - Fantasy models
1: SD soldier
2: SD magician
3: SD Monk
4: SD Queen
5: SD King
6: SD Skeleton
7: SD Goblin
8: SD Cyclops
9: SD Demon
10: SD Dragon

Model buffer 7 - Battle models
1: SD tank
2: SD truck
3: SD turret
4: SD attack helicopter
5: SD Fighter
6: SD Submarine
7: SD aircraft carrier
8: SD Battleship
9: SD Space Fighter
10: SD Robot

Model buffer 8 - Comical models
1: SD Dog
2: SD Cat
3: SD Rabbit
4: SD Monkey
5: SD Penguin
6: SD Whale
7: SD Dolphin
8: SD UFO
9: SD Martian
10: SD Gray

Initial sample music

BGM buffer	Song title
1	Casino song from Lunatic Dawn Tempest
2	Battle song from Lunatic Dawn Odyssey
3	ZEUS Carnage Heart Second Training School Song
4	Library song from Lunatic Dawn Tempest

Initial SE

SE number	SE content	SE number	SE content
1	ITEM1	20	FANTASY5
2	ITEM2	21	FANTASY6
3	ITEM3	22	BATTLE1
4	ITEM4	23	BATTLE2
5	ITEM5	24	BATTLE3
6	ITEM6	25	BATTLE4
7	ITEM7	26	BATTLE5
8	ITEM8	27	BATTLE6
9	EFFECT1	28	BATTLE7
10	EFFECT2	29	BATTLE8
11	EFFECT3	30	BATTLE9
12	EFFECT4	31	BATTLE 10
13	EFFECT5	32	BATTLE 11
14	EFFECT6	33	COMICAL1
15	EFFECT7	34	COMICAL2
16	FANTASY1	35	COMICAL3
17	FANTASY2	36	COMICAL4
18	FANTASY3	37	COMICAL5
19	FANTASY4	38	COMICAL6

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